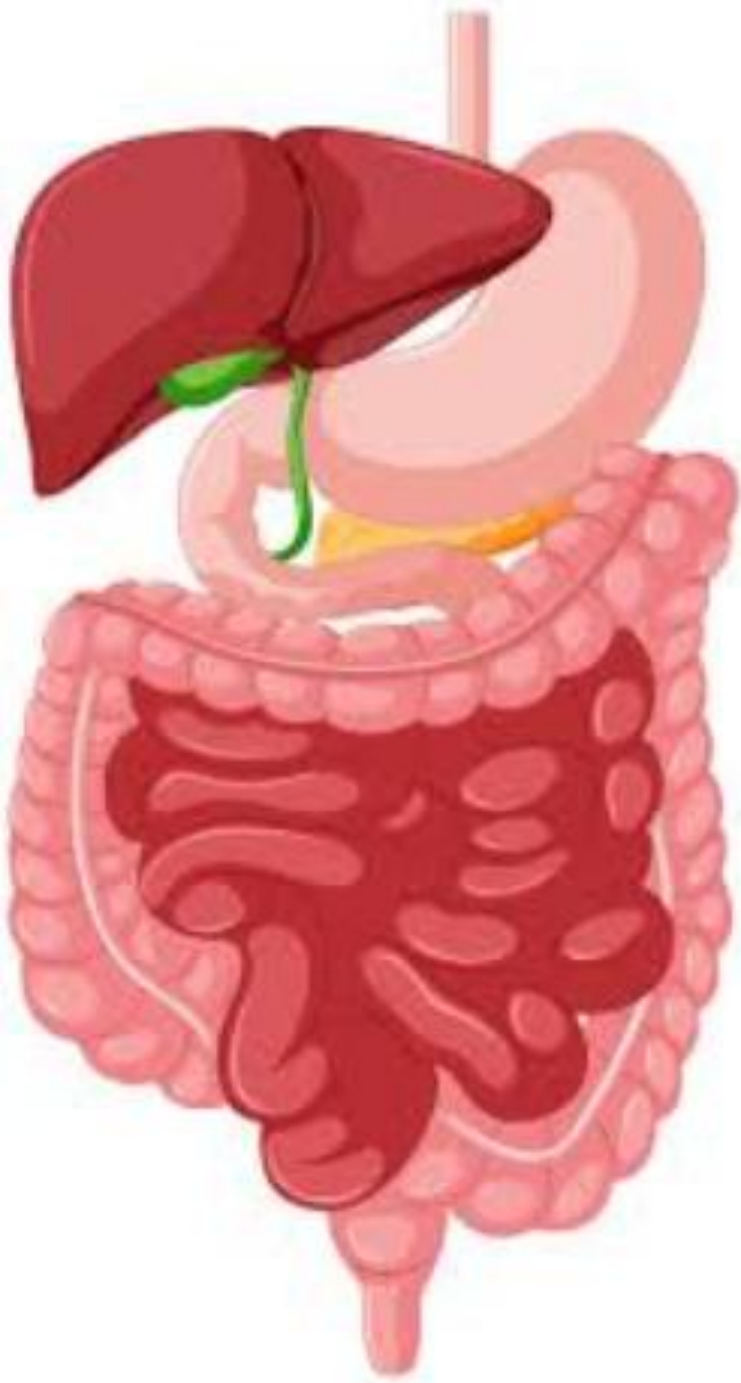




PATHOLOGY

GI TRACT

Dosen Pembimbing:

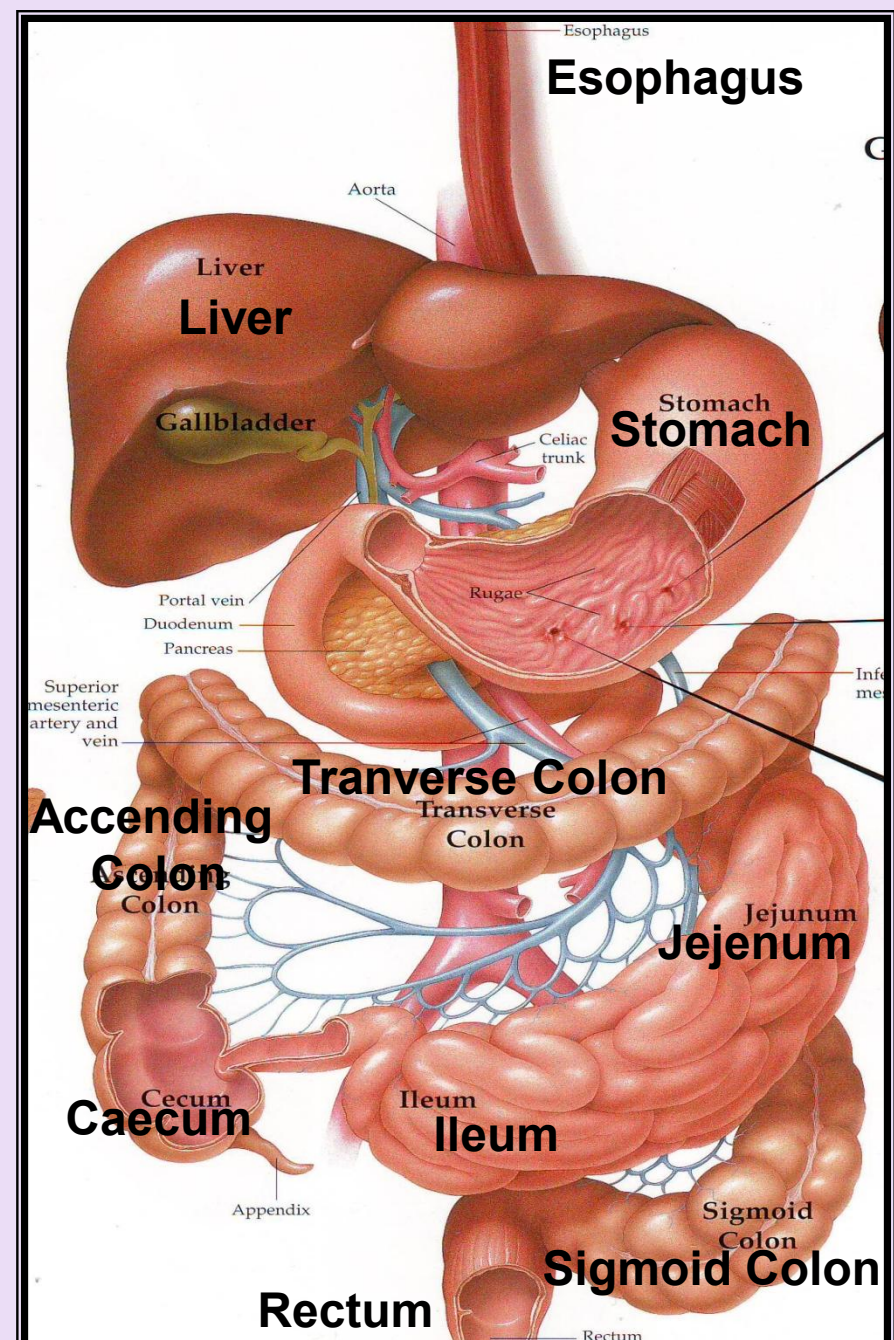
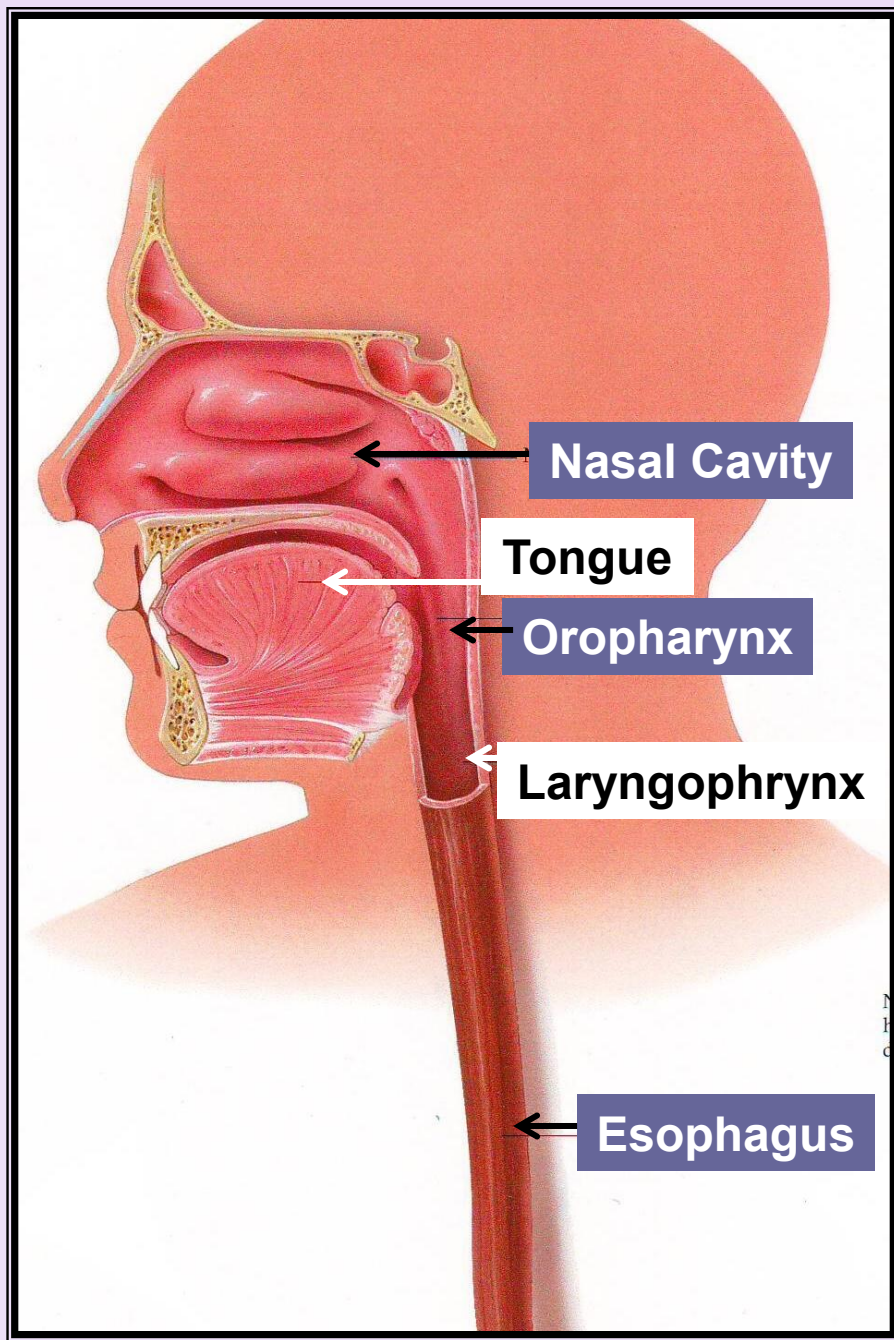
Prof. dr. Syarifuddin Wahid, Ph.D., Sp.PA(K)

Prof. Dr. Upik A. Miskad, PhD., Sp.PA(K)

dr. Amalia, M.Kes., Sp.PA(K)

Sub CPMK

- Mahasiswa mampu menjelaskan gambaran patologi kelainan system gastroenterohepatologi



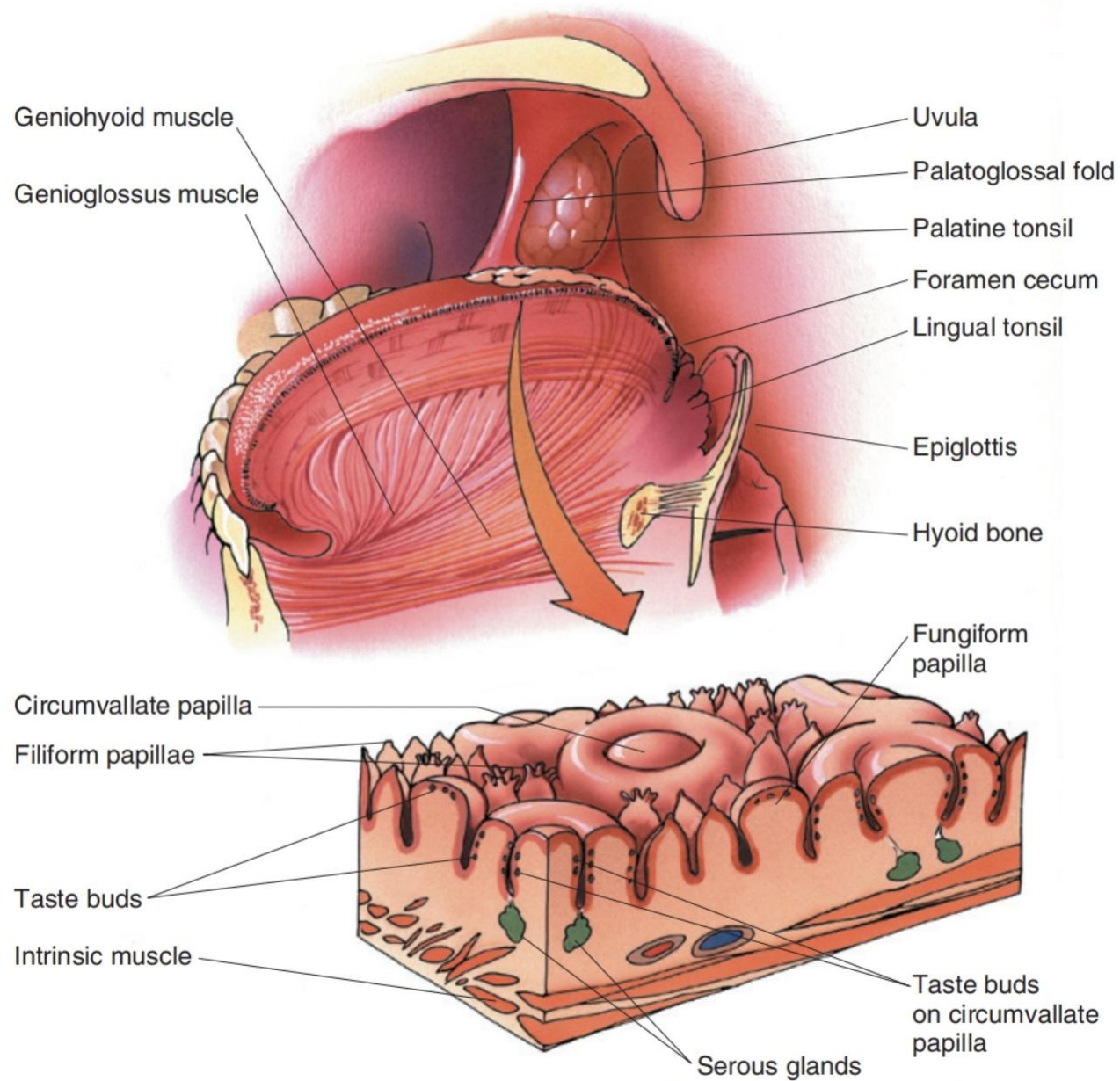
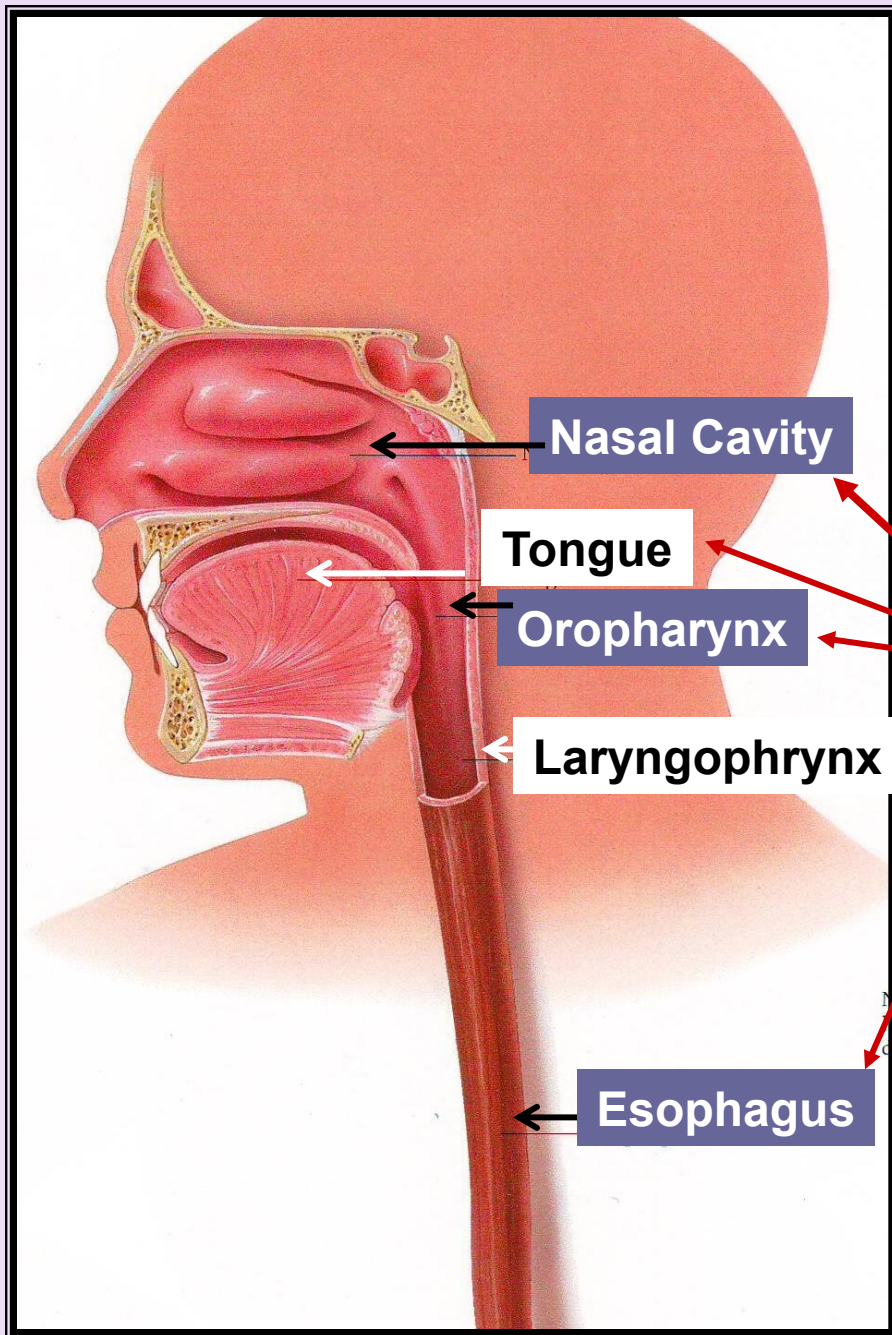
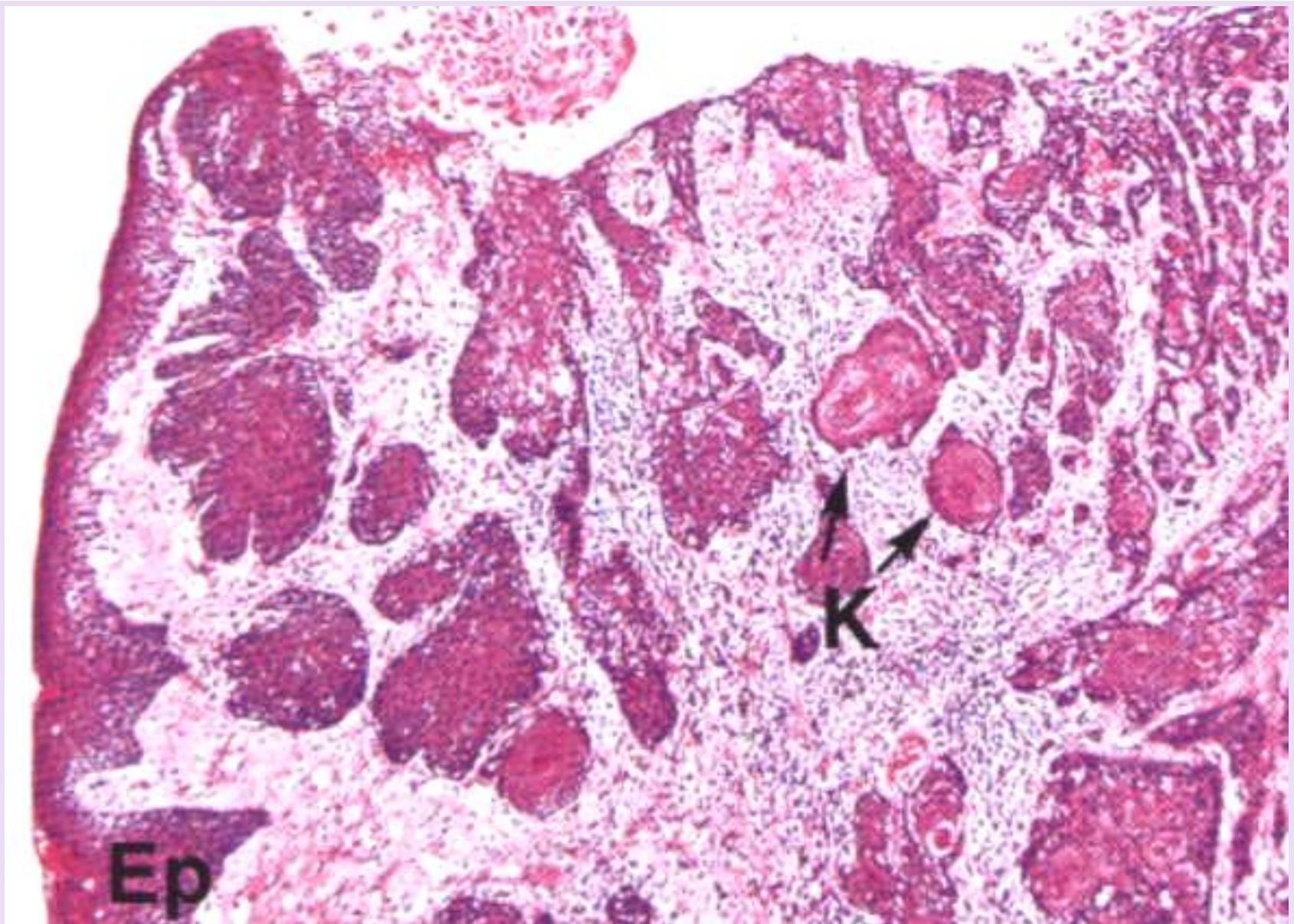


Figure 16-9 Schematic diagram of the tongue and its lingual papillae.



Squamous Cell Carcinoma



CARCINOMA OF THE TONGUE
(Squamous cell carcinoma)

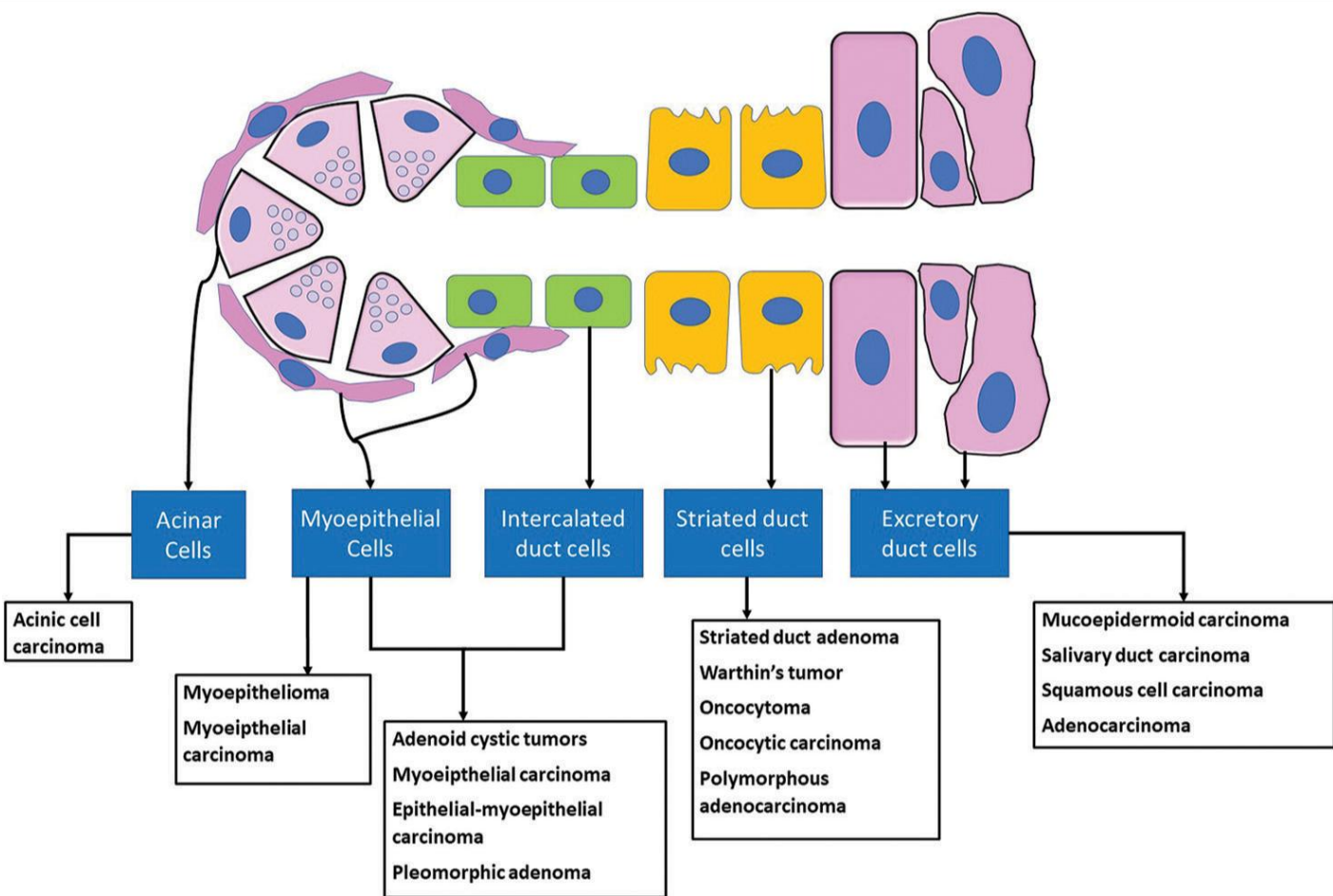
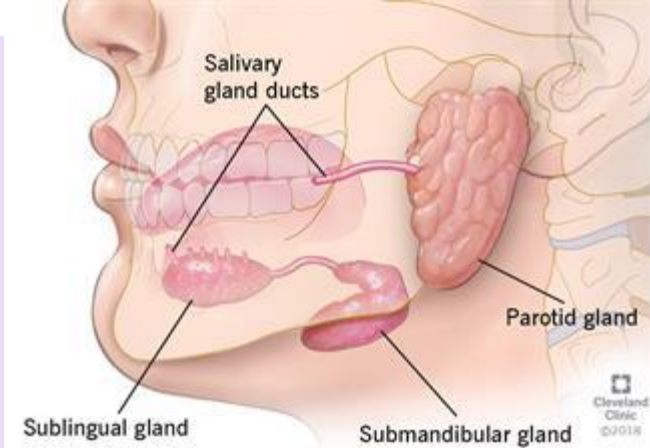
Ep= Epitel
K = Keratin

CARCINOMA OF THE TONGUE (Squamous cell carcinoma)



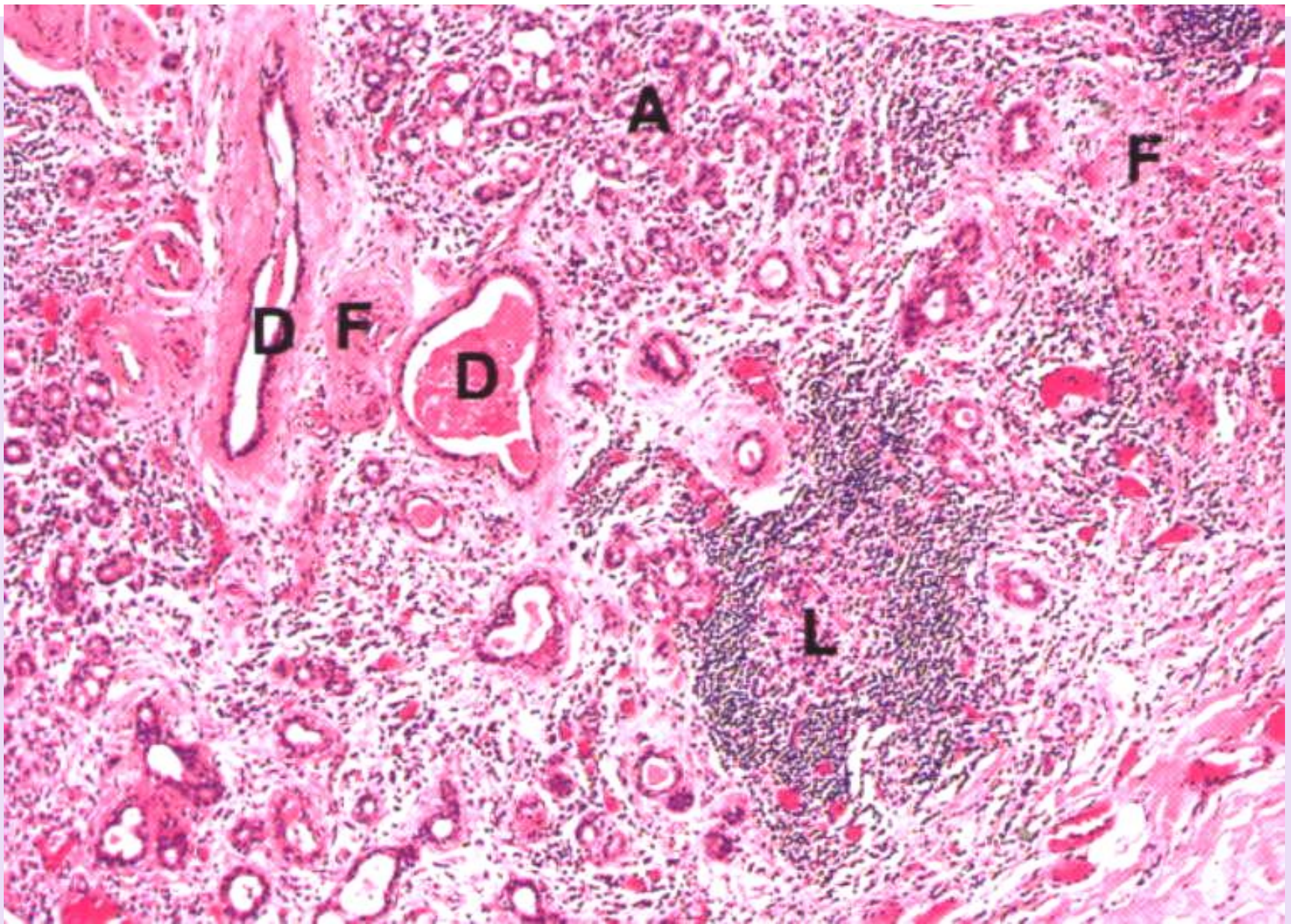
Figure 2: A 73-year-old woman presented with a 1-year history of a persistent ulcer and white patch on the right margin of the tongue and soreness while eating.

SALIVARY GLAND



CHRONIC SIALADENITIS

- Chronic obstruction of a large salivary gland duct by a calculus (*sialolith*) .
- *chronic inflammation* and *atrophy* in the gland
- Histopathology :
 - *Dilatation* of duct (D)
 - Periductul *fibrosis* (F)
 - Formation of *lymphoid follicles* (L)
 - *Atrophy* of the secretory acini (A).

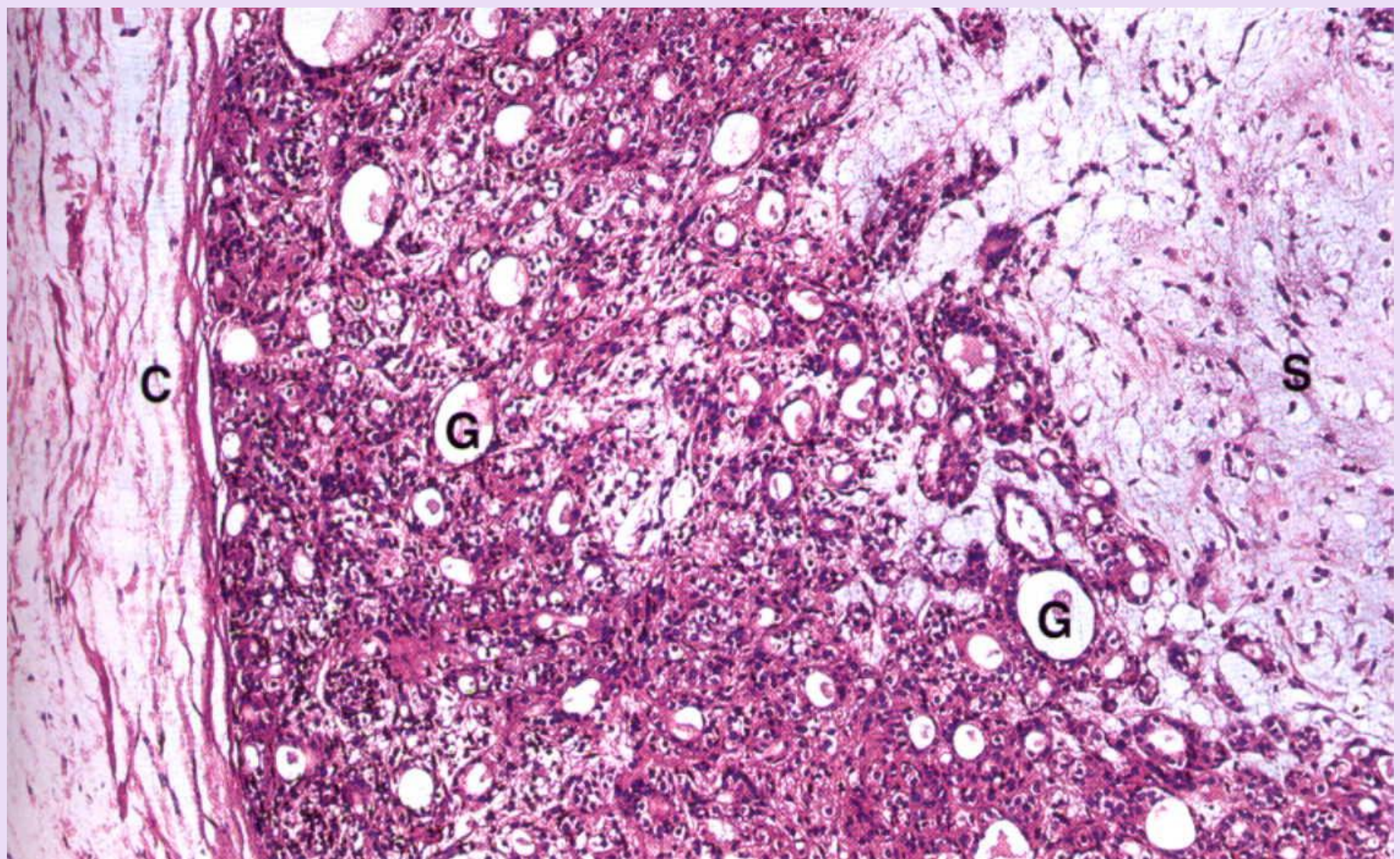


Chronic sialadenitis

D= Duct branches F = Periductal fibrosis
L= Lymphoid follicles A = Secretory acini

Pleomorphic Adenoma (Mixed Tumour) of the Salivary Gland

- Most commonly in the parotid gland
- Usually Encapsulated.
- Almost benign with local recurrence.
- Histopathology:
 - Neoplastic epithelial element (G)
 - Loose connective tissue stroma (S).
 - Fibrous Capsule (C).

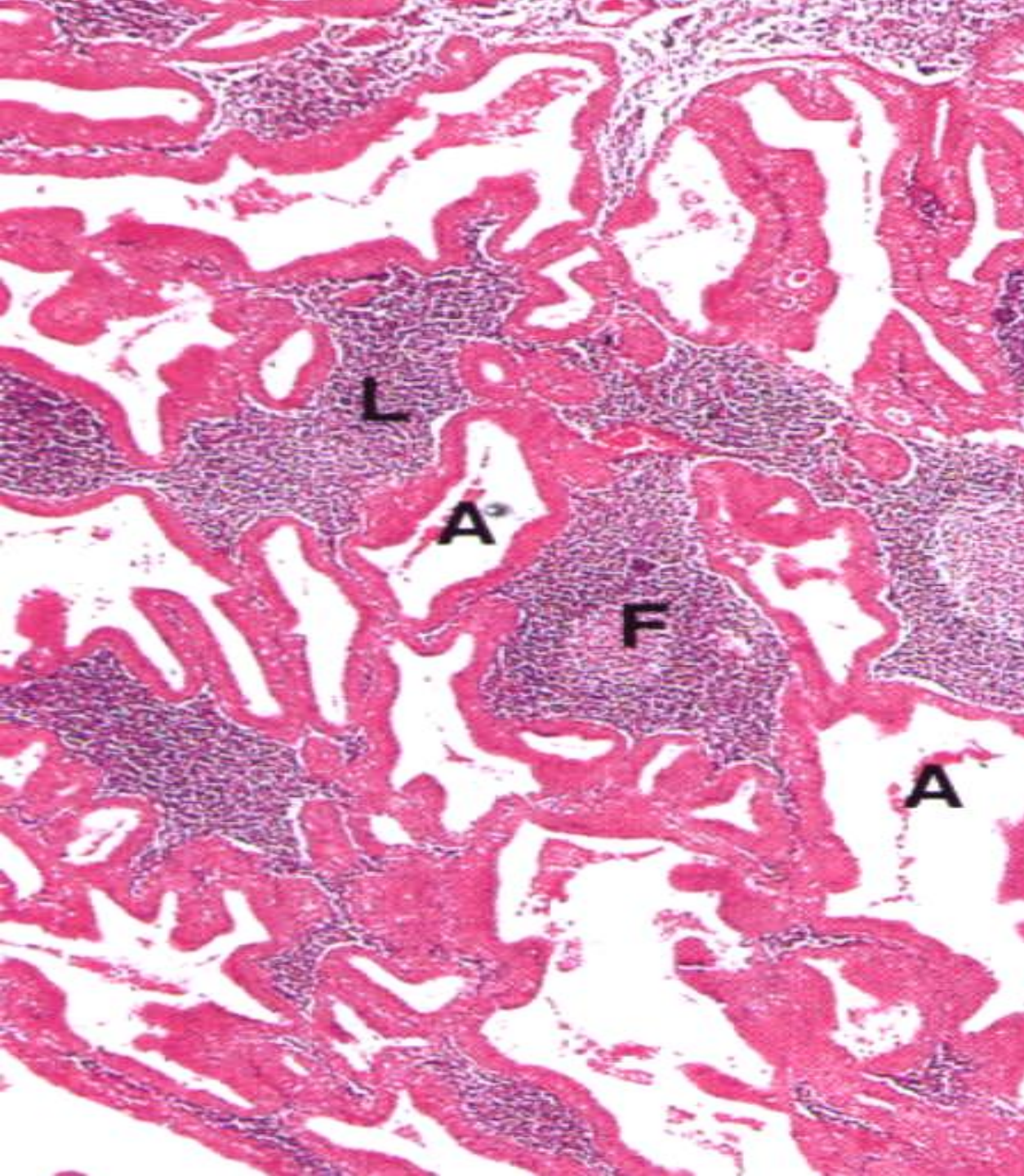


**Pleomorphic adenoma of the salivary glands
(Typical pleomorphic form)**

C= Fibrous capsule
G= Glands
S= Stroma

Adenolymphoma (Warthin's Tumour)

- Benign tumour.
- Almost occurs in and around the parotid gland.
- It commonly arises in middle-aged and older men.
- Histopathology:
 - Large glandular acini (A).
 - Dense Lymphoid Tissue (L) with atypical Lymphoid Follicle (F)



Warthin Tumor

A= Glandular Acini

F= Lymphoid Follicles

L= Lymphoid tissue

ADENOCYSTIC CARCINOMA

- Most common malignant tumour in the salivary tissue.
- Uncommon in parotid gland, it is seen in other mayor glands and minor salivary glands.
- Histopathology:
 - **Cribriform patterns** with **small space** (S).
 - **Fibrous tissues**(F) separates the arrangement of tumour cells.



Adenoid cystic carcinoma

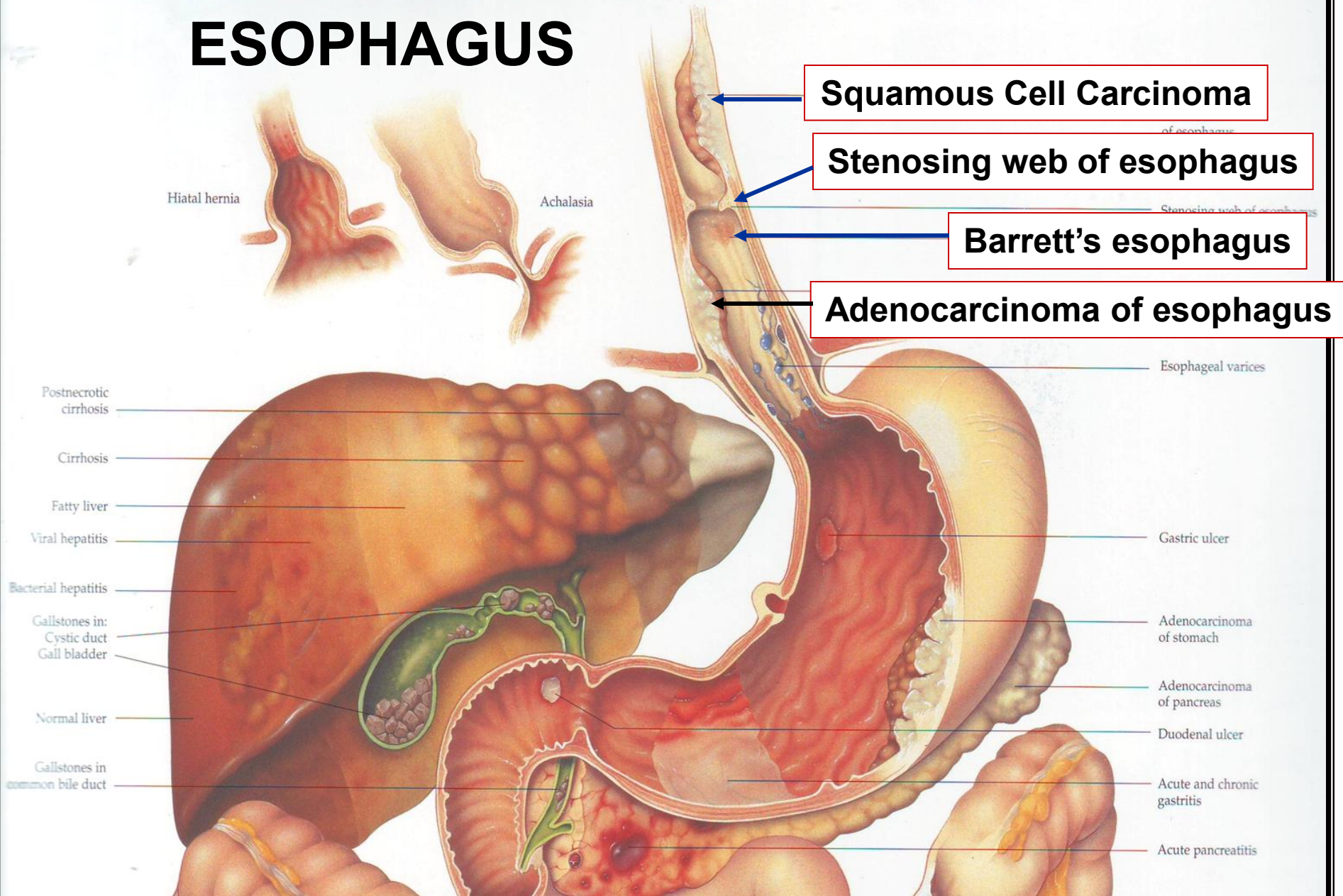
F= Fibrous stroma

S= Small spaces

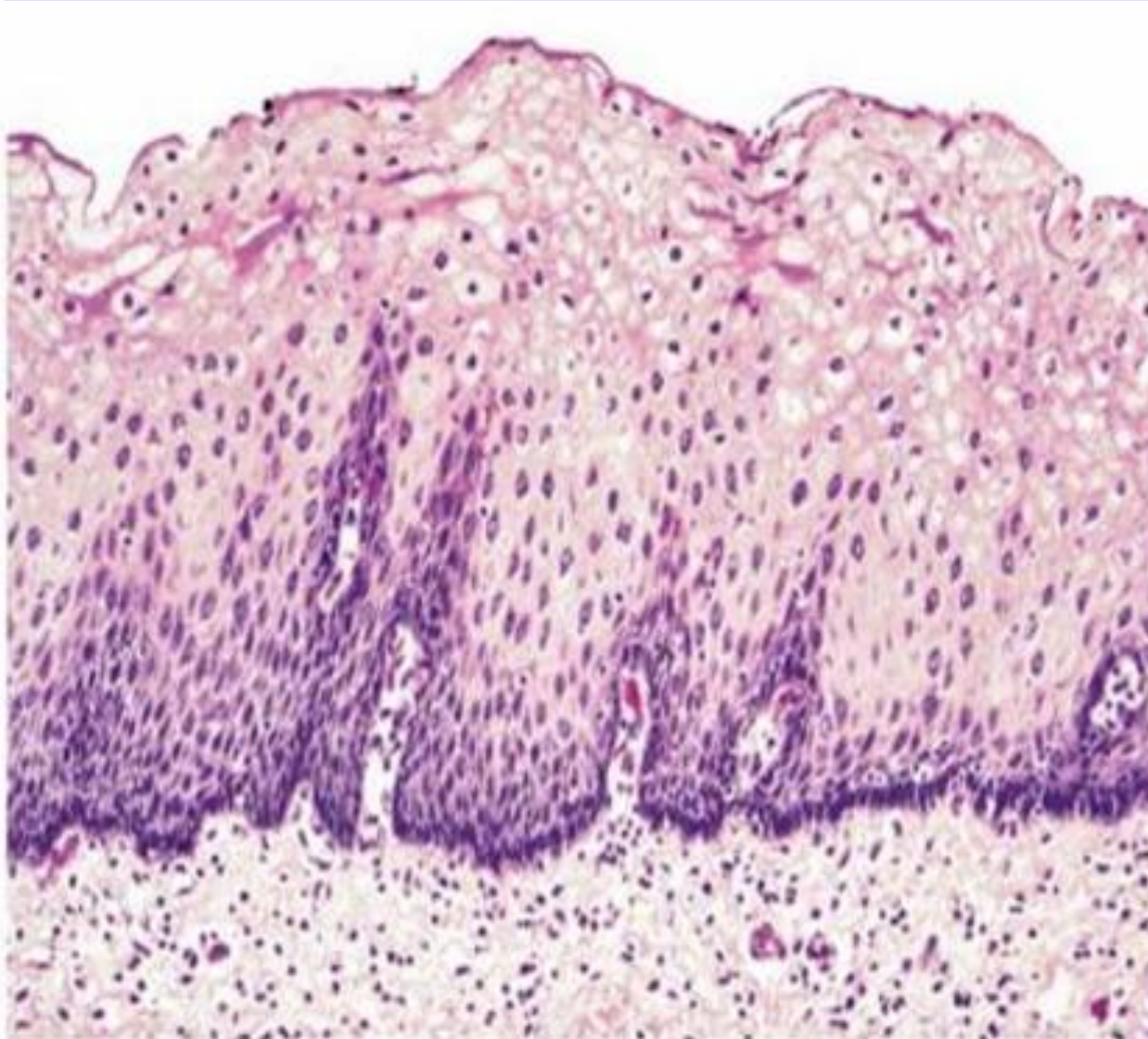
OESOPHAGUS

Diseases of the Digestive System

ESOPHAGUS

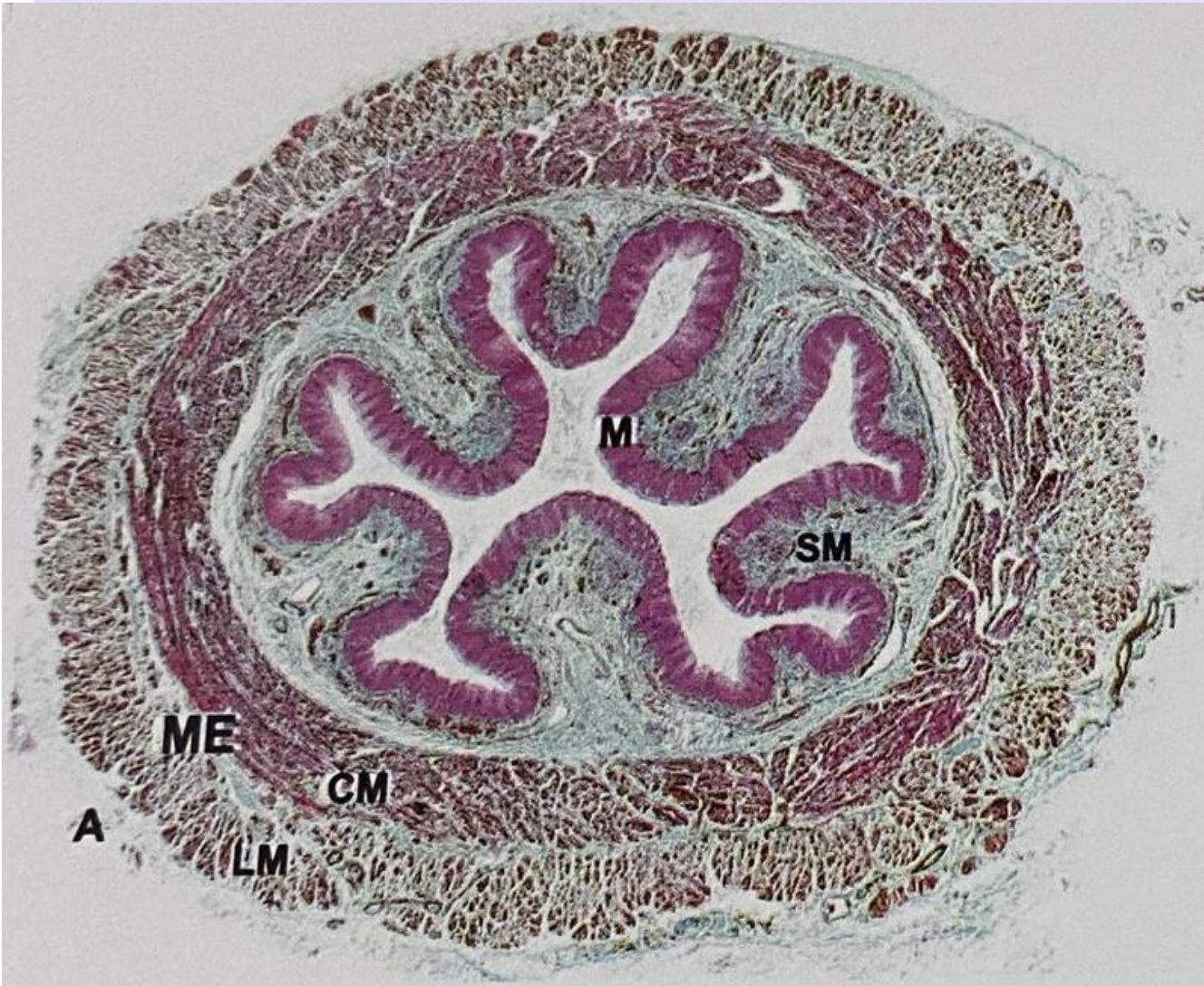


Histology esophagus



Stratified nonkeratinizing squamous mucosa exhibits a basal zone regenerating cells, rete pegs that extend partially into epithelial layer, and scattered intraepithelial lymphocytes

Histology esophagus

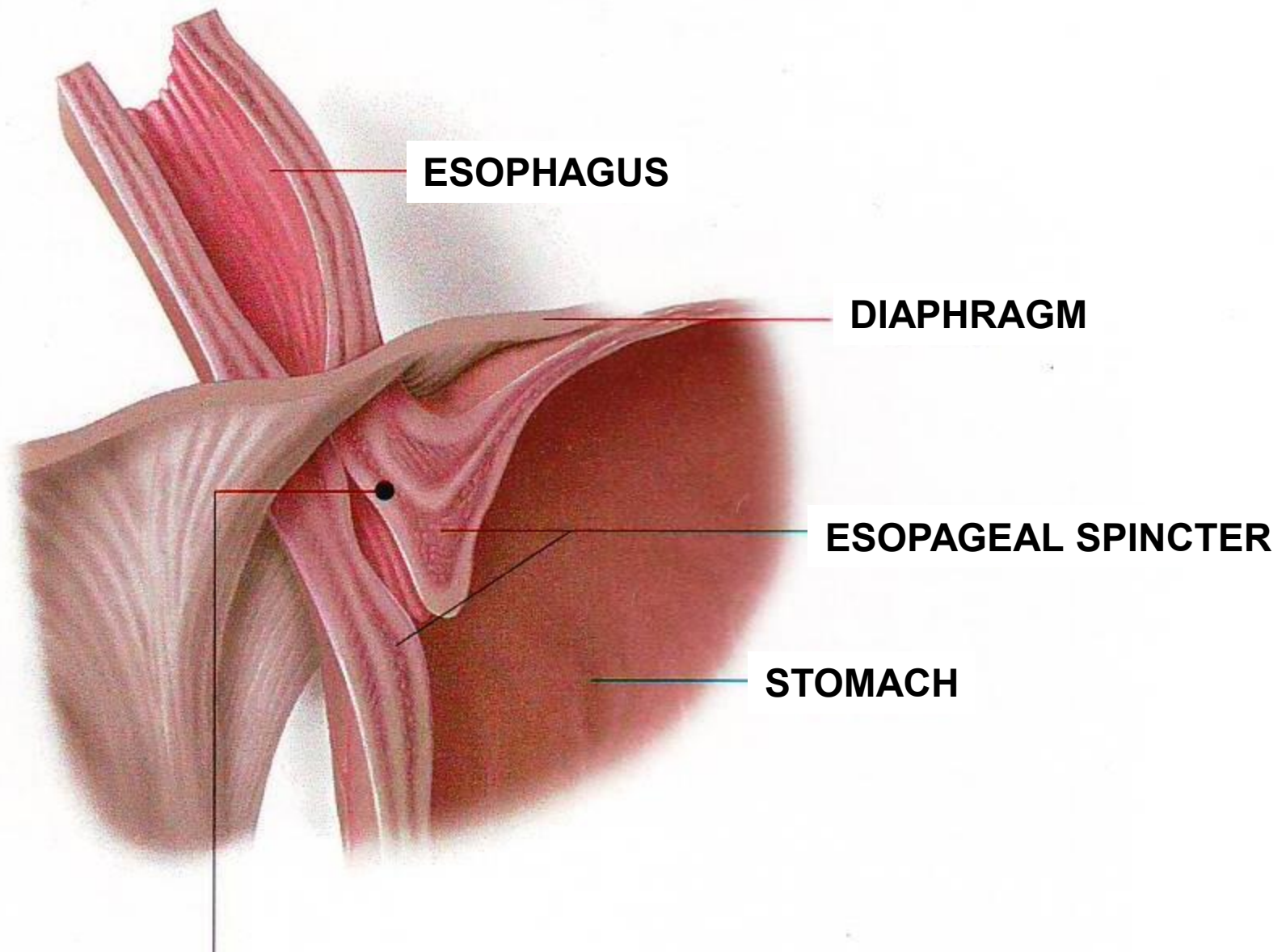


Mukosa (M)
Submukosa (SM)
Muskularis eksternal (ME)
Adventisia (A)

BARRET OESOPHAGUS

- **Metaplastic change** in the lower end of oesophagus due to to the long-standing **acid reflux** from the stomach.
- Epithelium exhibit a **gastric (G)** or/and **intestinal (In)** type pattern.
- **Displastic change** may occure with high risk of adenocarcinoma.
- Histopathology:
 - There is abrupt **transition** from squamous epithelium (S) to gastric epithelium (G).
 - A dense infiltration of **plasme calls(P)** are seen in the lamina propria.

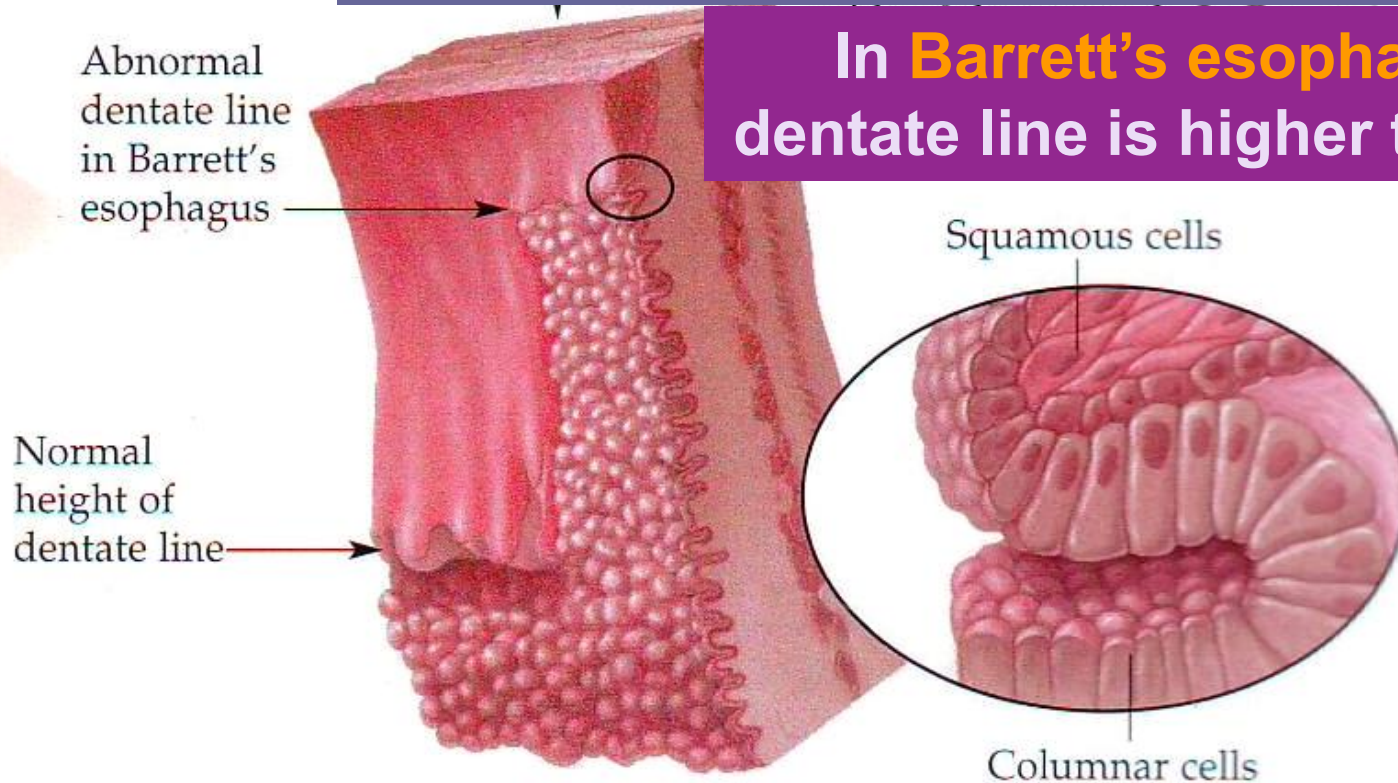
Gastroesophageal Junction

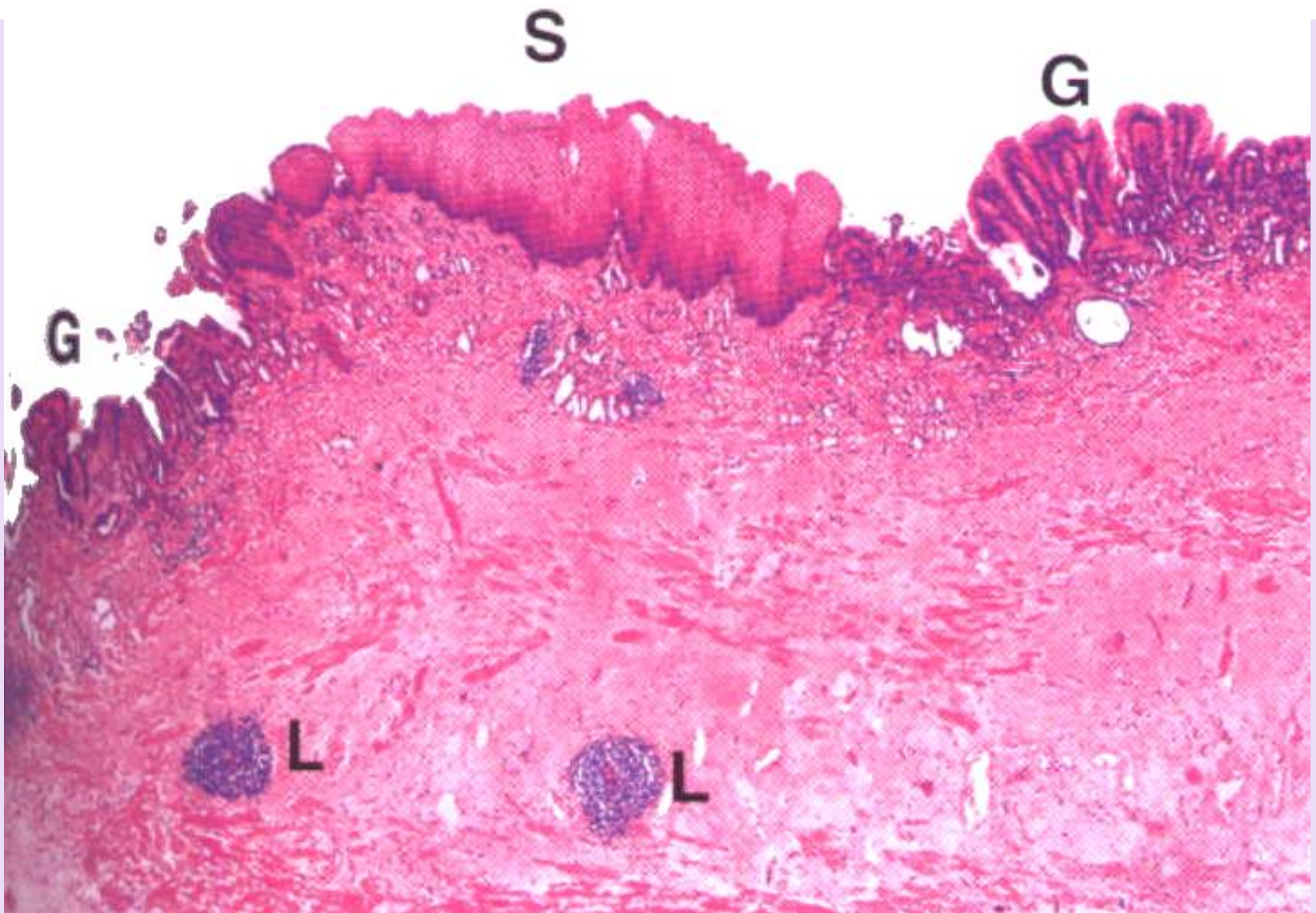


Dentate Line

The junction at which esophageal squamous cells meet gastric columnar cells

In **Barrett's esophagus**, the dentate line is higher than normal



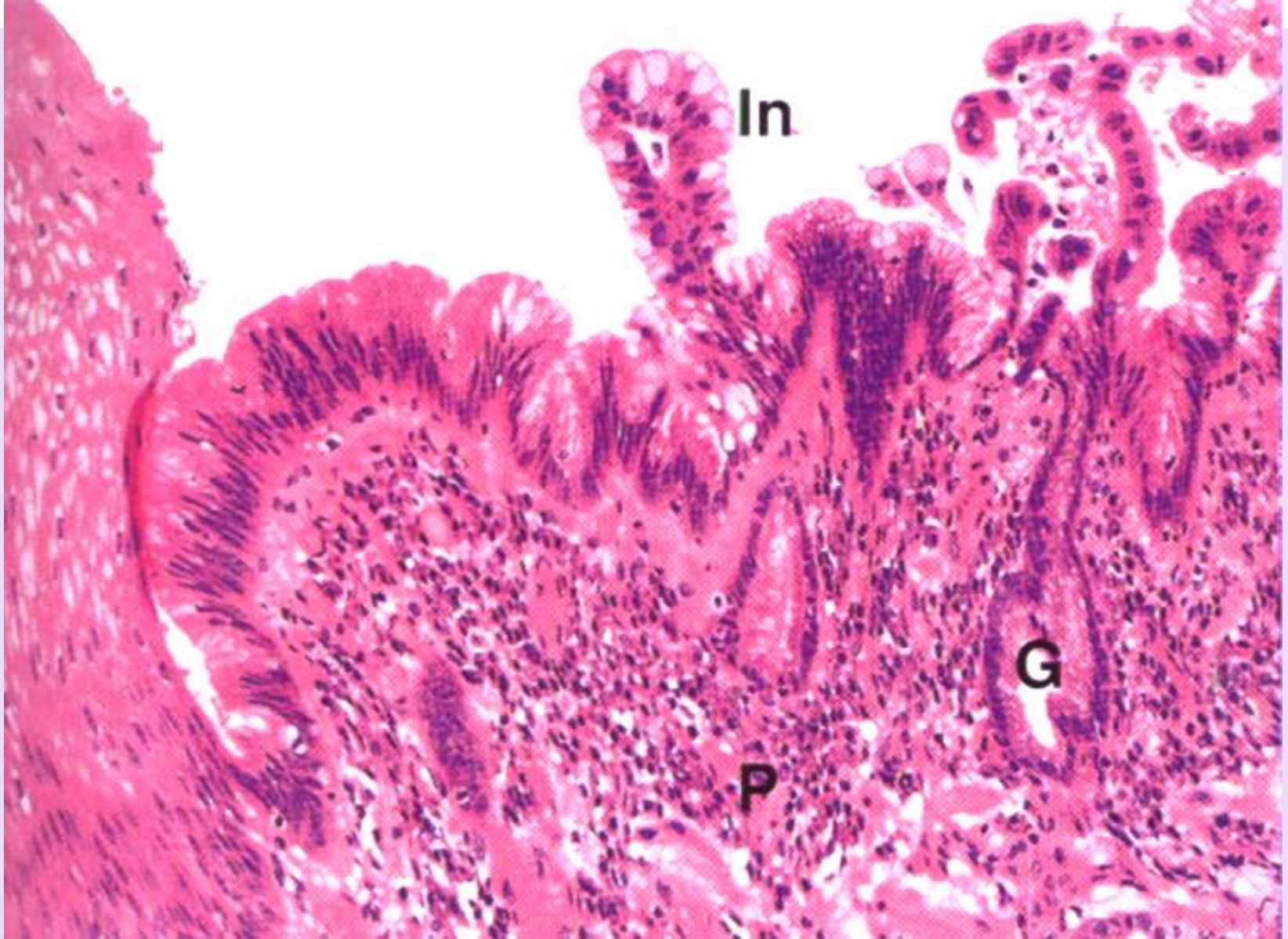


Barrett's oesophagus

G= Metaplastic Gastric epithelium

L= Lymphoid aggregates,

S= Squamous epithelium

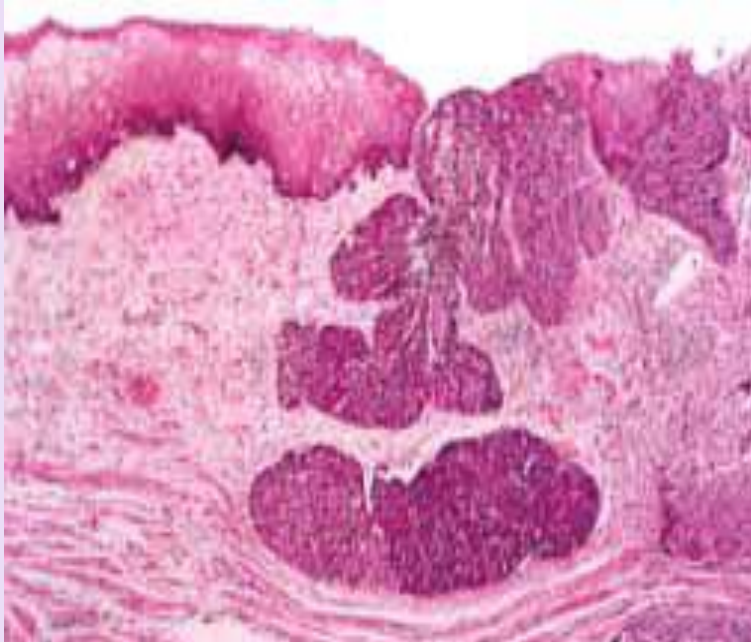
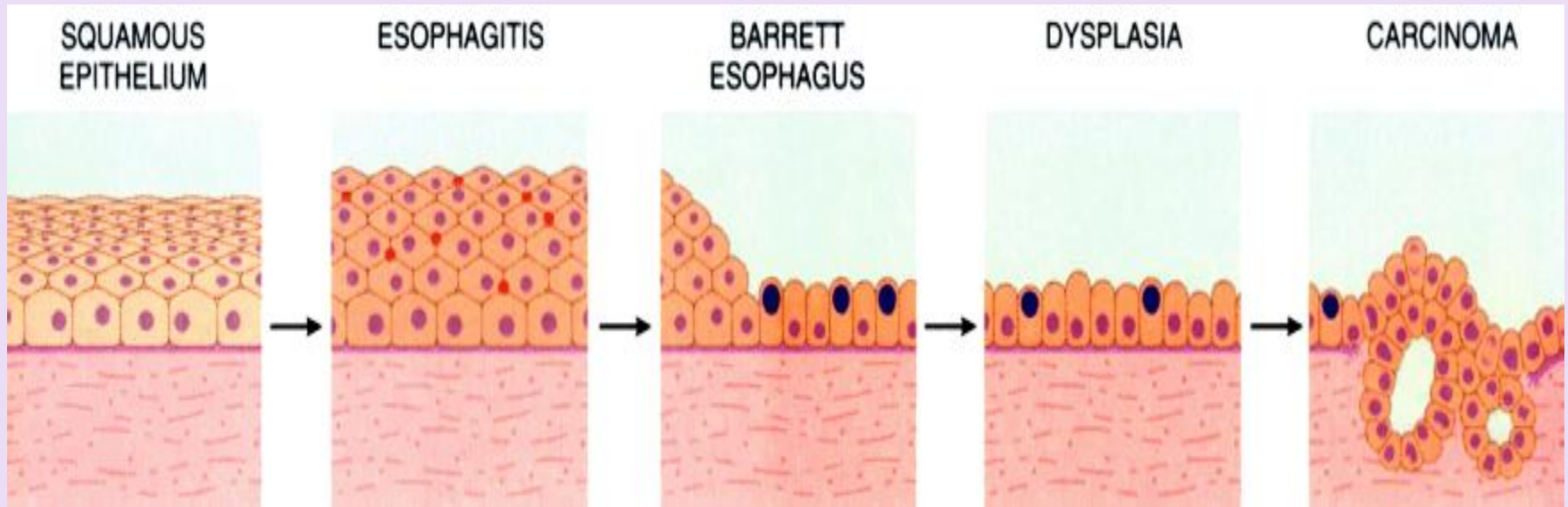


Barrett's oesophagus

G= Gastric epithelium

In= Intestinal epithelium

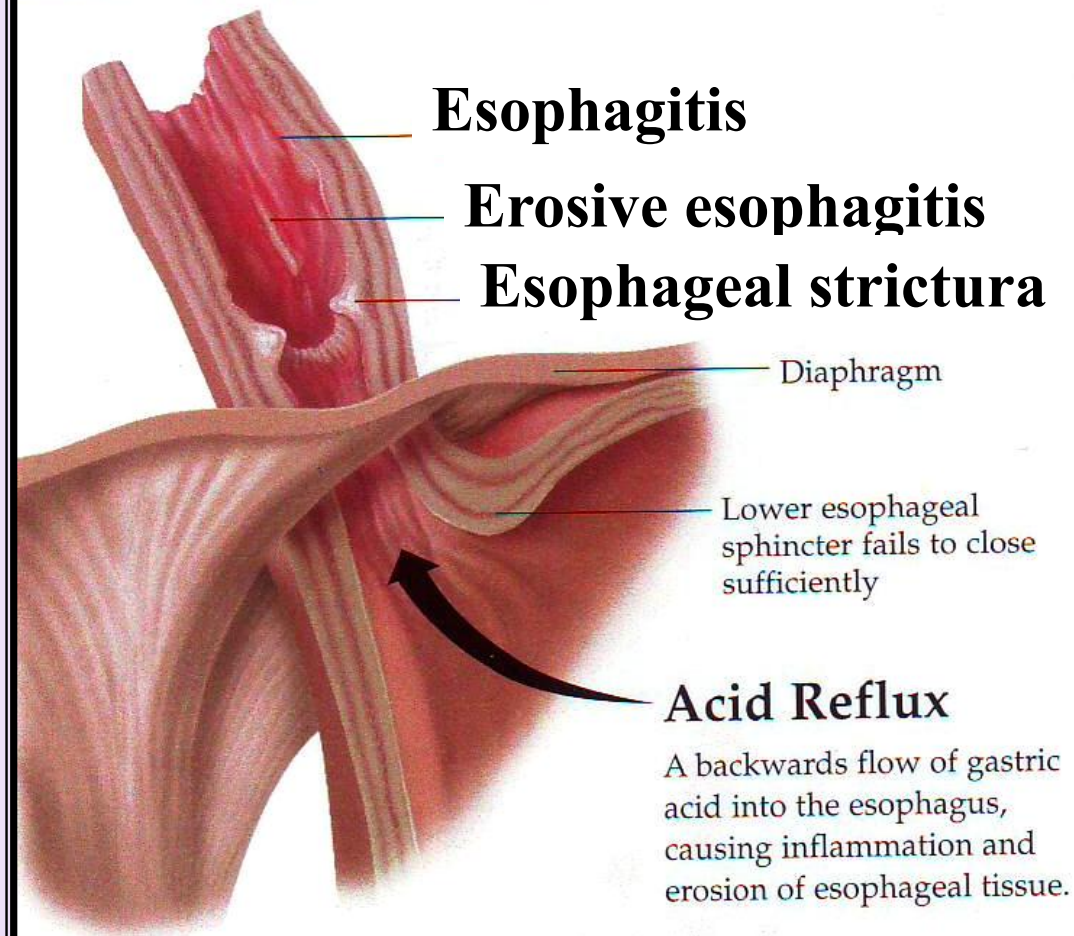
P= Plasma cell



Transition from Barrett esophagus
to adenocarcinoma

GERD

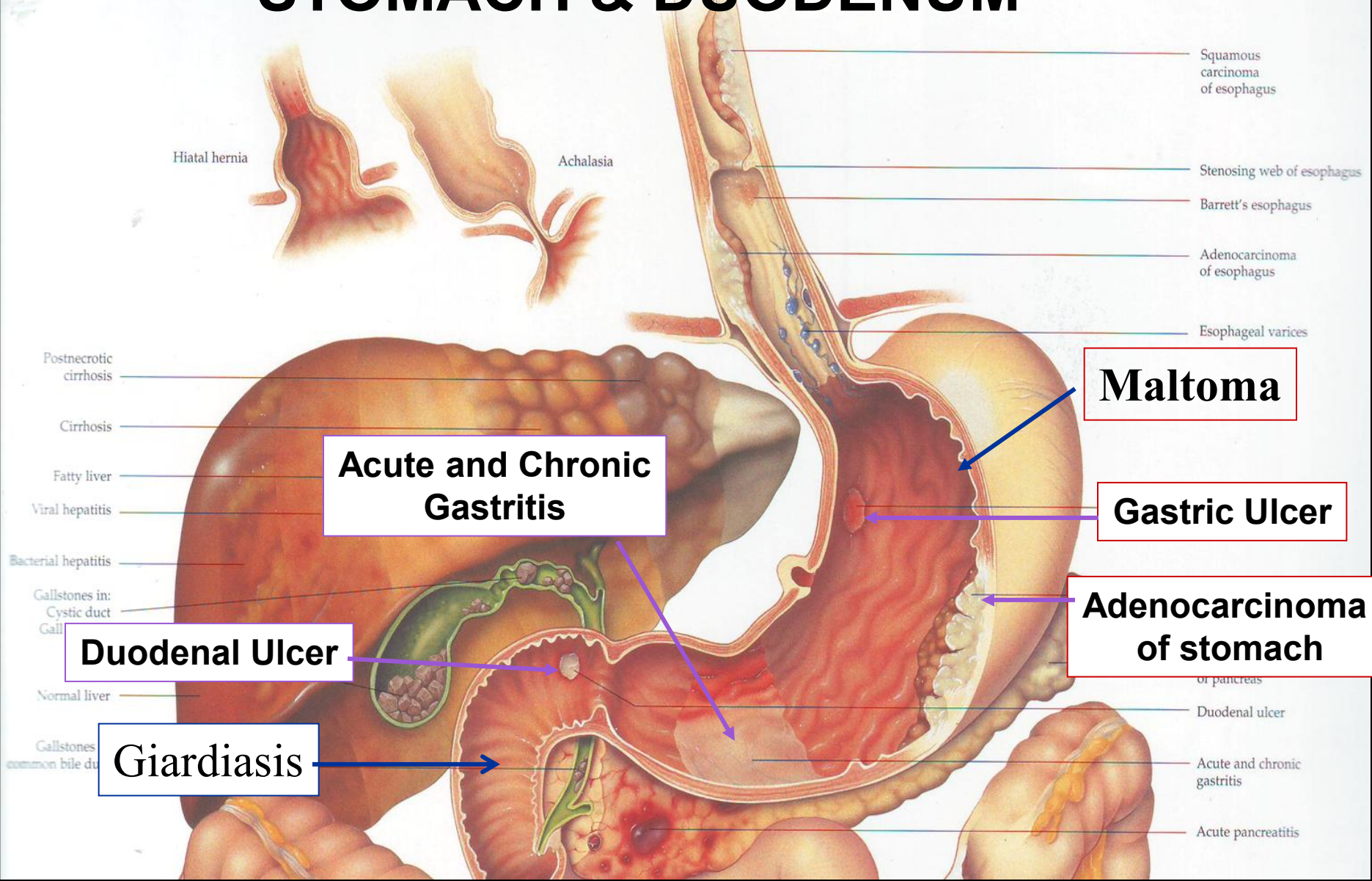
Gastroesophageal Reflux Disease



STOMACH

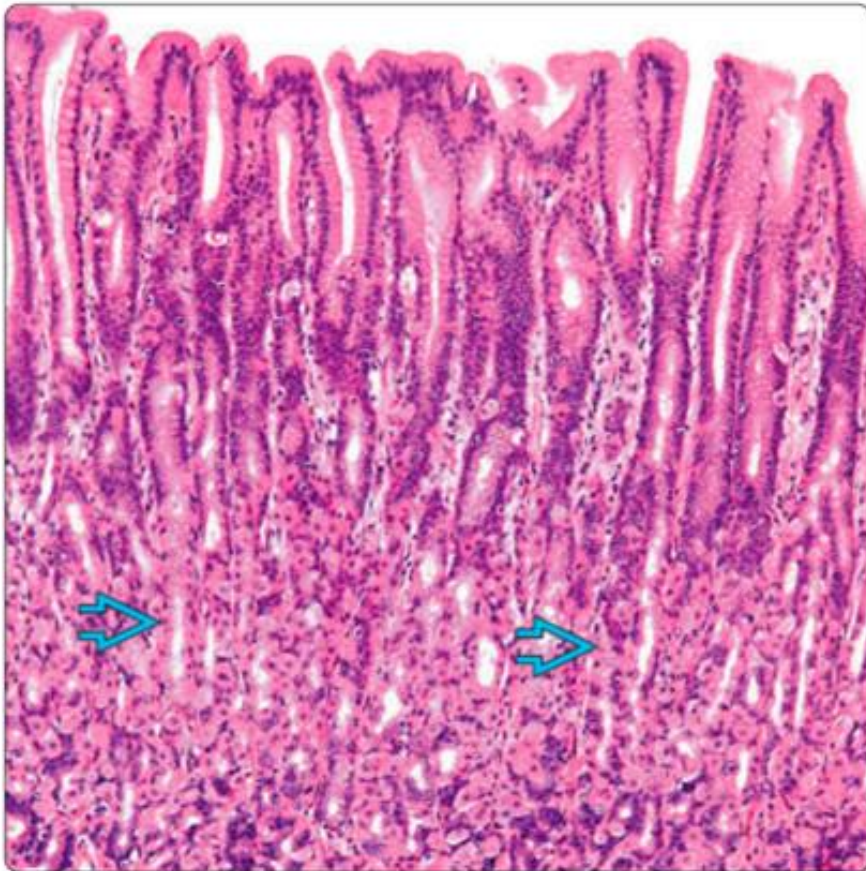
Diseases of the Digestive System

STOMACH & DUODENUM

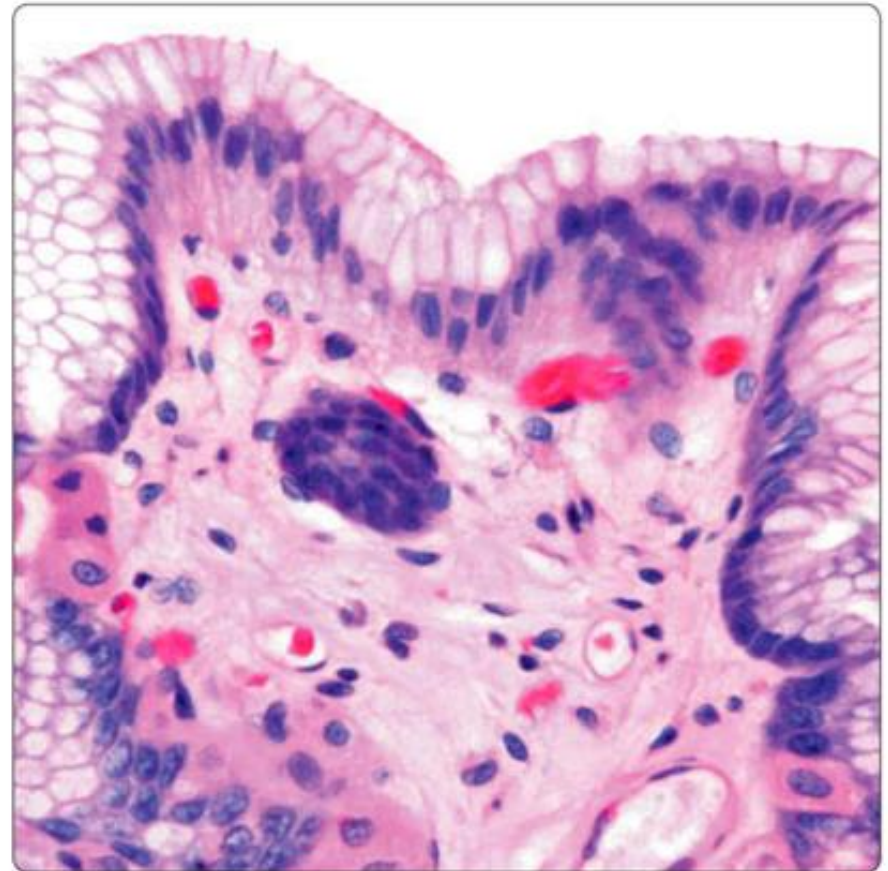


Histology

Glands

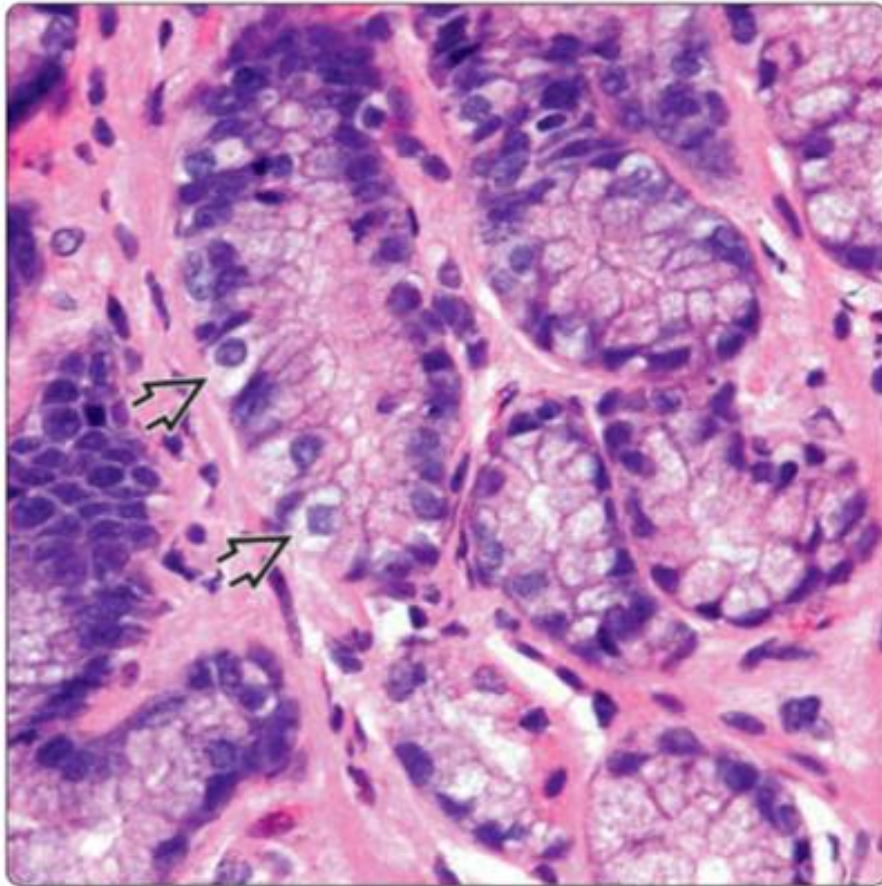


Pit/Surface Epithelium



Histology

Endocrine Cells



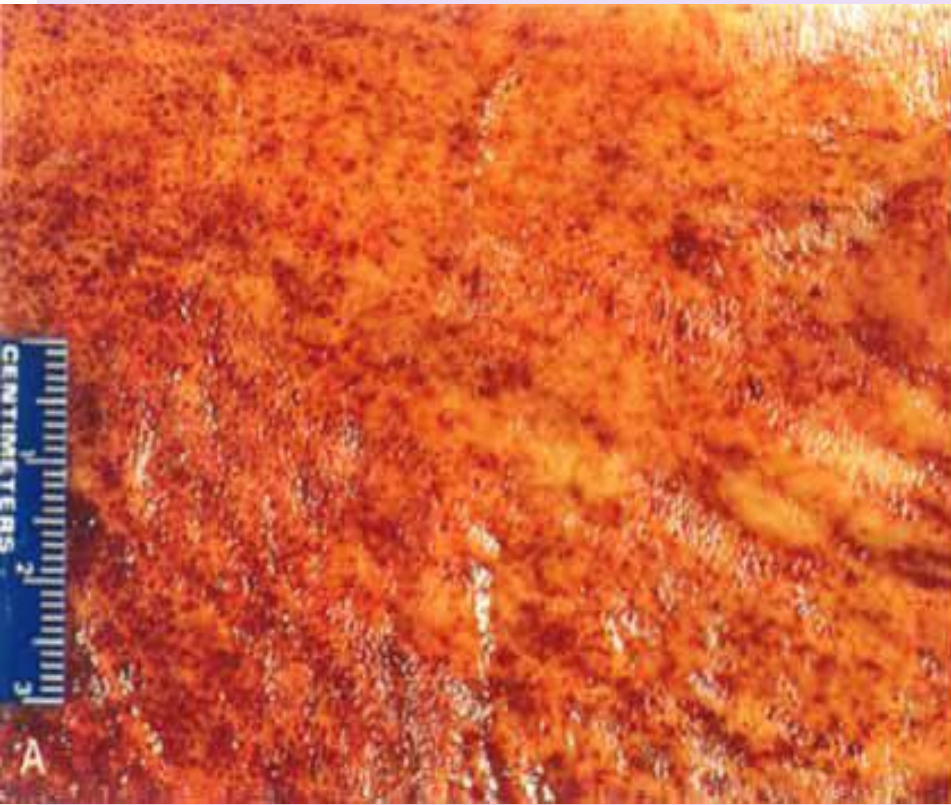
The type of endocrine cell differs with the region of the stomach. In the antrum, the pale, ovoid cells with gray cytoplasm are the gastrin-producing or G cells. The body contains ECL cells, which produce histamine, in the lower 1/3 of the mucosa near the chief cells.

GASTRITIS

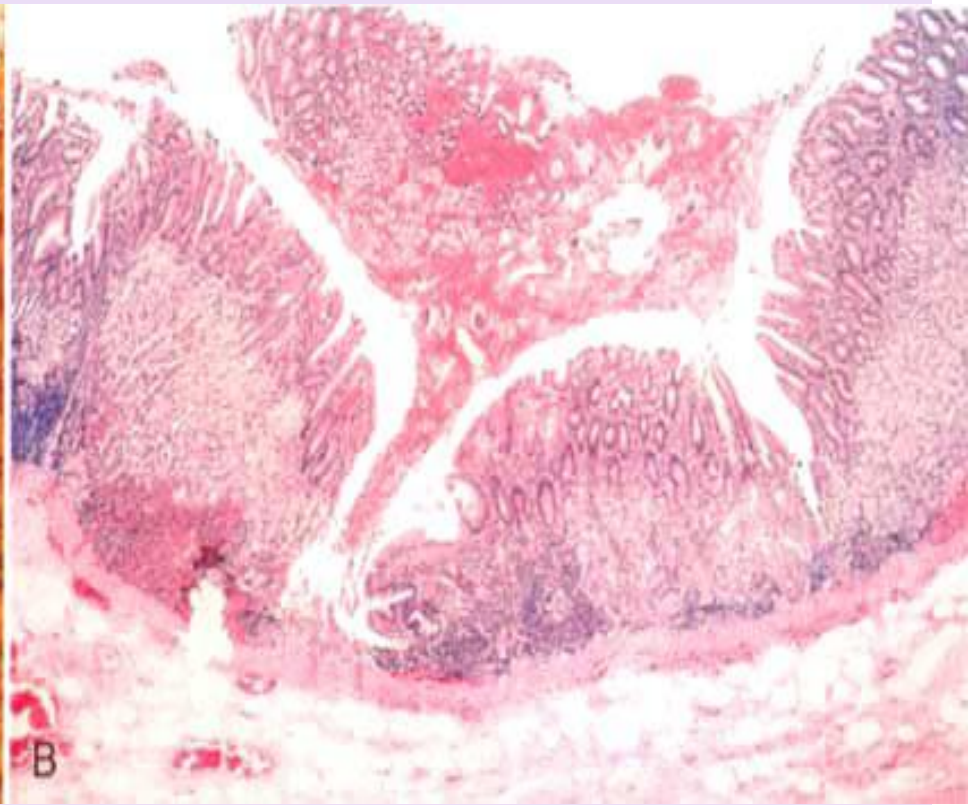
(Inflammation of the stomach)

1. **Acute Gastritis** (aspirin, other antiinflammatory drugs, alcohol and severe stress).
2. **Chronic Gastritis**:
 - a. **Chronic Infection** (*Helicobacter pylori*)
 - b. **Chronic Autoimmune Gastritis** (antibody to parietal cells with anemia perniciosa).
 - c. **Chronic Chemical Gastritis** (reflux of bile).
 - d. **Gastric outlet obstruction.**

Acute gastritis



A, Gross view showing punctate erosions in an otherwise unremarkable mucosa; adherent blood is dark due to exposure to gastric acid.



B, Low-power microscopic view of focal mucosal disruption with hemorrhage; the adjacent mucosa is normal

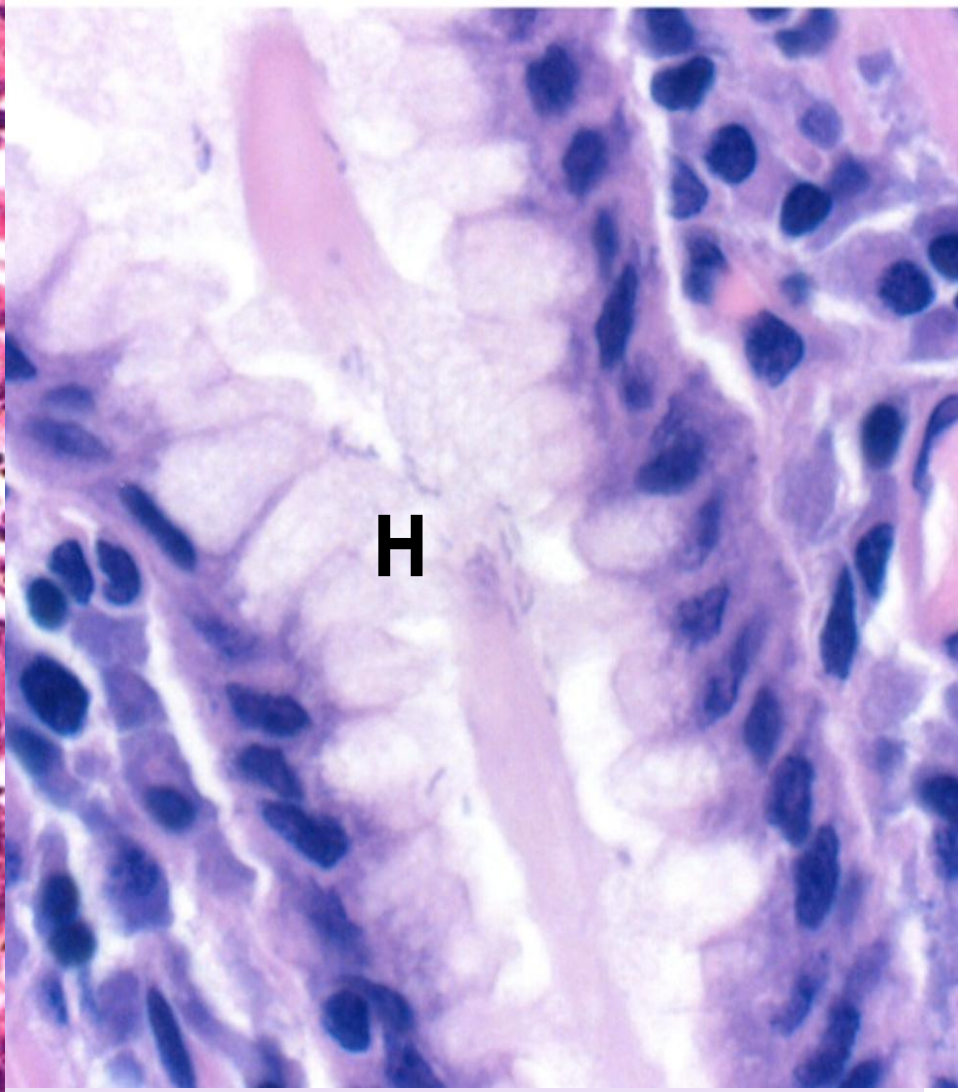
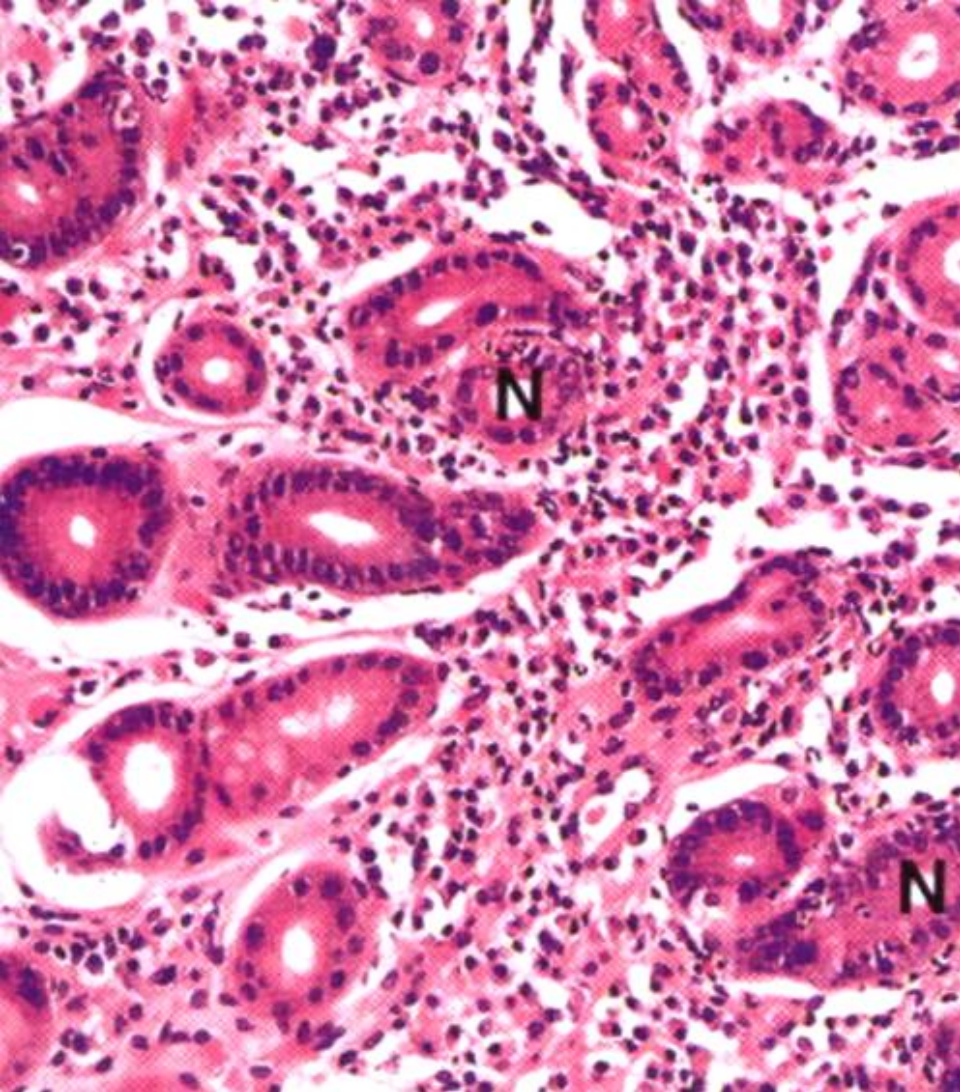
HELICOBACTER INFECTION

- The most common cause of chronic gastritis.
- Affecting the mucosa of the gastric antrum and extending to the body of stomach in severe cases.
- Histopathology:
 - A mixed inflammatory infiltrate (Neut, Pl cell, Lymph & Eos) in the lamina propria.
 - Intestinal metaplasia and dysplasia are commonly seen.

H. Pylori **Bacteria**

(a factor in gastric disorders)





Chronic gastritis (H. Pylory infection)

N= Neutrophils
H= Organism

AUTOIMMUNE GATRITIS

- Primarily affect the body of stomach.
- Histopathology :
 - Atrophy of the gastric gland (small).
 - Loss of parietal cells.
 - The mucosa is thin.
 - Inflammatory infiltrates (Lymph & Pl Cells) in lamina propria.
 - Intestinal metaplasia of the gastric mucosa.
- Late stages: Chronic Atrophic Gastritis.

CHEMICAL GASTRITIS (Reactive Gastritis)

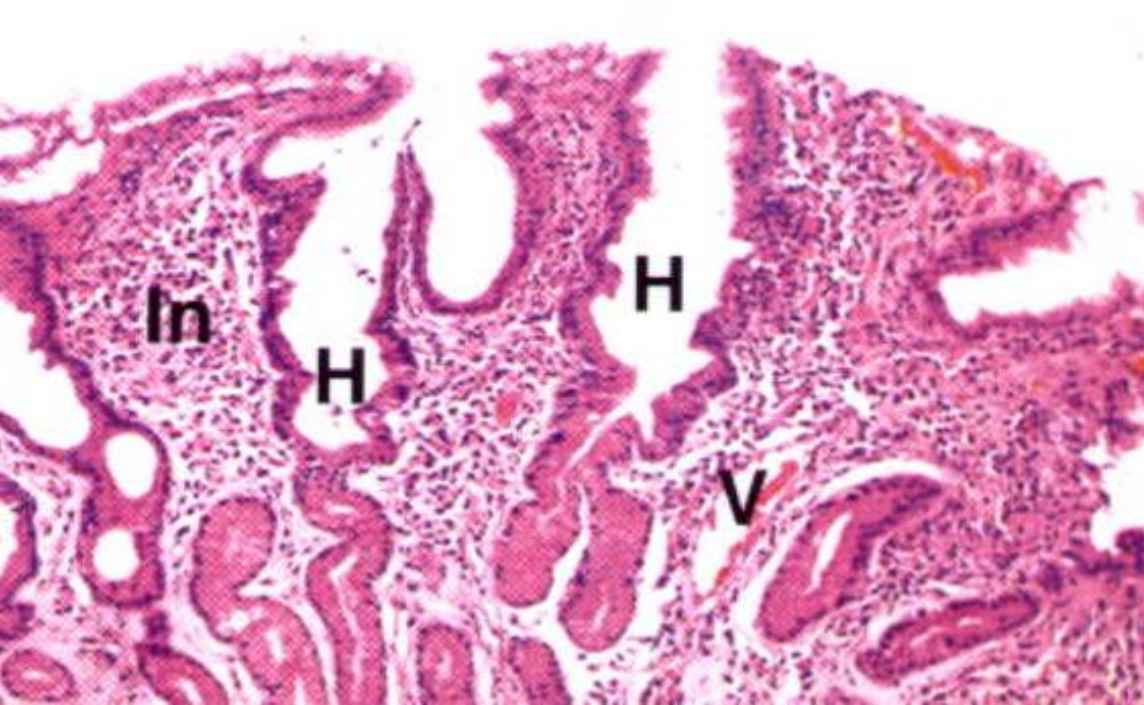
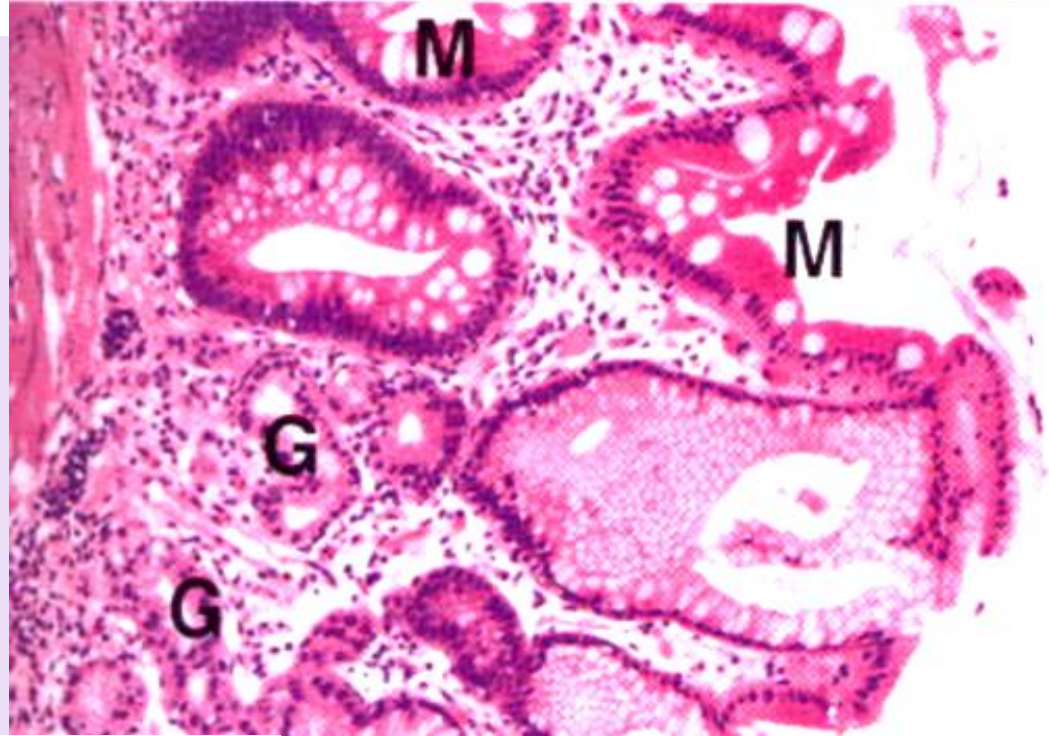
Histopathology:

- Inflammation of the superficial mucosa with scanty infiltrate of inflammatory cells (Lymph & Plasma Cells).
- The neck of gastric glands show hyperplasia
- Lamina propria is oedematous with marked vasodilatation (characteristic features)

Gastritis chronic (Autoimmun gastritis)

G= glands

M= Intestinal metaplastic



Gastritis chronic (Chemical gastritis)

H= Hyperplastic

In= Inflammatory cell

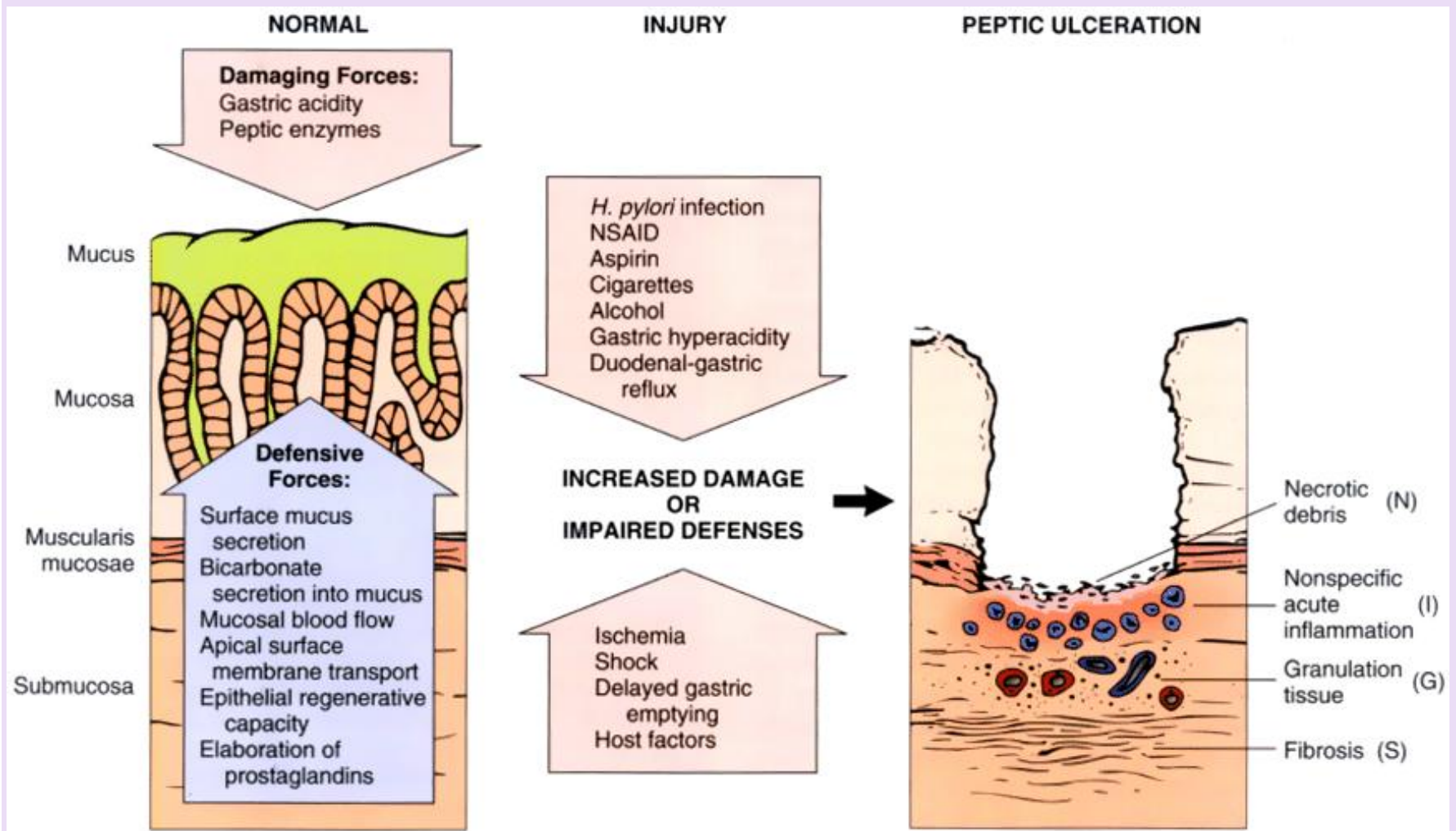
V= Vasodilation

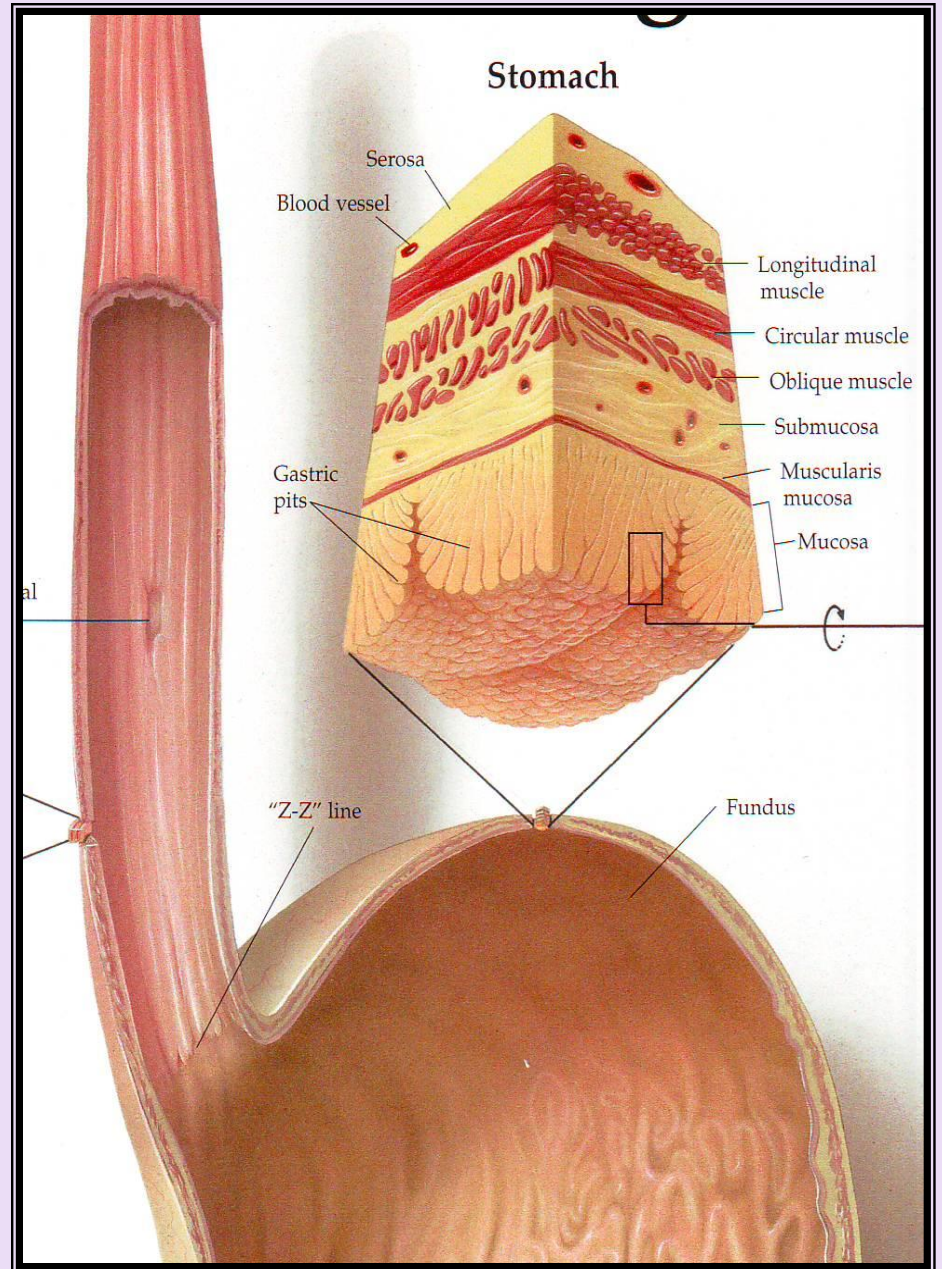
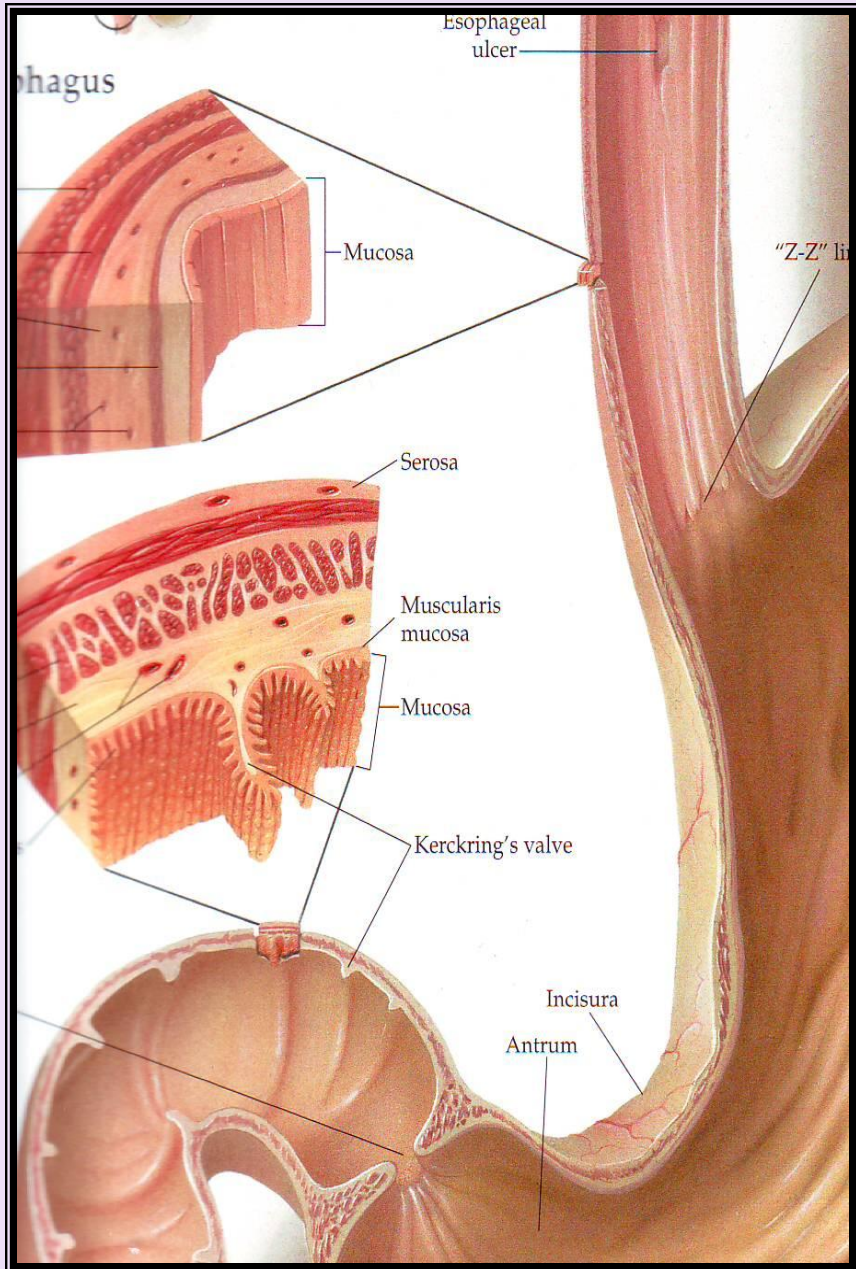
PEPTIC ULCERATION

(Stomach and Duodenum)

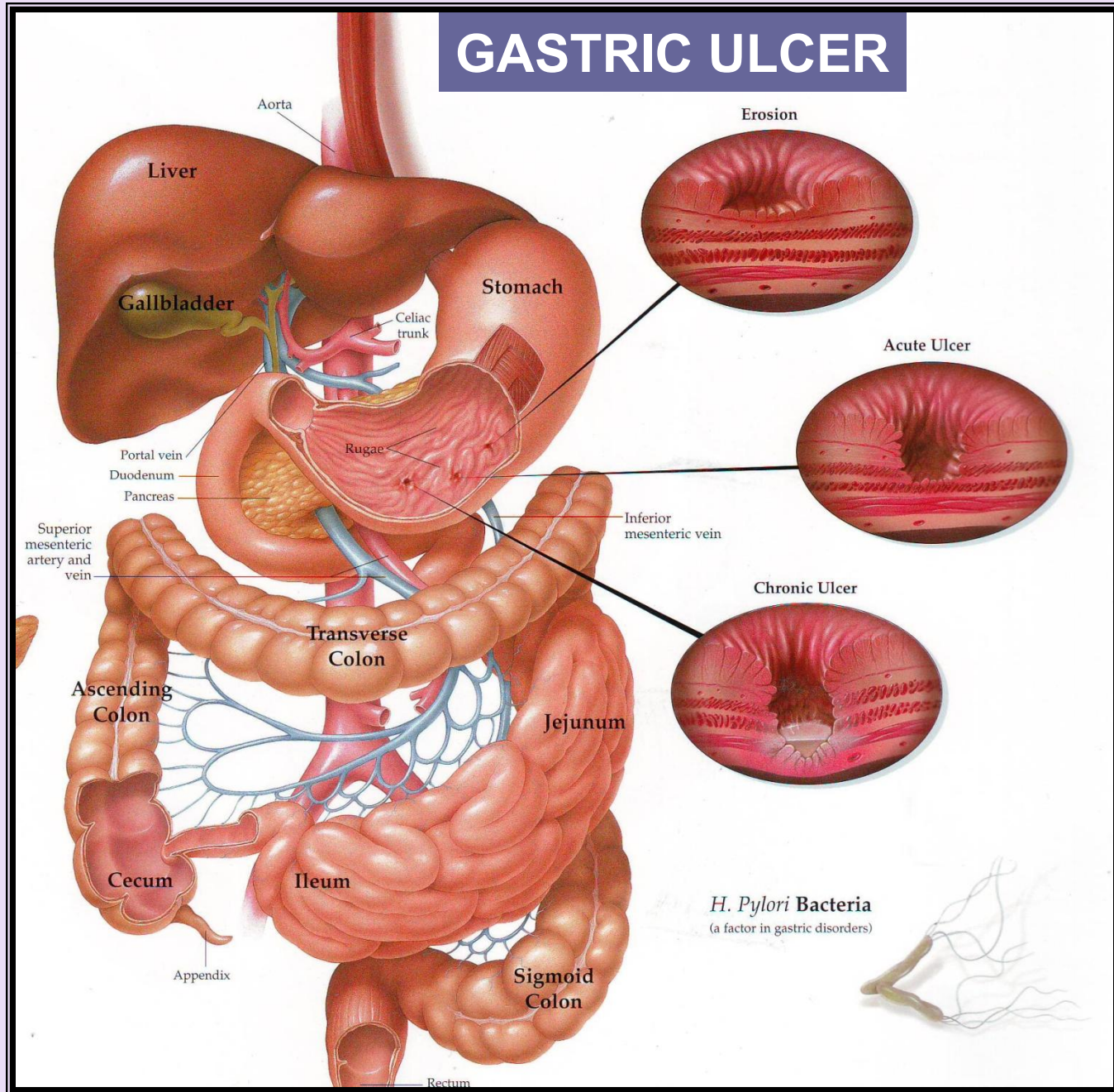
1. Haemorrhage (Haematemesis and Melaena).
2. Perforation → Acute peritonitis.
3. Obstruction (inflammation → progressive fibrous scarring → strictura → narrowing → obstruction)

Diagram of causes of, and defense mechanisms against, peptic ulceration. Diagram of the base of a nonperforated peptic ulcer, demonstrating the layers of necrosis (N), inflammation (I), granulation tissue (G), and scar (S), moving from the luminal surface at the top to the muscle wall at the bottom.





GASTRIC ULCER



Types of Ulcers

Erosion

Penetration of only the superficial layer

Mucosa

Muscularis mucosa

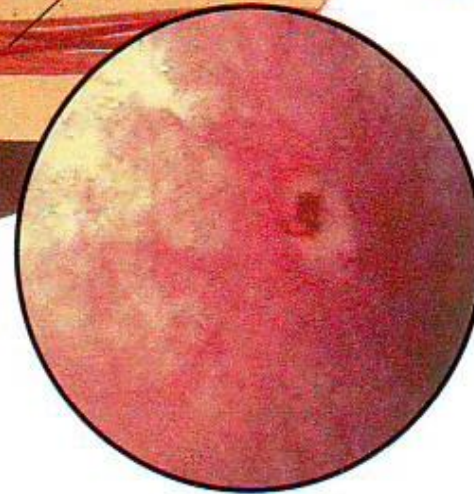
Submucosa

Oblique muscle

Circular muscle

Longitudinal muscle

Serosa



Acute Ulcer

Penetration into the muscular layer

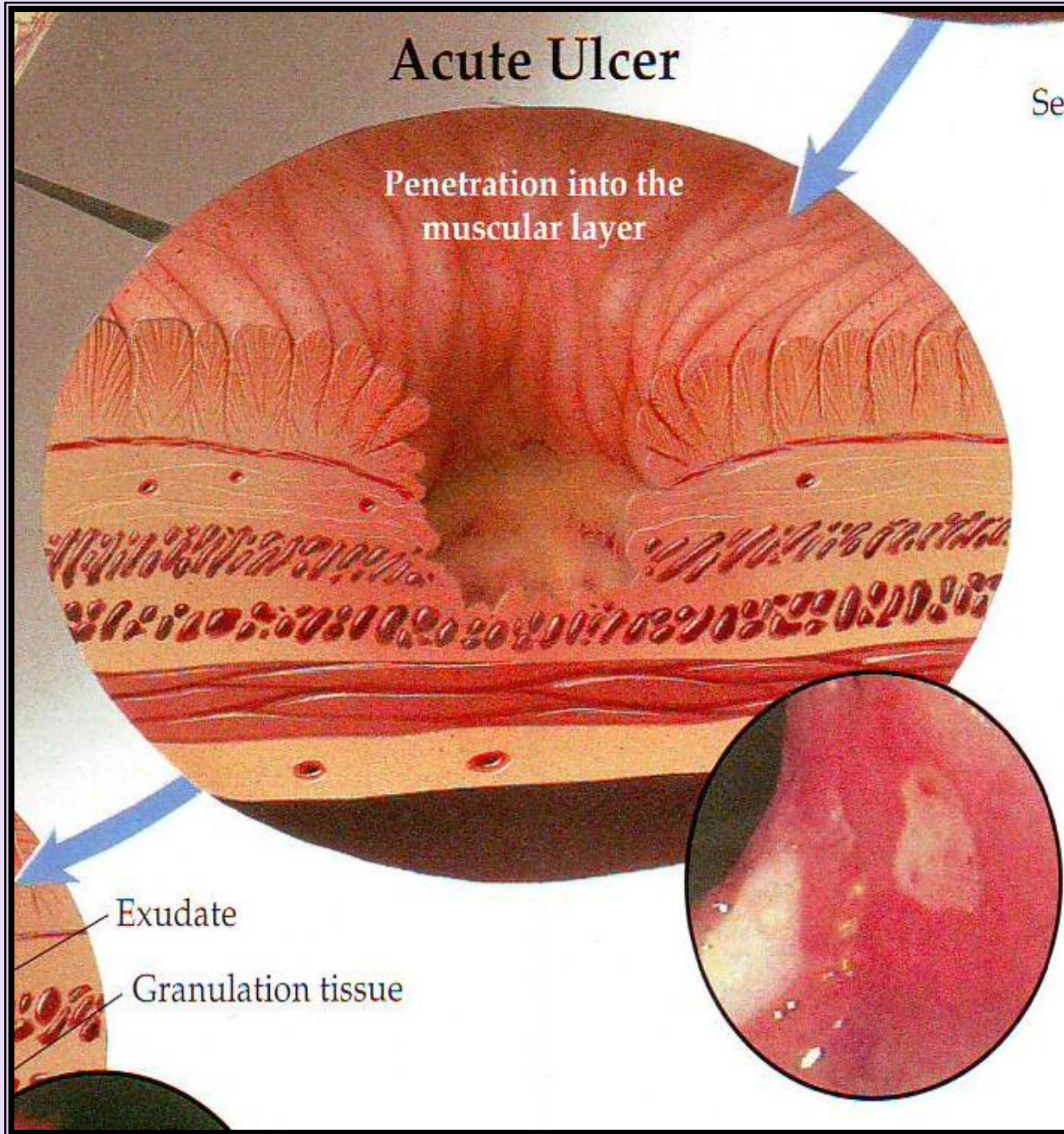
Acute Ulcer

Ser

Penetration into the
muscular layer

Exudate

Granulation tissue

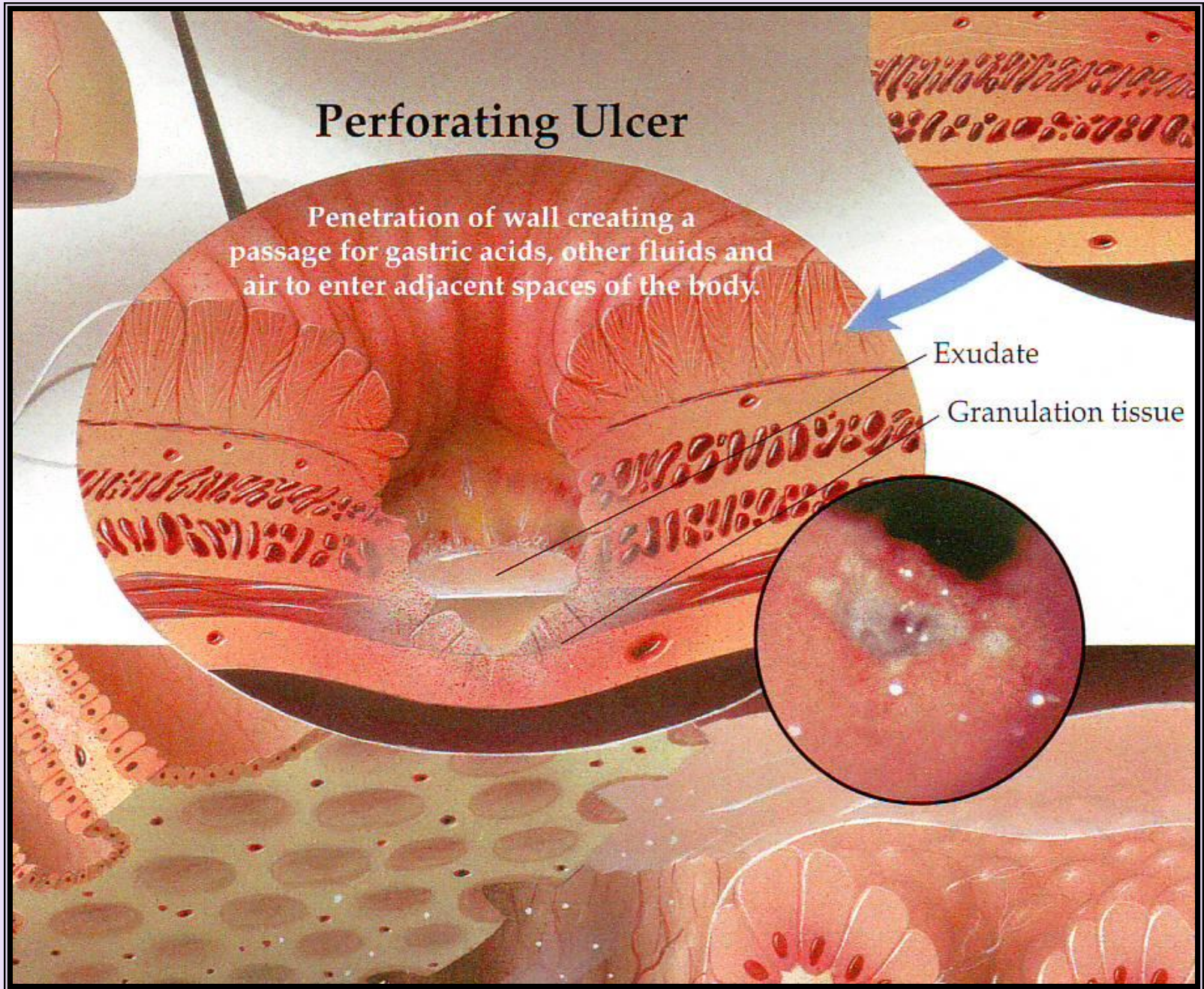


Perforating Ulcer

Penetration of wall creating a passage for gastric acids, other fluids and air to enter adjacent spaces of the body.

Exudate

Granulation tissue



Complication of peptic ulceration

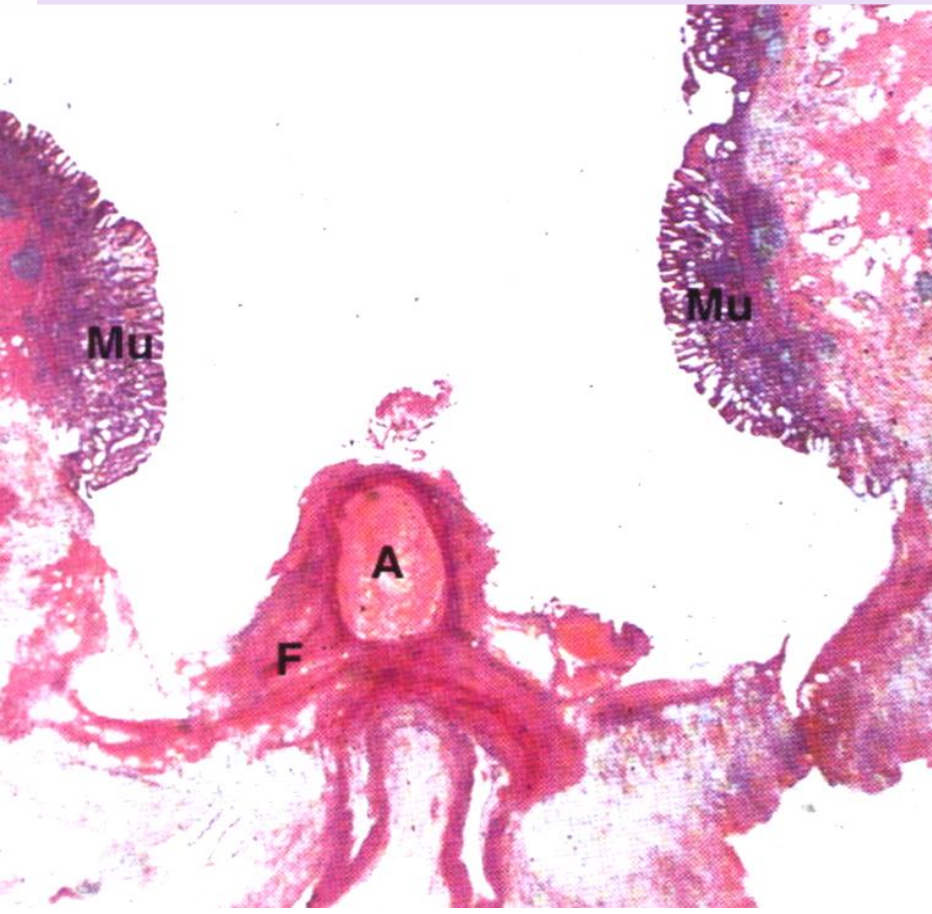
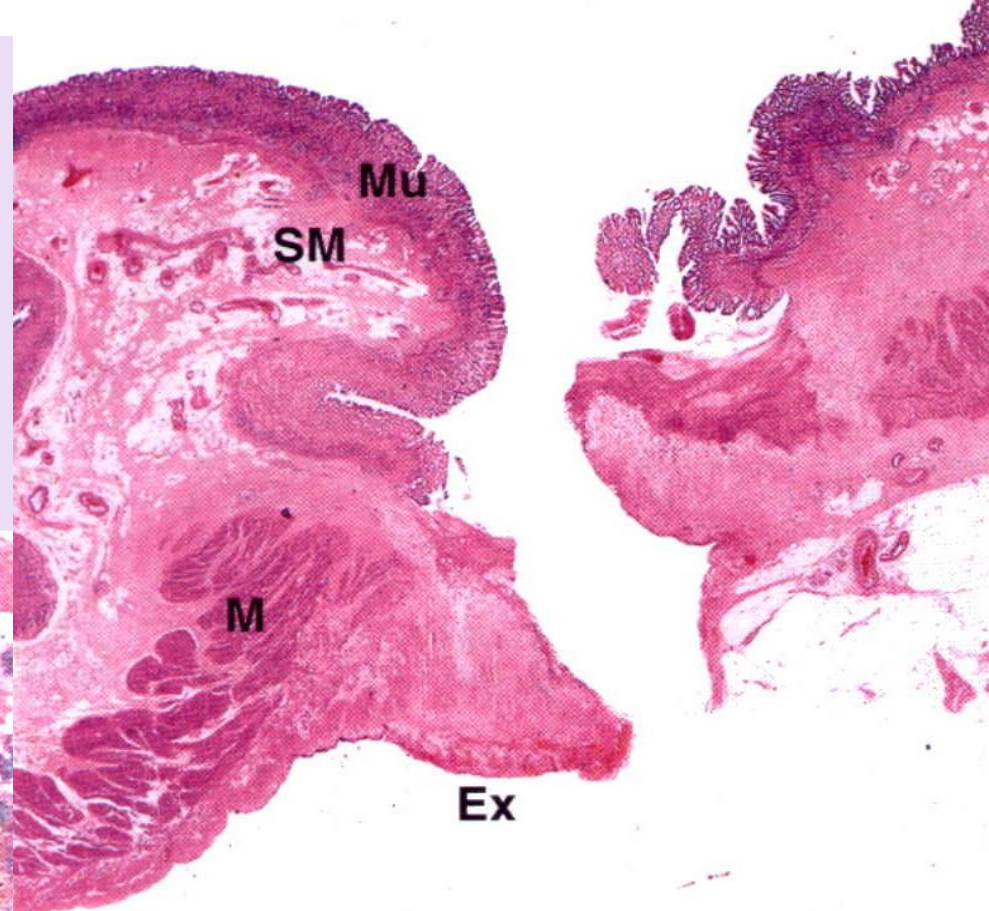
Perforated gastric ulcer

Mu= Mucosae

SM= Submucosae

M = Muscle layers

Ex= Inflammatory exudat



Bleeding gastric ulcer

A= Artery

F = Fibrous scar tissue

GASTRIC DYSPLASIA (Precancer lesion)

- Chronic Gastritis → Gastric Displasia
- Low grade dysplasia
- High grade dysplasia:
 - Marked irregularity of the gland with nuclear enlargement and Hyperchromacity.
 - Crowding of the epithelial cells.
 - No mucous production by epithelial cells.
 - Cellular polarity (basal alignment of nuclei) is completely lost.



Gastric dysplastic

G= High grade dysplastic

Gastric Carcinoma

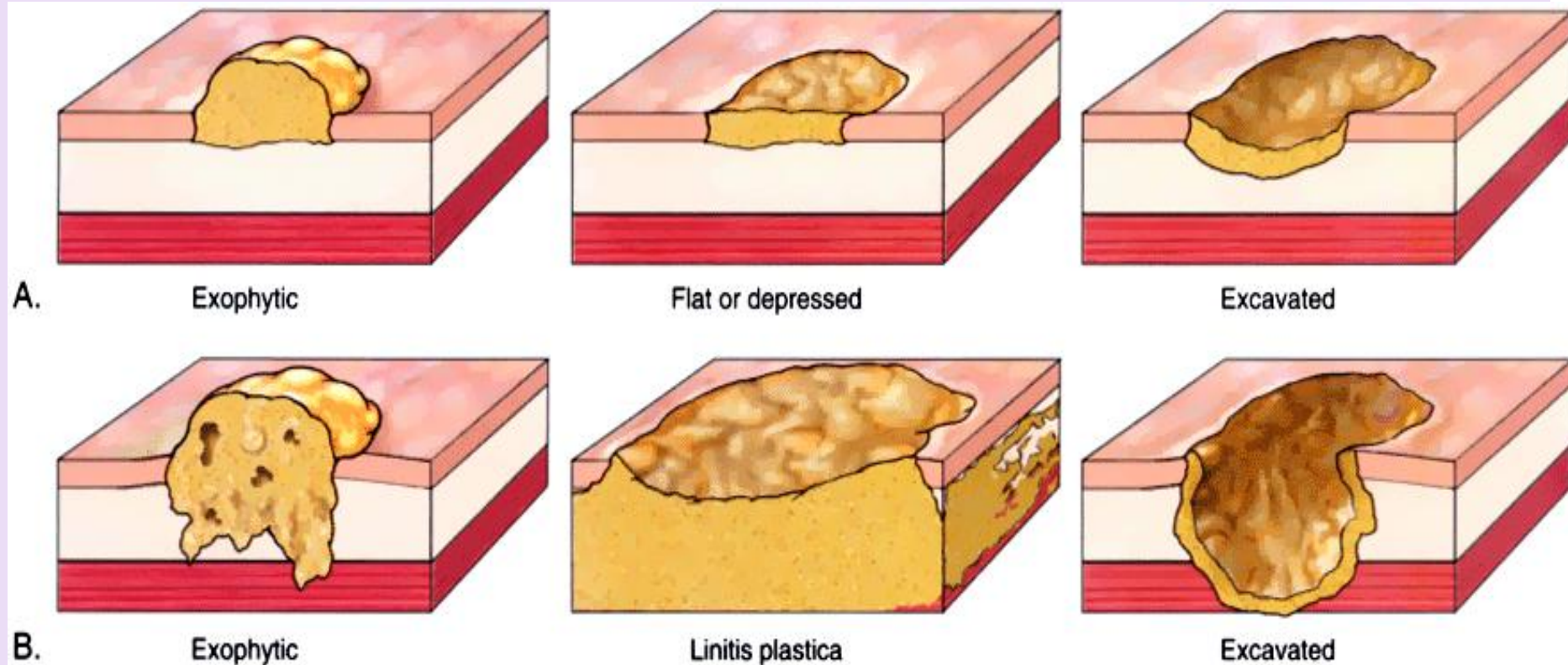
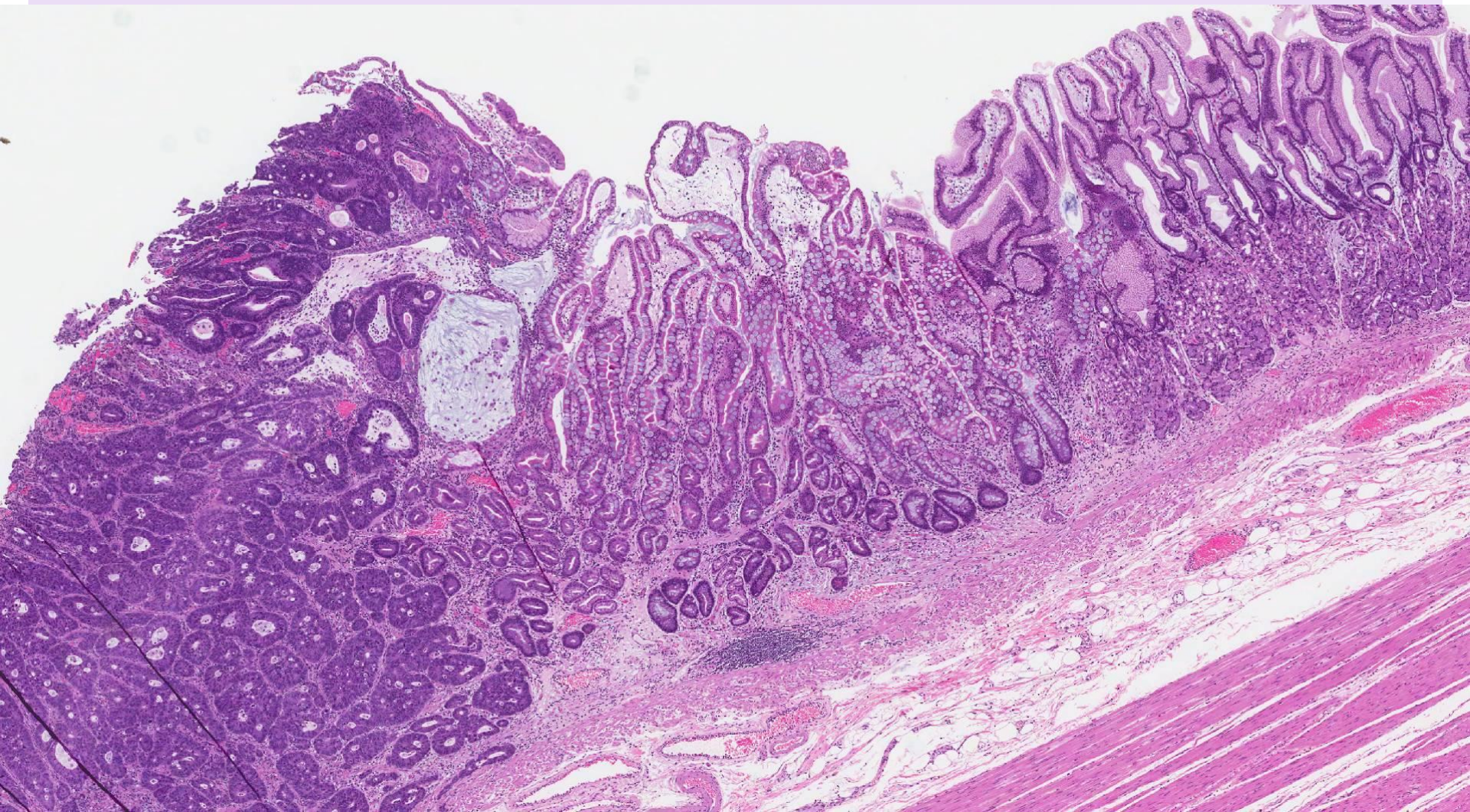


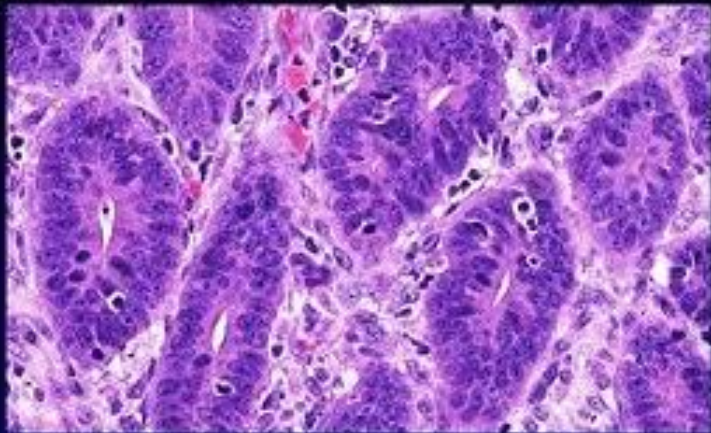
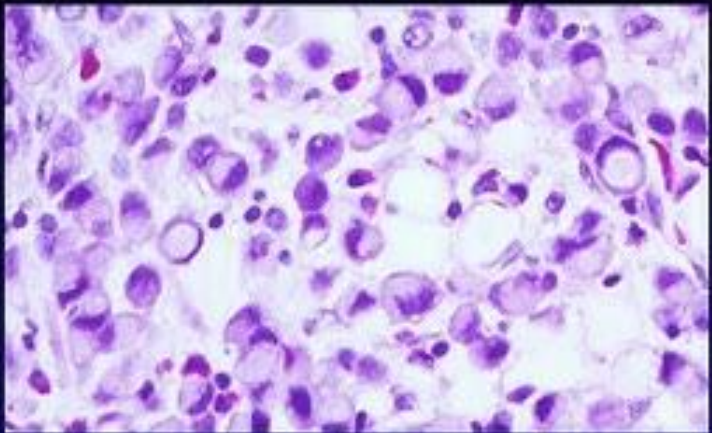
Diagram of growth patterns and spread of gastric carcinoma. In early gastric carcinoma (A), the tumor is confined to the mucosa and submucosa and may exhibit an exophytic, flat or depressed, or excavated conformation. Advanced gastric carcinoma (B) extends into the muscularis propria and beyond. Linitis plastica is an extreme form of flat or depressed advanced gastric carcinoma.

Adenocarcinoma gaster



Lauren Classification of Gastric Adenocarcinoma

SARAWUT - PATHOLOGY TO THE POINT

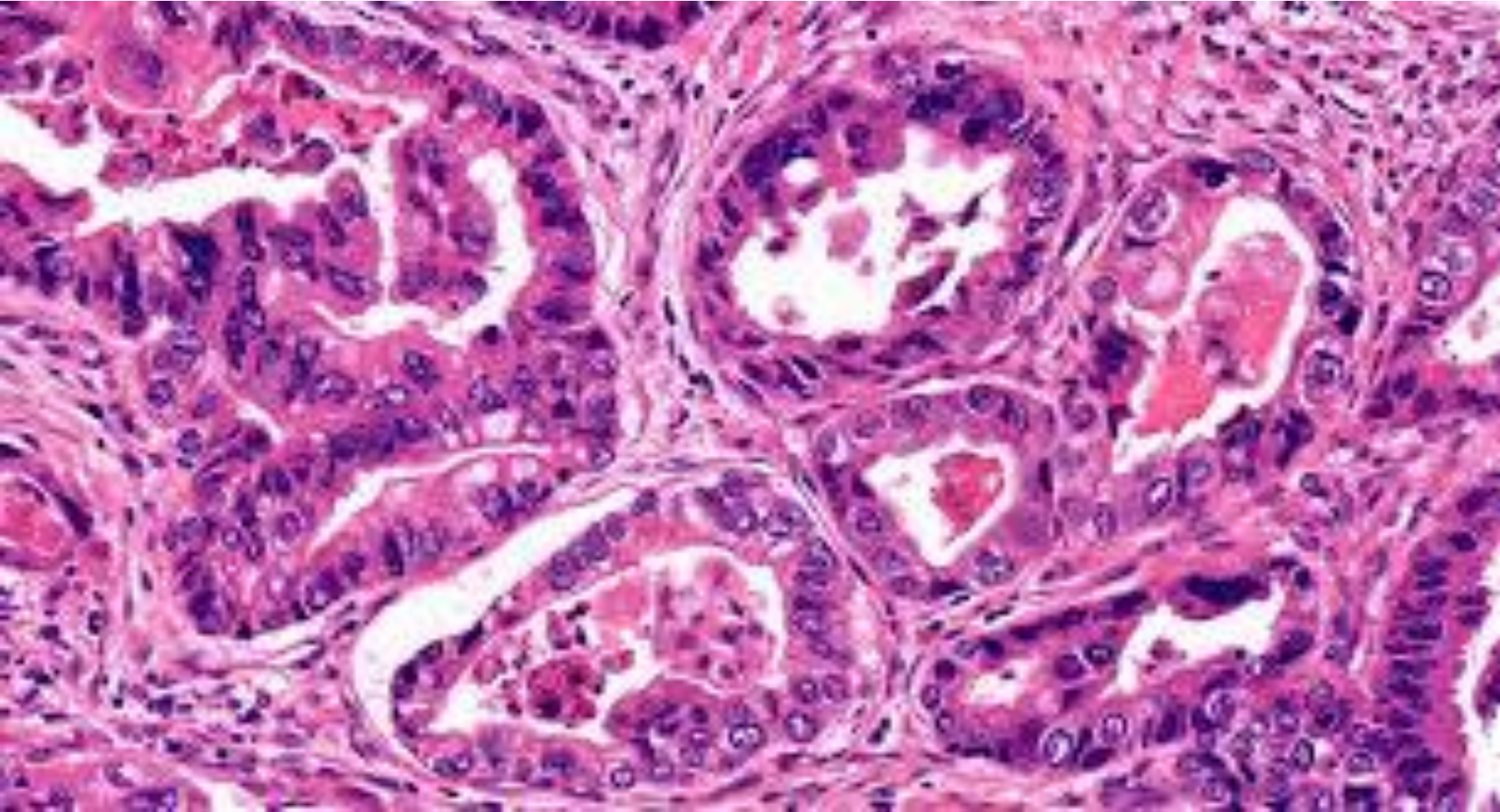
	Intestinal type	Diffuse type
Microscopic features	 Columnar, gland-forming cells infiltrating through desmoplastic stroma	 Signet-ring cells with cytoplasmic mucin vacuoles and peripherally crescent nuclei
WHO classification (2019)	Well- or moderately-differentiated ▪ Tubular (most common) ▪ Papillary Poorly differentiated	Poorly cohesive carcinoma ▪ Signet-ring cell phenotype (formerly <i>signet-ring cell carcinoma</i>) ▪ Other cell types
Macroscopic features	Commonly fungating mass with headped-up borders and ulceration	Linitis plastica – a leather bottle appearance with thickened gastric wall

GASTRIC CARCINOMA

Chronic Gastritis → Displasia → Gastric Carcinoma
Histopathology :

- **Intestinal Type:**

- well or moderately differentiated.
- usually form a **polypoid tumour or an ulcer** with “heaped-up edges”
- Usually found in association with **intestinal metaplasia**
- **Late stage** : Malignant glands infiltrating the gastric walls (lamina propria to muscularis propria).



GASTRIC CARCINOMA : INTESTINAL TYPE

G= Malignant Gland

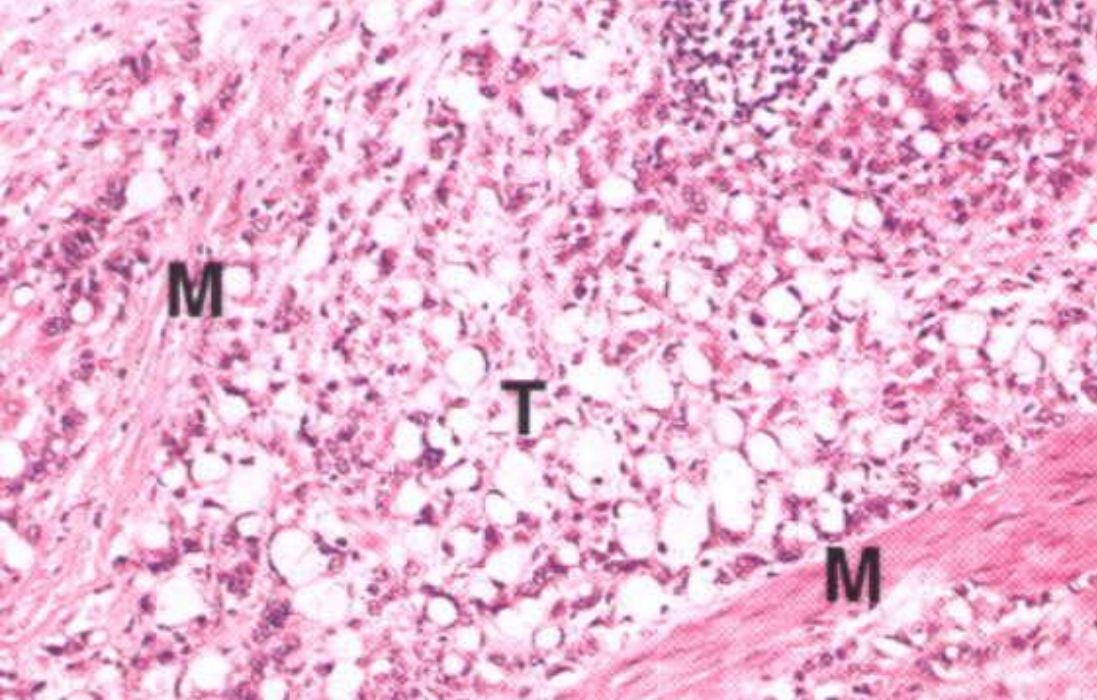
M= focal intestinal metaplasia/ or muscle

GASTRIC CARCINOMA (lanjutan)

Histopathology:

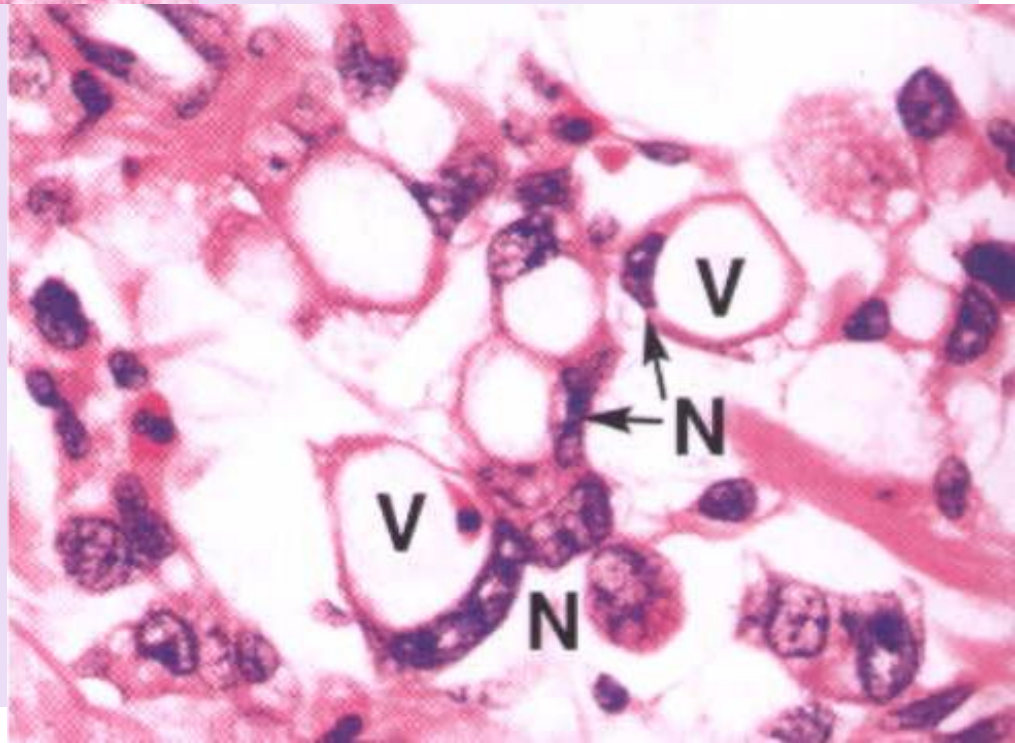
- **Diffuse Type (Linitis Plastica):**

- Poorly differentiated adenocarcinoma with little or no discernible gland formation.
- The tumour cells forming a diffuse sheet infiltrating between bundle of smooth muscle.
- Tumour cells : **Signet Ring Cells.**

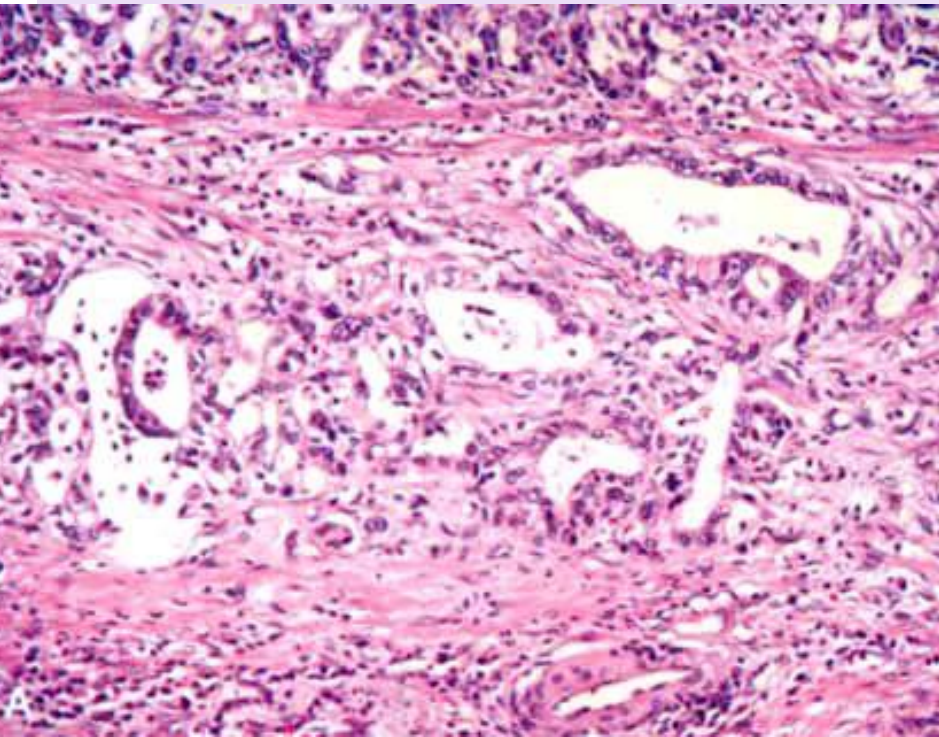


M: Smooth Muscle
T : Tumour Cells
V : Vacuoles
N : Nucleous

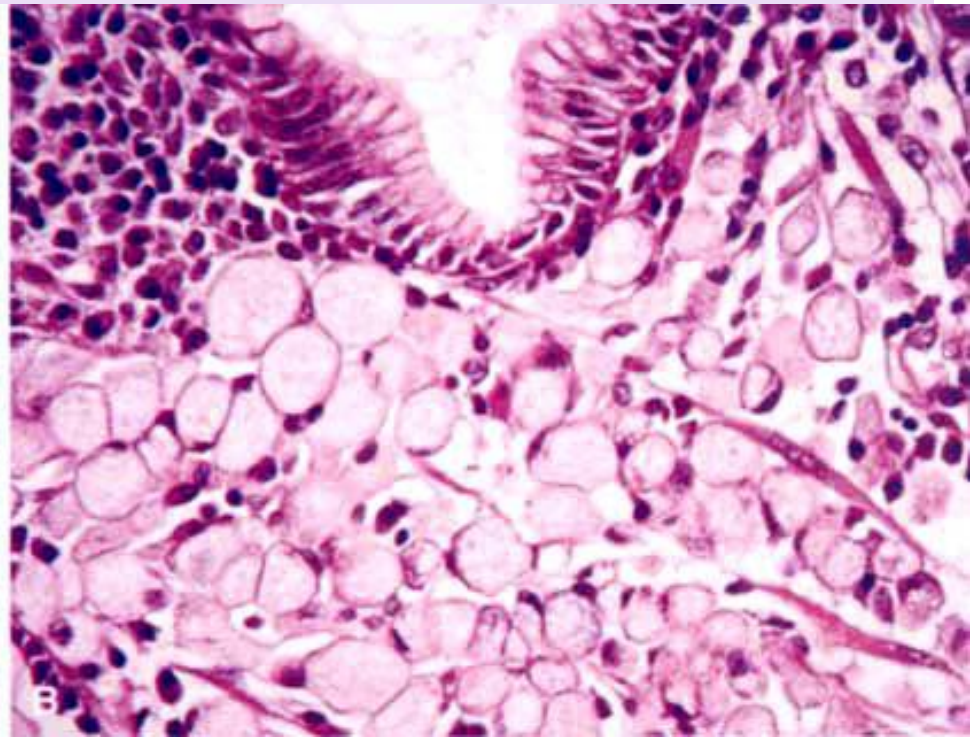
**Gastric carcinoma =
 DIFFUSE TYPE**



Gastric carcinoma

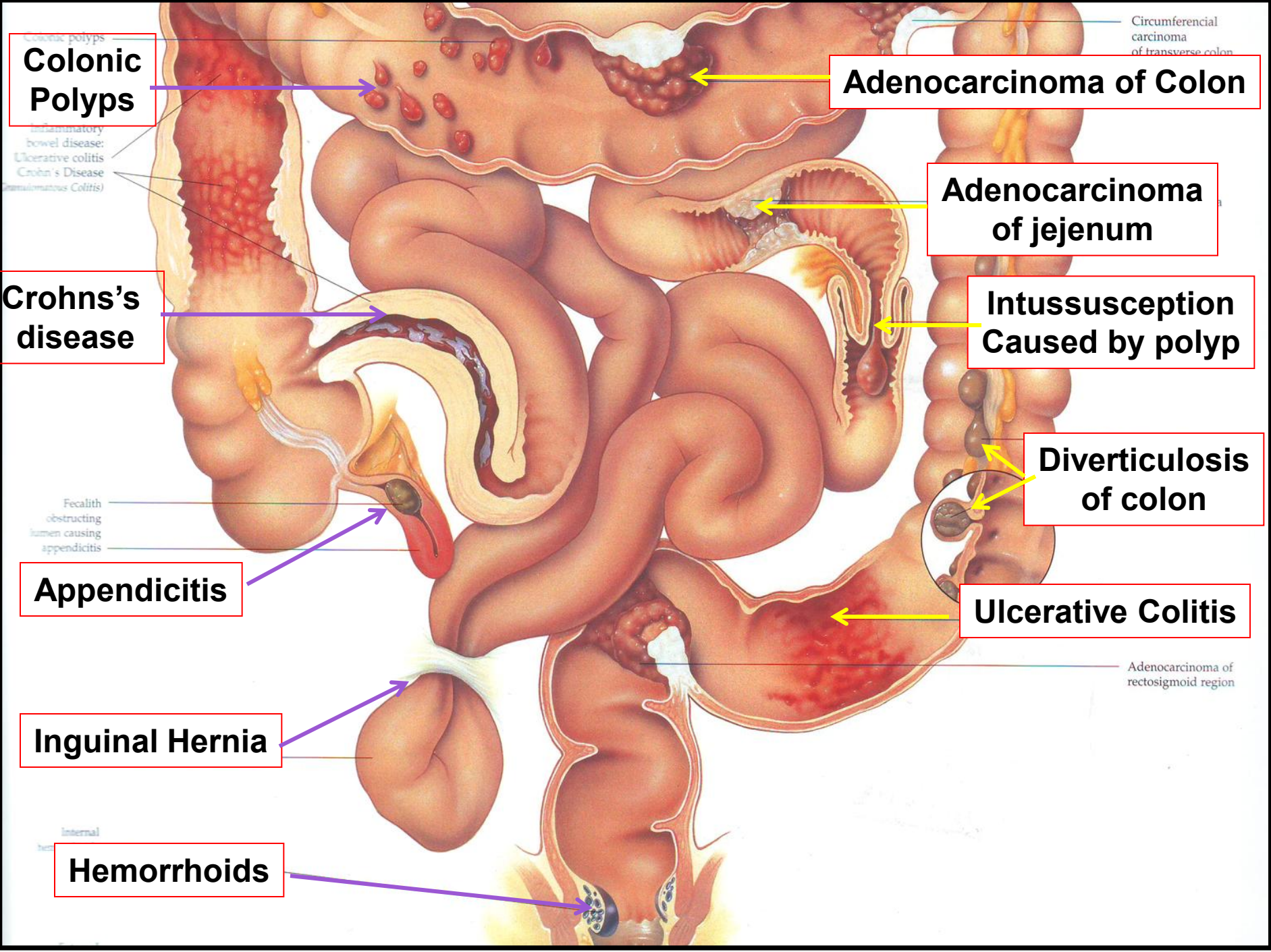


A, Intestinal type demonstrating gland formation by malignant cells, which are invading the muscular wall of the stomach



B, Diffuse type demonstrating signet-ring carcinoma cells

INTESTINUM



Colonic polyps
Colonic Polyps

Inflammatory bowel disease:
Ulcerative colitis
Crohn's Disease
(Granulomatous Colitis)

Crohn's disease

Appendicitis

Inguinal Hernia

Hemorrhoids

Adenocarcinoma of Colon

Adenocarcinoma of jejunum

Intussusception Caused by polyp

Diverticulosis of colon

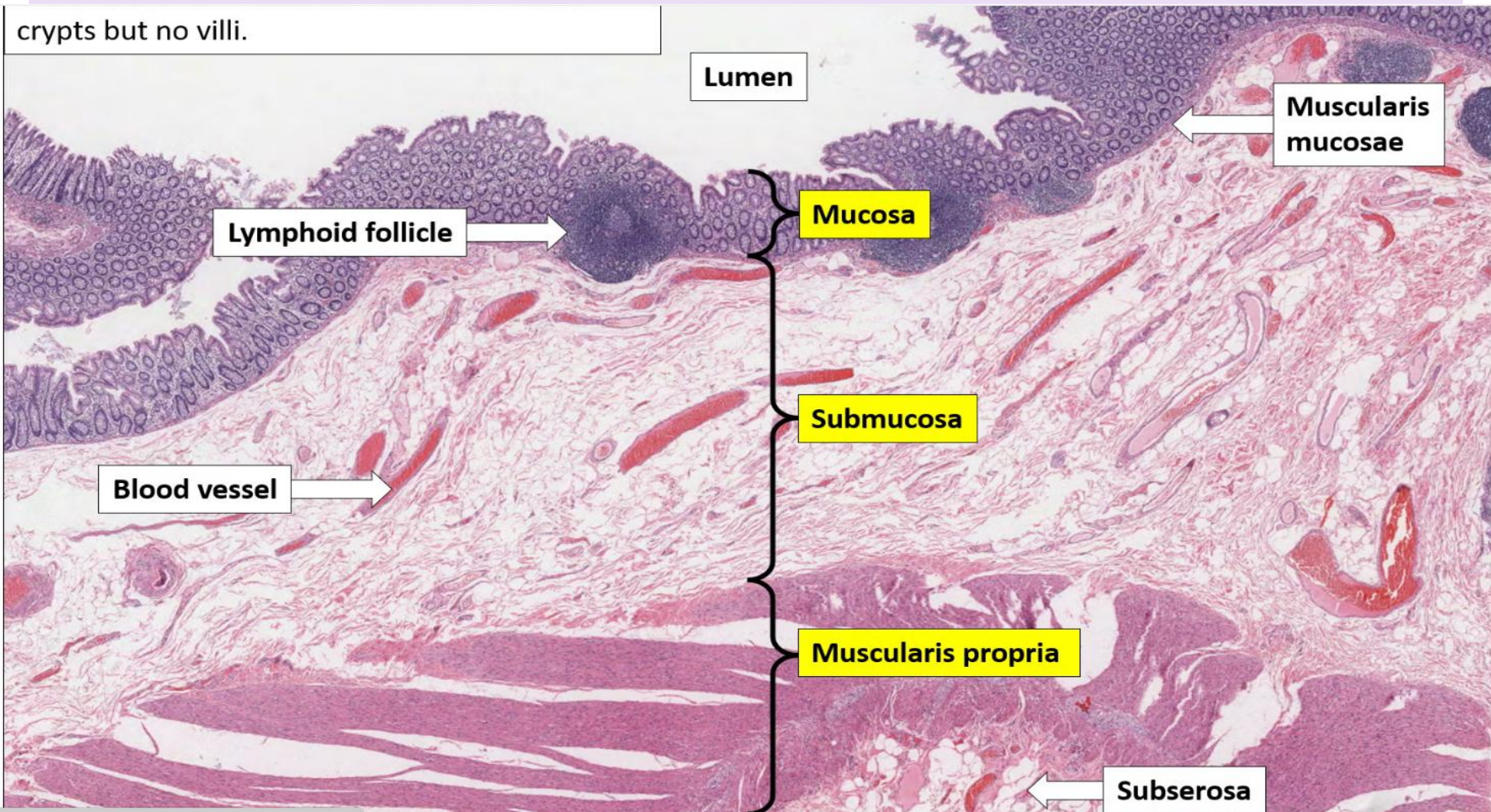
Ulcerative Colitis

Circumferencial carcinoma of transverse colon

Adenocarcinoma of rectosigmoid region

Fecalith obstructing lumen causing appendicitis

Histology: Large Intestinal

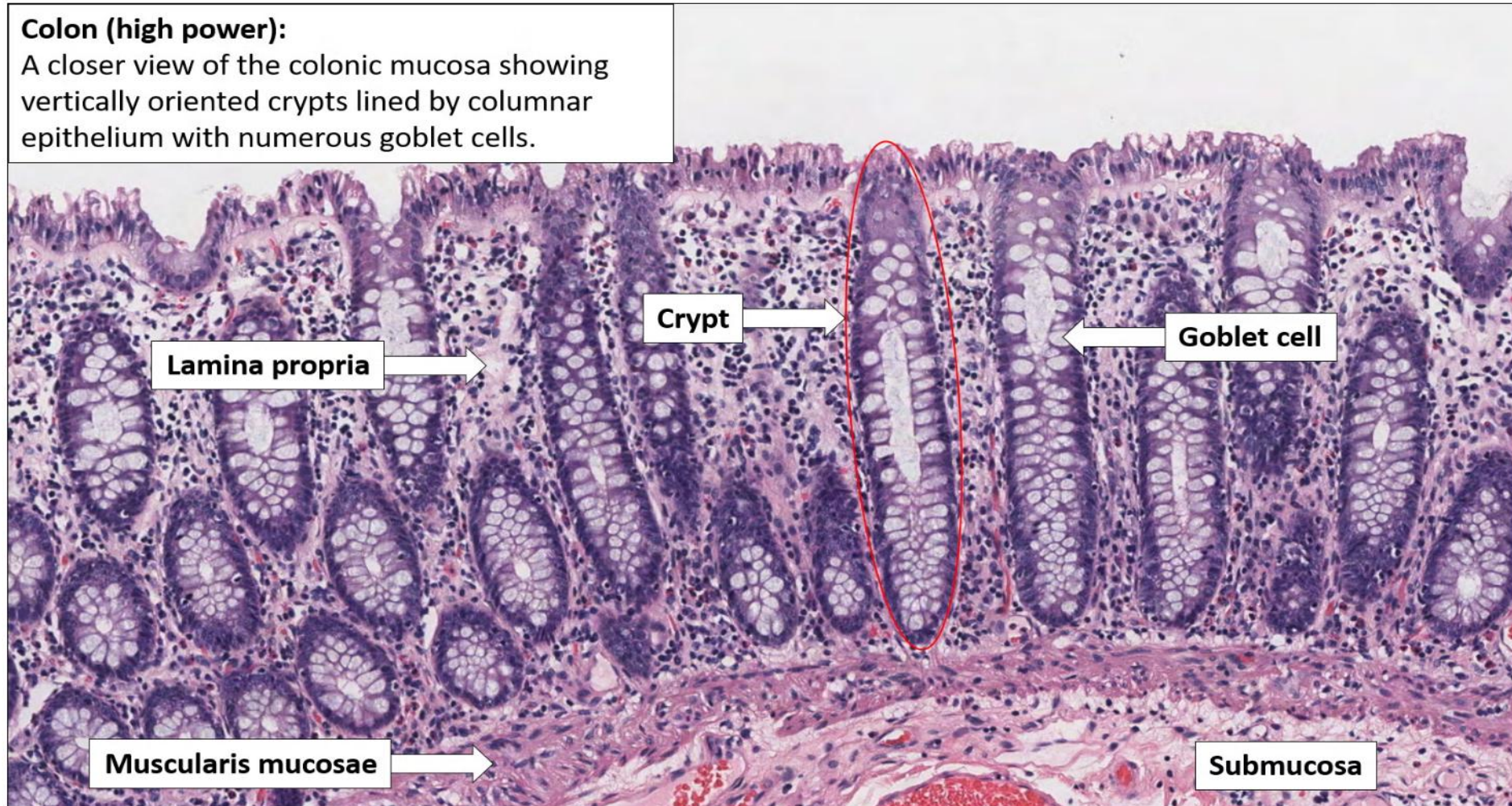


Histology: Large Intestinal

Annotations

Colon (high power):

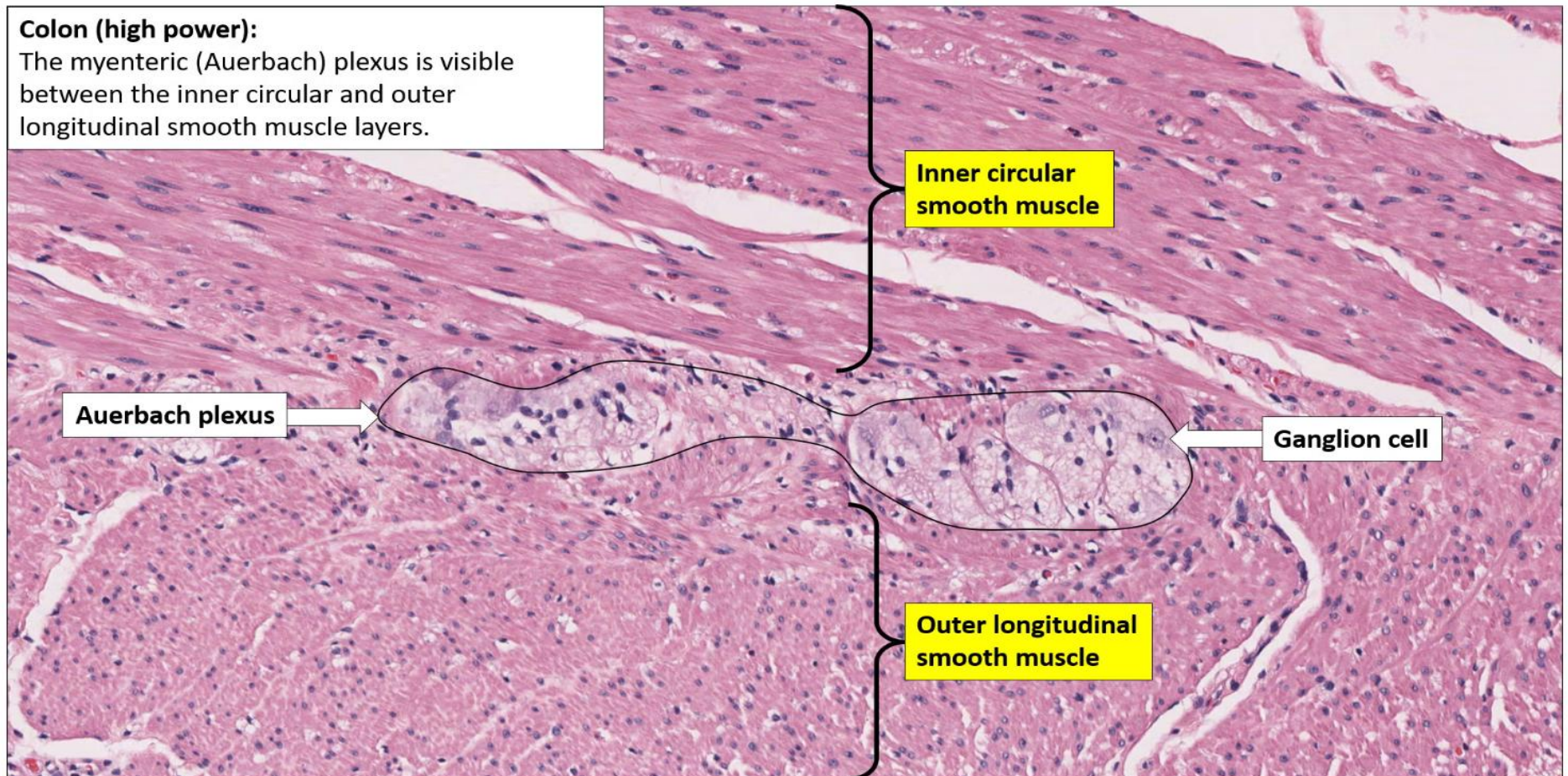
A closer view of the colonic mucosa showing vertically oriented crypts lined by columnar epithelium with numerous goblet cells.



Histology: Large Intestinal

Colon (high power):

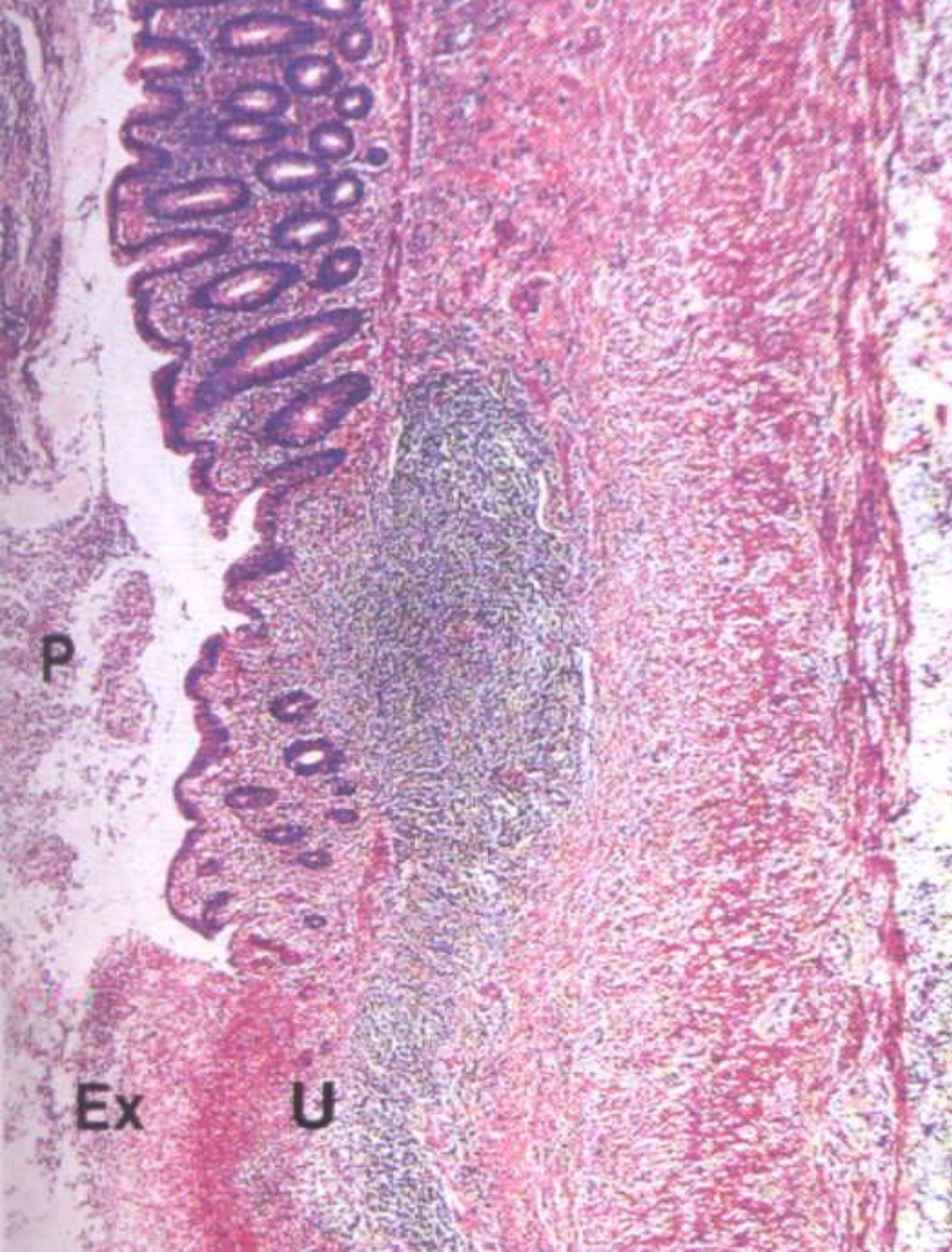
The myenteric (Auerbach) plexus is visible between the inner circular and outer longitudinal smooth muscle layers.



ACUTE APPENDICITIS

Early Acute Appendicitis

- Ulceration of the mucosa with overlying acute fibrinopurulent inflammatory exudate.
- Purulent exudate within the lumen.



ACUTE APPENDICITIS (I)

P : Purulent exudate

Ex : Exudate

U : Ulceration of the mucosa

ACUTE APPENDICITIS

Later Acute Appendicitis

- The inflammation spreads throughout all layers of the wall of the appendix
- Mucosal ulceration more extensive.
- Large numbers of neutrophils have infiltrated through the submucosa and muscular layer to the serosa.
- One point of fibrinous exudate is beginning to form on the peritoneal surface (Peritonitis)



ACUTE APPENDICITIS (II)

SM : Submucosa

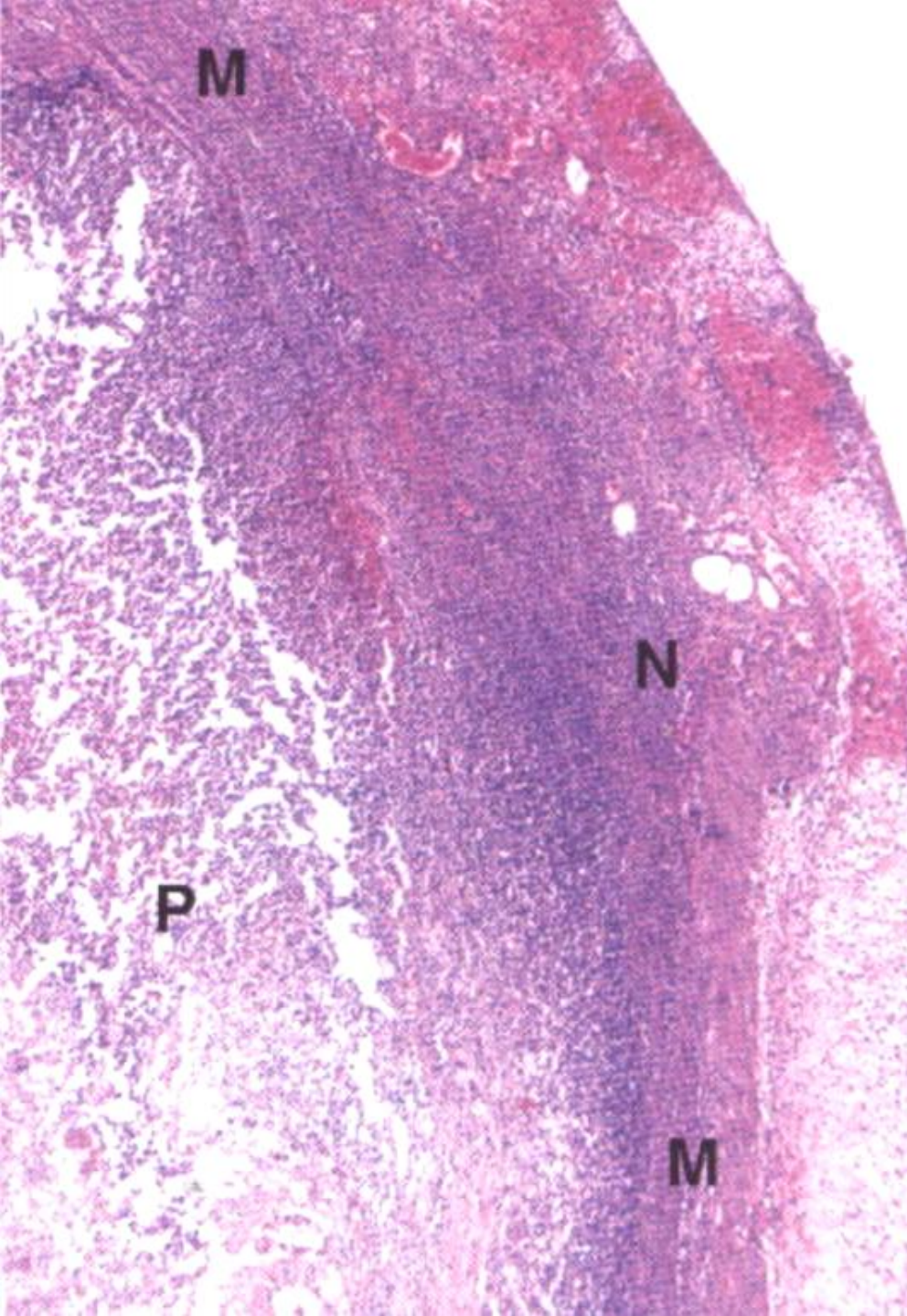
M : Mucosa

S : Serosa

G : Gland

GANGRENOUS APPENDICITIS

- Severe continuing inflammation of the appendix wall often leads to extensive **necrosis of the muscle layer (gangren)**
- **Perforation** is imminent.
- **Pus (in the lumen) → peritoneal cavity → severe and extensive peritonitis**
- **Other complications: absces, septicaemia, shock & death.**



ACUTE APPENDICITIS (III)

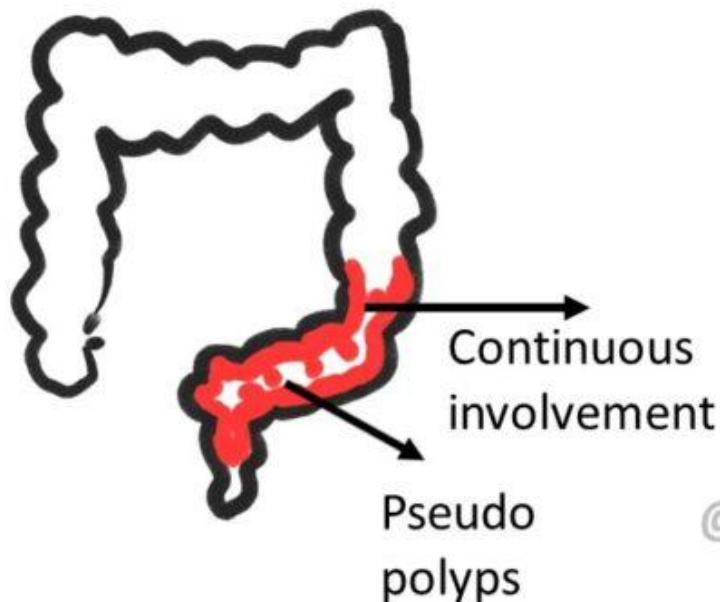
M : Muscle

N : Neutrophyl

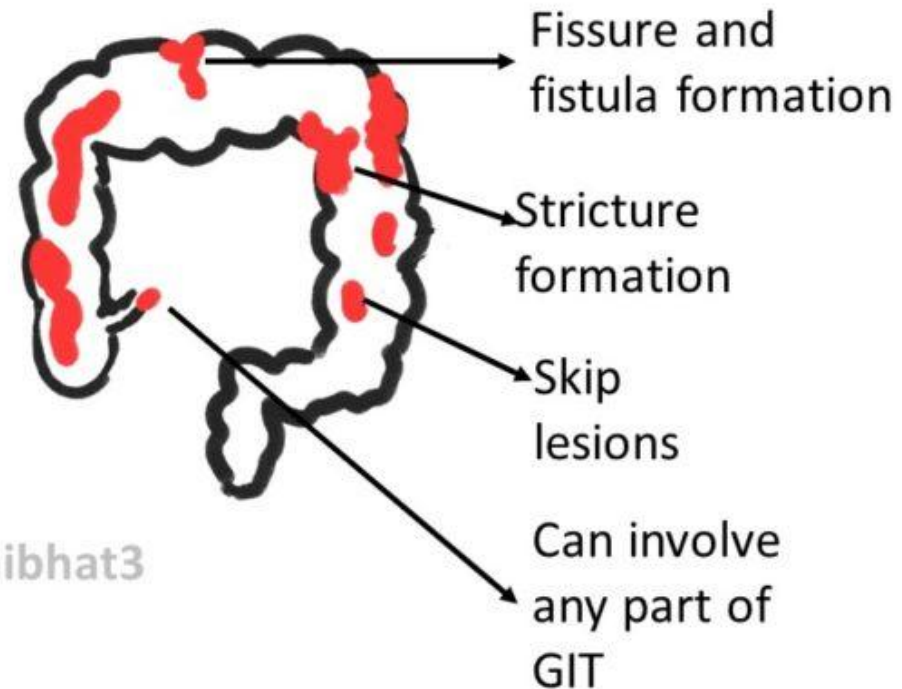
P : Purulent Exudate

IBD, or Inflammatory Bowel Disease, refers to a group of chronic inflammatory conditions affecting the digestive tract, primarily the colon and small intestine. The two main types of IBD are Crohn's disease and ulcerative colitis.

Ulcerative colitis



Crohn's disease

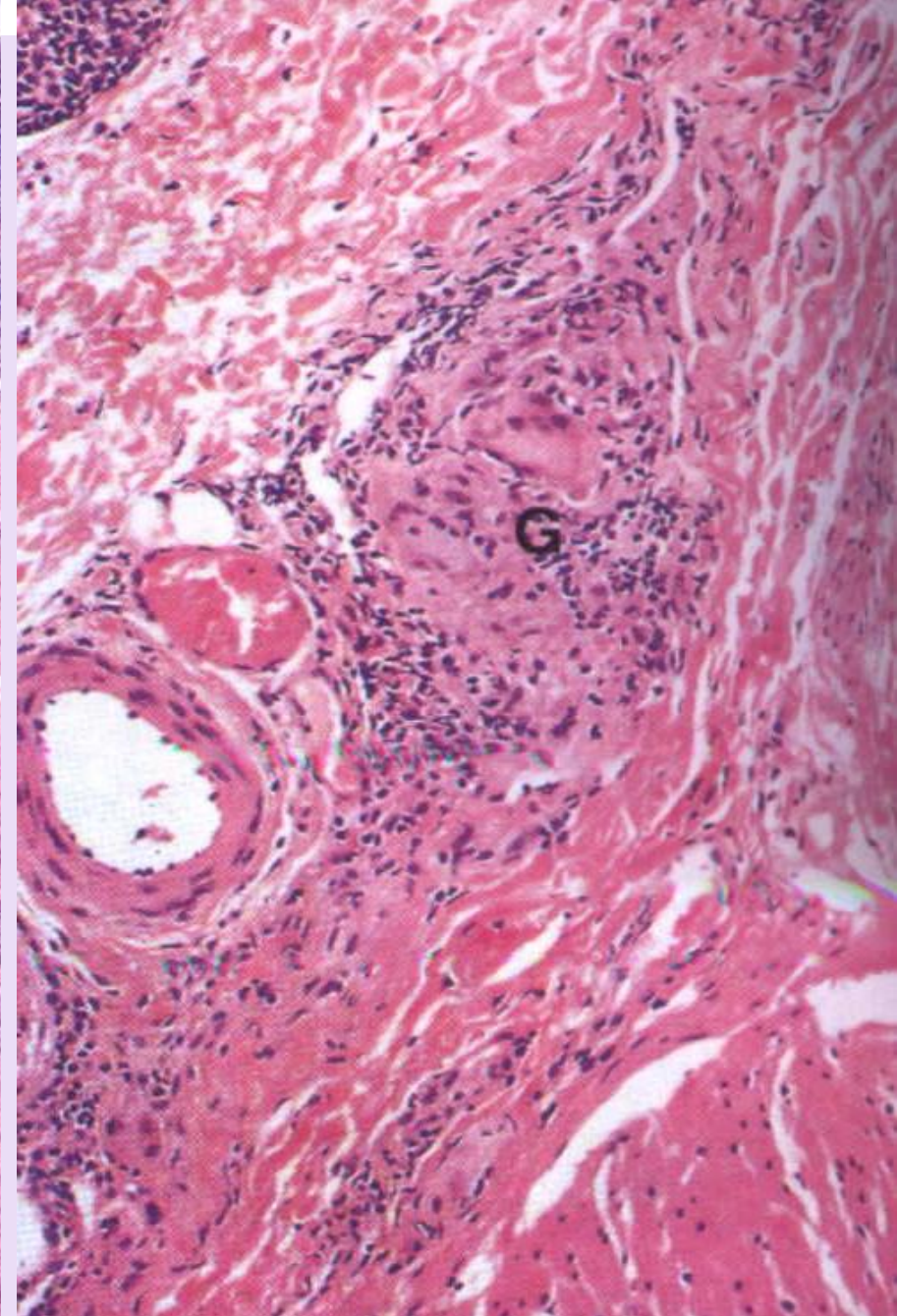
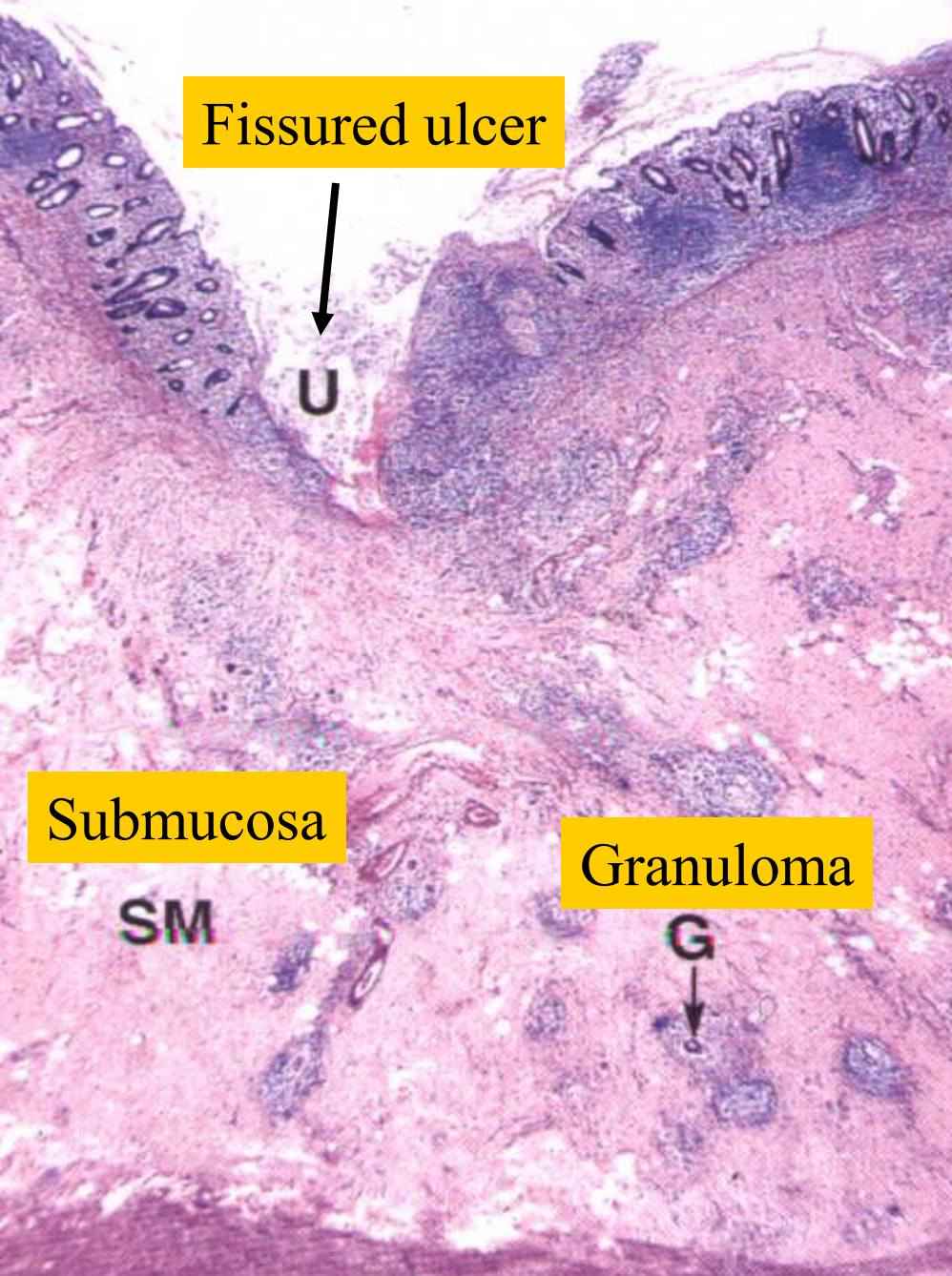


@archibhat3

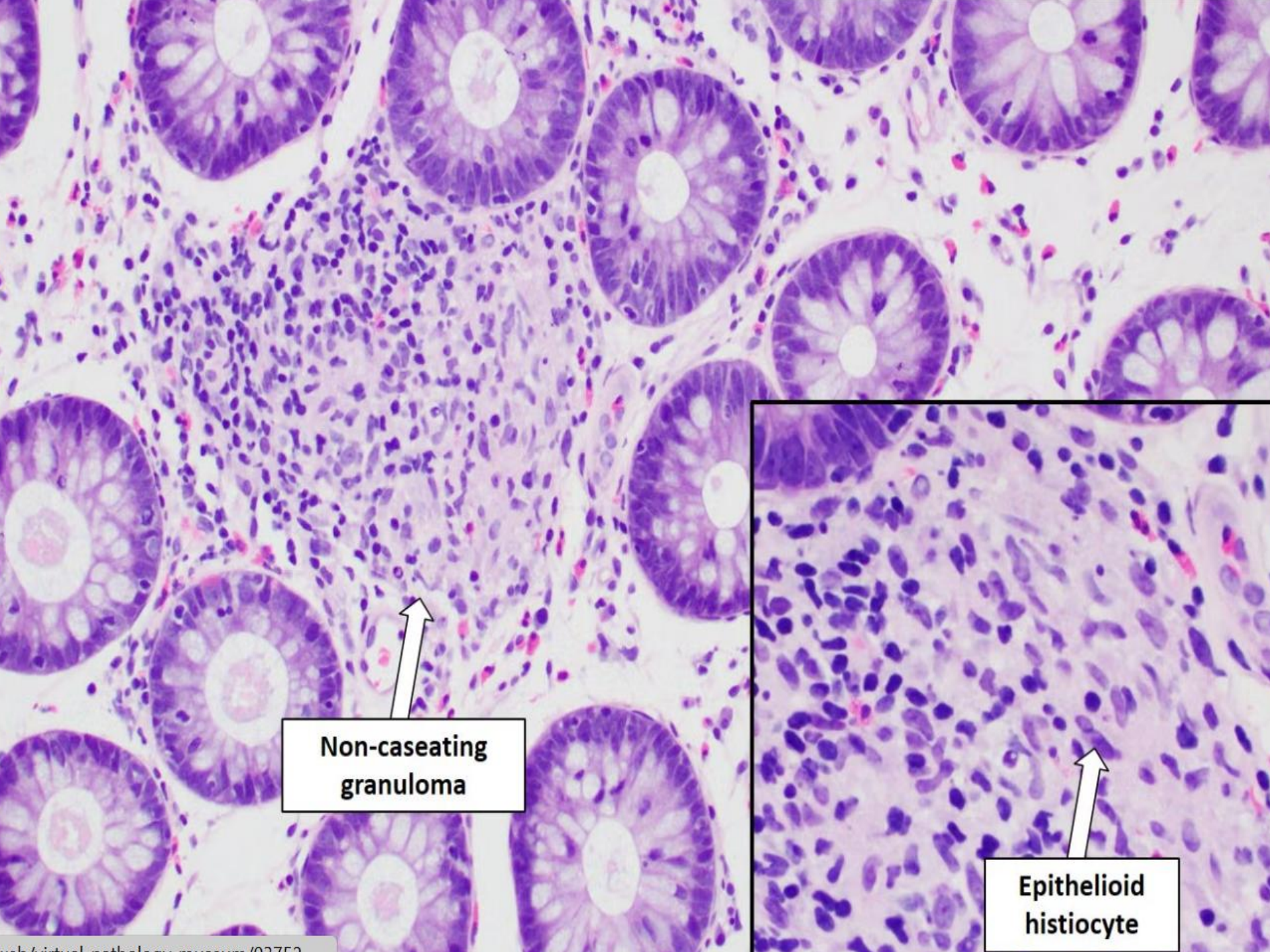
CROHN'S DISEASE

Histopathology:

- Marked **oedema and inflammation** of the submucosa → gross thickening of the intestinal wall.
- Between thickening → **fissured ulcer formation of the fistulae**
- Chronic inflammatory changes are **transmural** widespread fibrosis → **bowel obstruction**.
- **Granuloma** formation with **giant cells**



CHRON'S DISEASE (A= Fissured ulcer, B= Chron's granuloma)

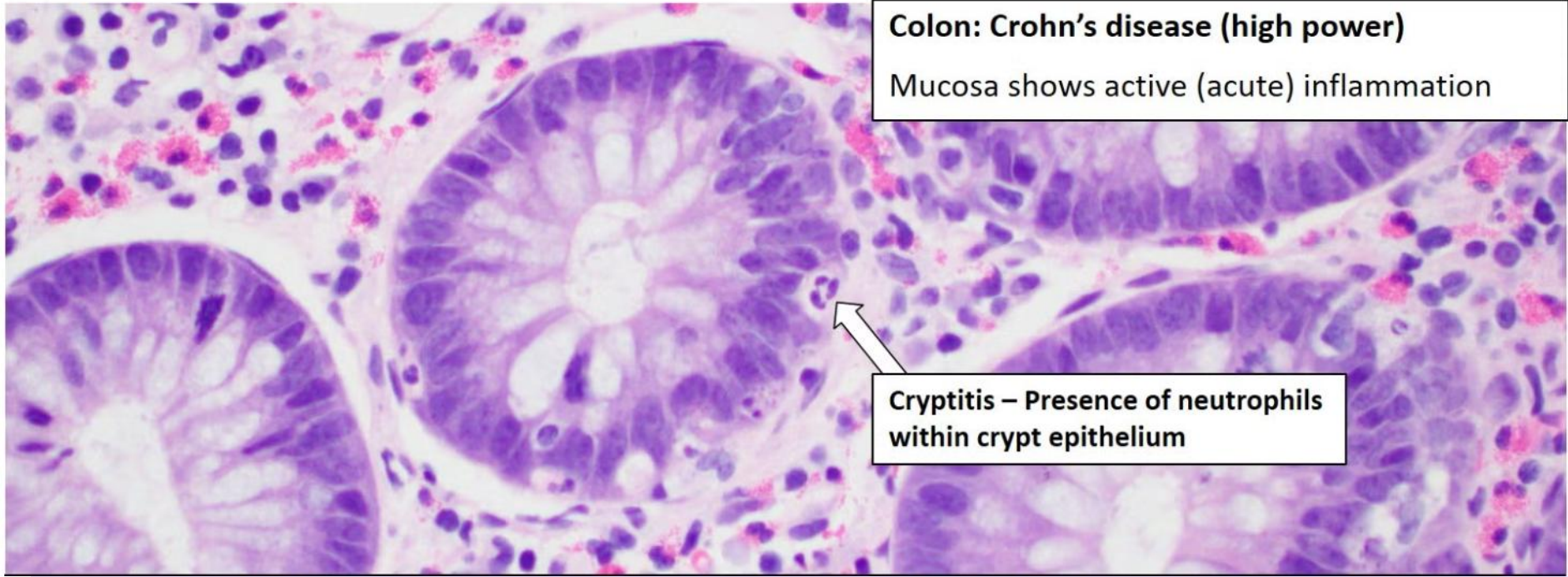


**Non-caseating
granuloma**

**Epithelioid
histiocyte**

Colon: Crohn's disease (high power)

Mucosa shows active (acute) inflammation



Cryptitis – Presence of neutrophils within crypt epithelium

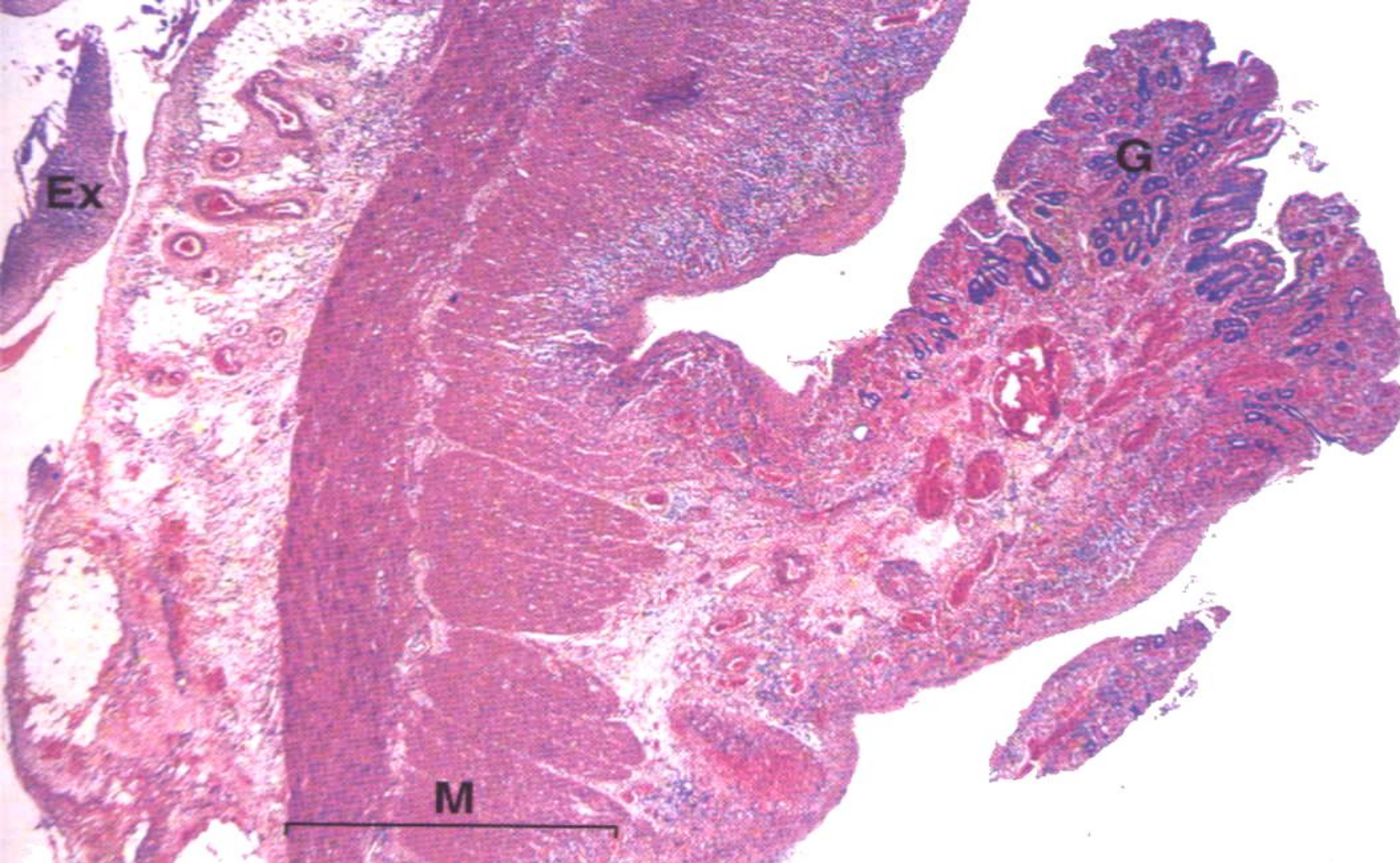
This histological image shows a cross-section of colonic mucosa. Several crypts are visible, lined by columnar epithelial cells. In the central crypt, an arrow points to a neutrophil (a cell with a multi-lobed nucleus) that has migrated into the crypt epithelium, a condition known as cryptitis. The surrounding lamina propria contains numerous other inflammatory cells, including lymphocytes and plasma cells, indicating active inflammation.

ULCERATIVE COLITIS

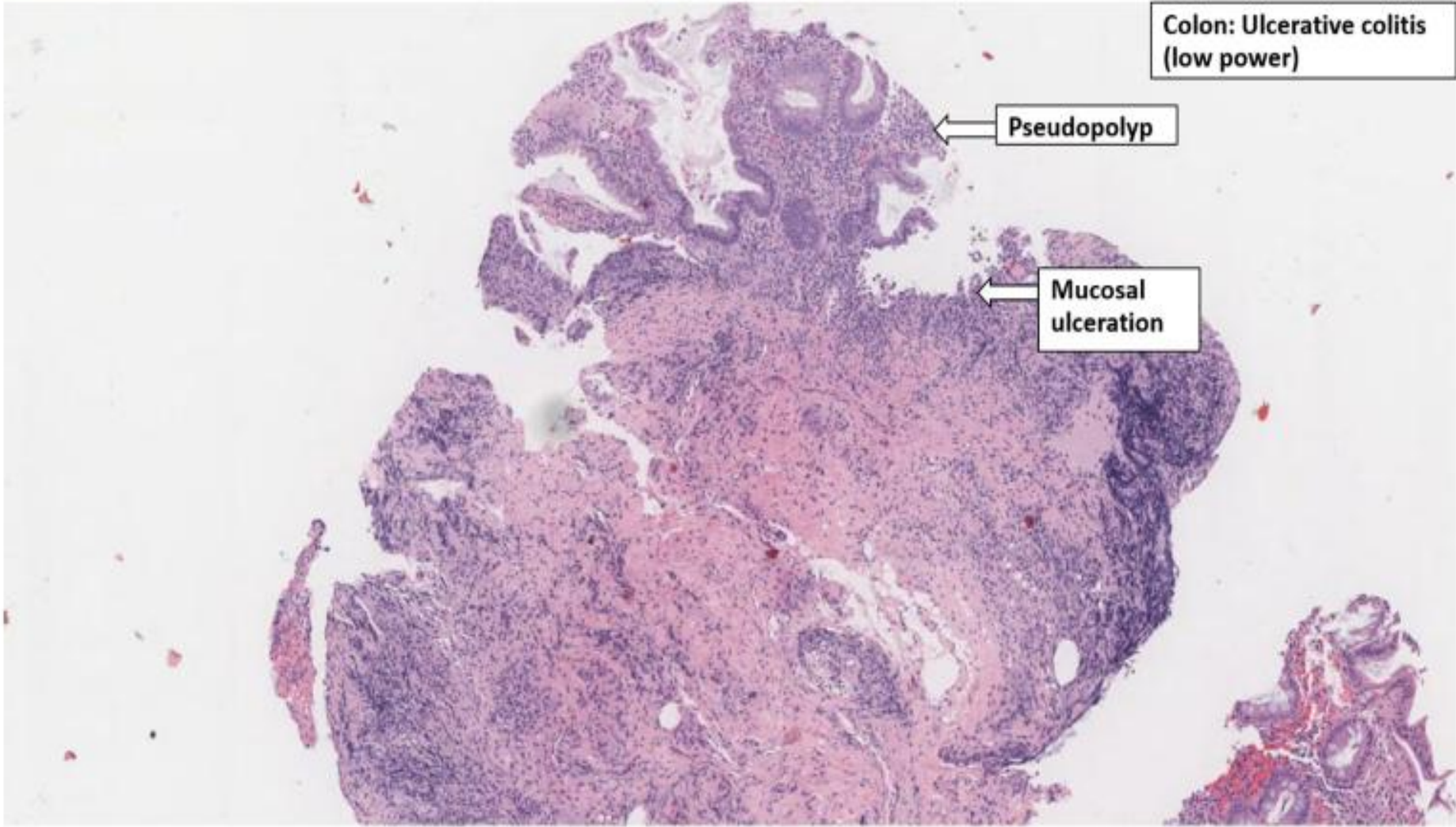
- Chronic relapsing inflammatory disease
- Affecting the large bowel
- Unknown cause

Histopathology:

- Ulcerative process destroyed much of the mucosa and submucosa (crypts abscesses).
- Non ulcerative mucosa → inflammatory pseudopolyps.



ULCERATIVE COLITIS (I)
G: Gland, M: Muscle, Ex: Exudate

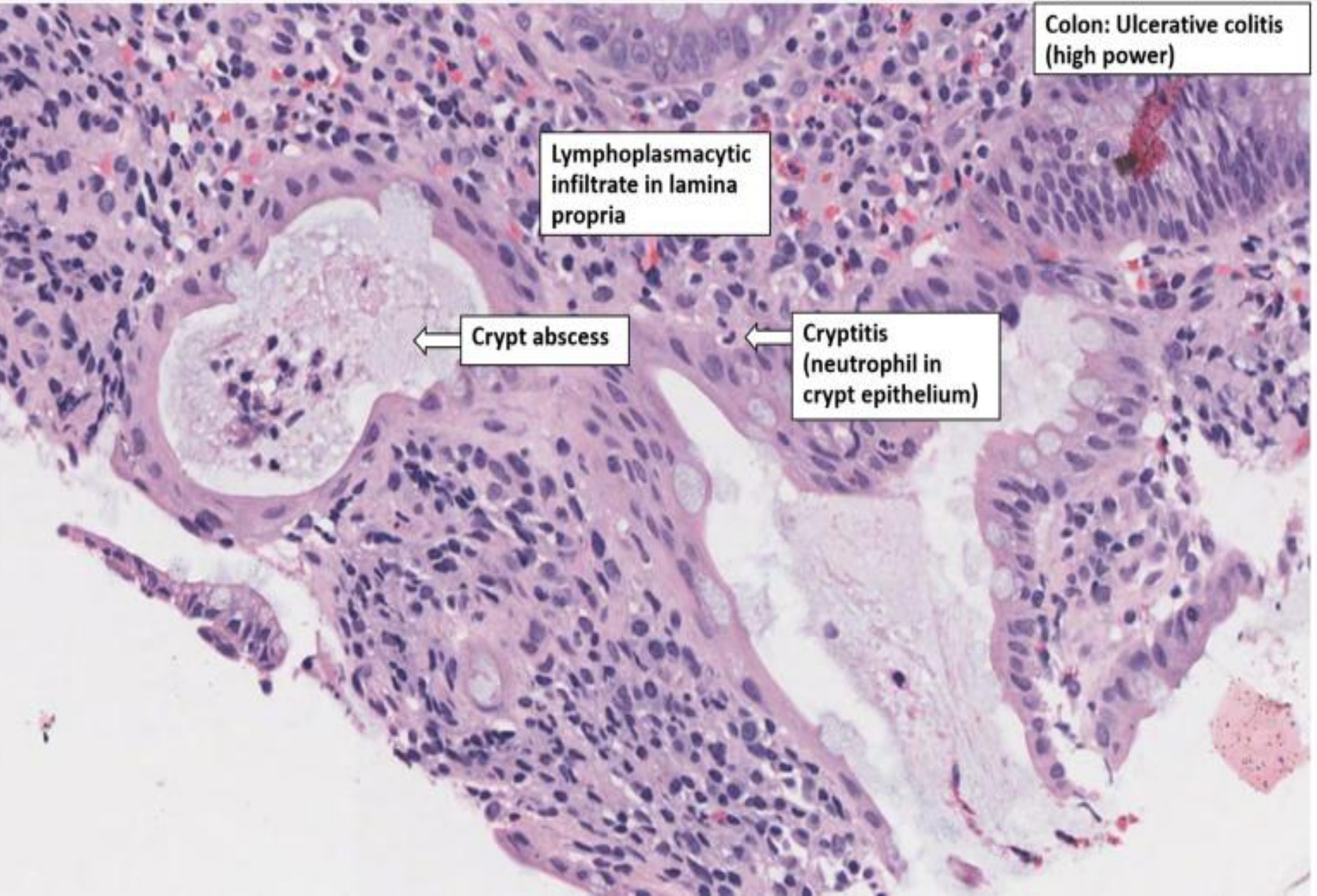


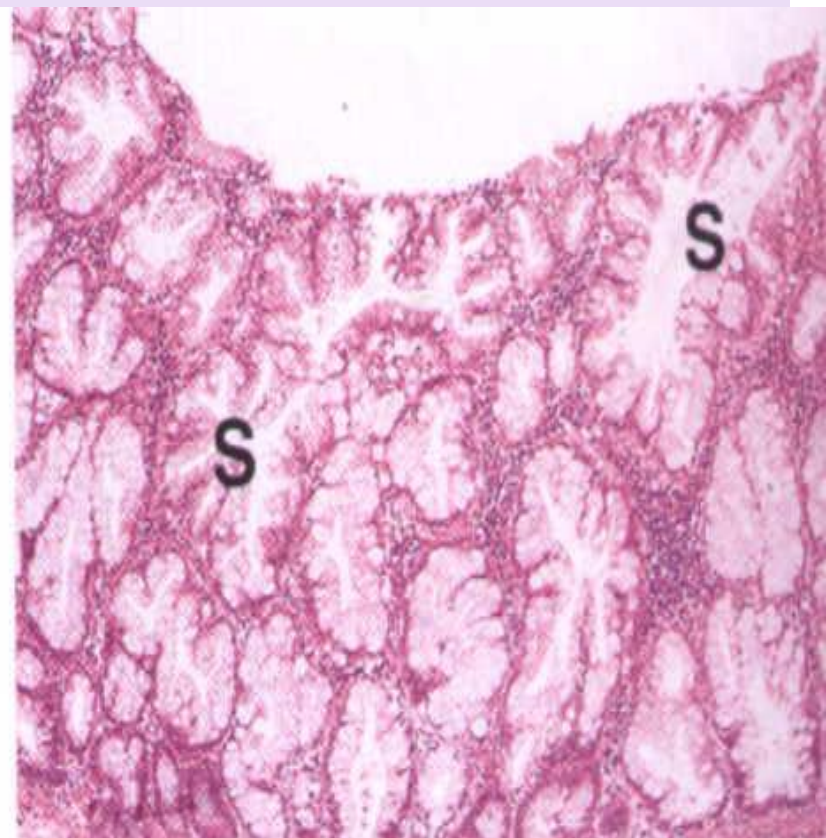
Colon: Ulcerative colitis
(high power)

Lymphoplasmacytic
infiltrate in lamina
propria

← Crypt abscess

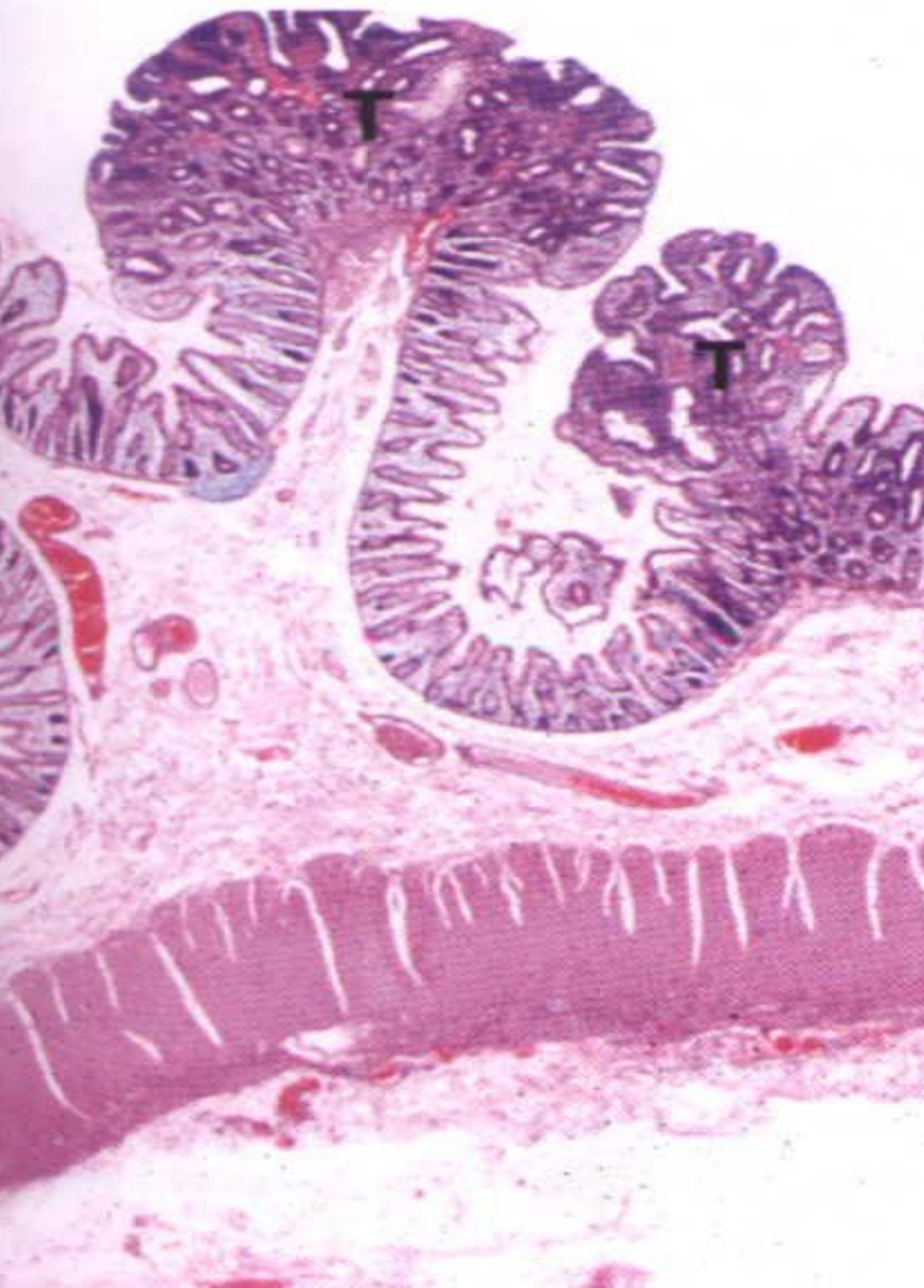
← Cryptitis
(neutrophil in
crypt epithelium)





HYPERPLASTIC POLYP (large bowel)
Small, often multiple and occasionally bleed

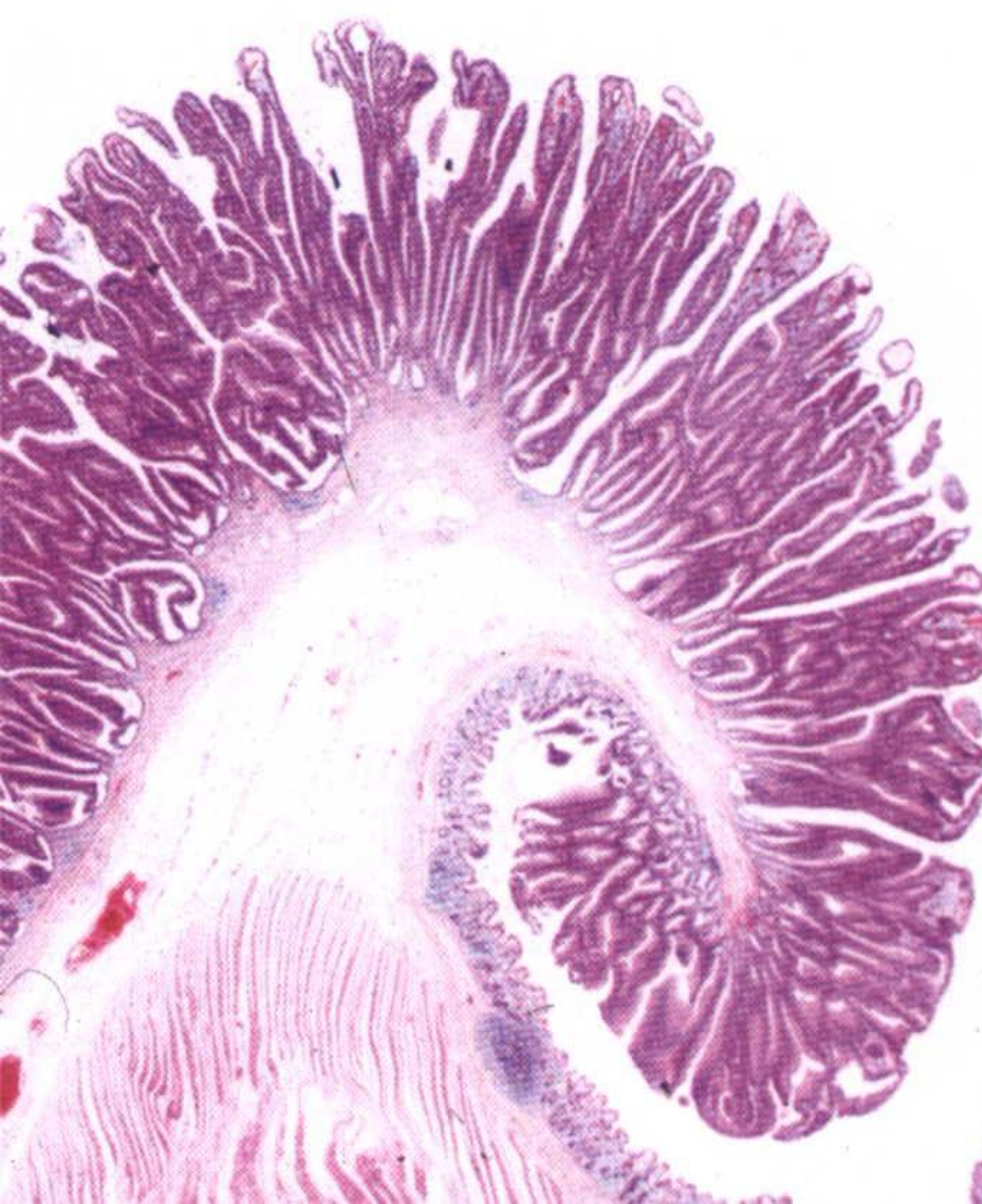
S : “Sawtooth” outline



COLONIC ADENOMATOUS POLYP

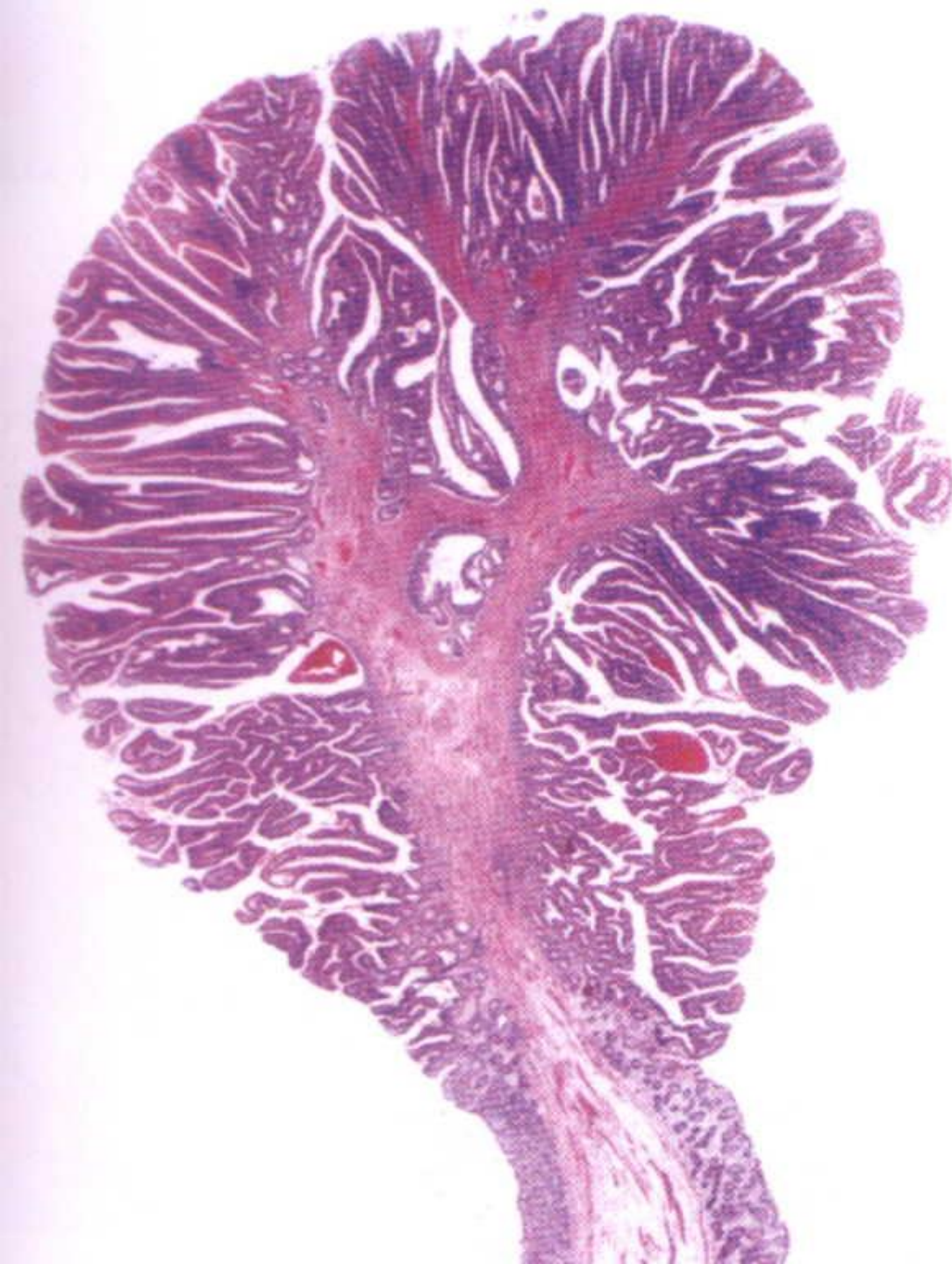
Tubular Adenoma (T)

- Displastic colonic epithelium arranged in straight tubular gland
- Solitary or multiple (familial polyposis coli)



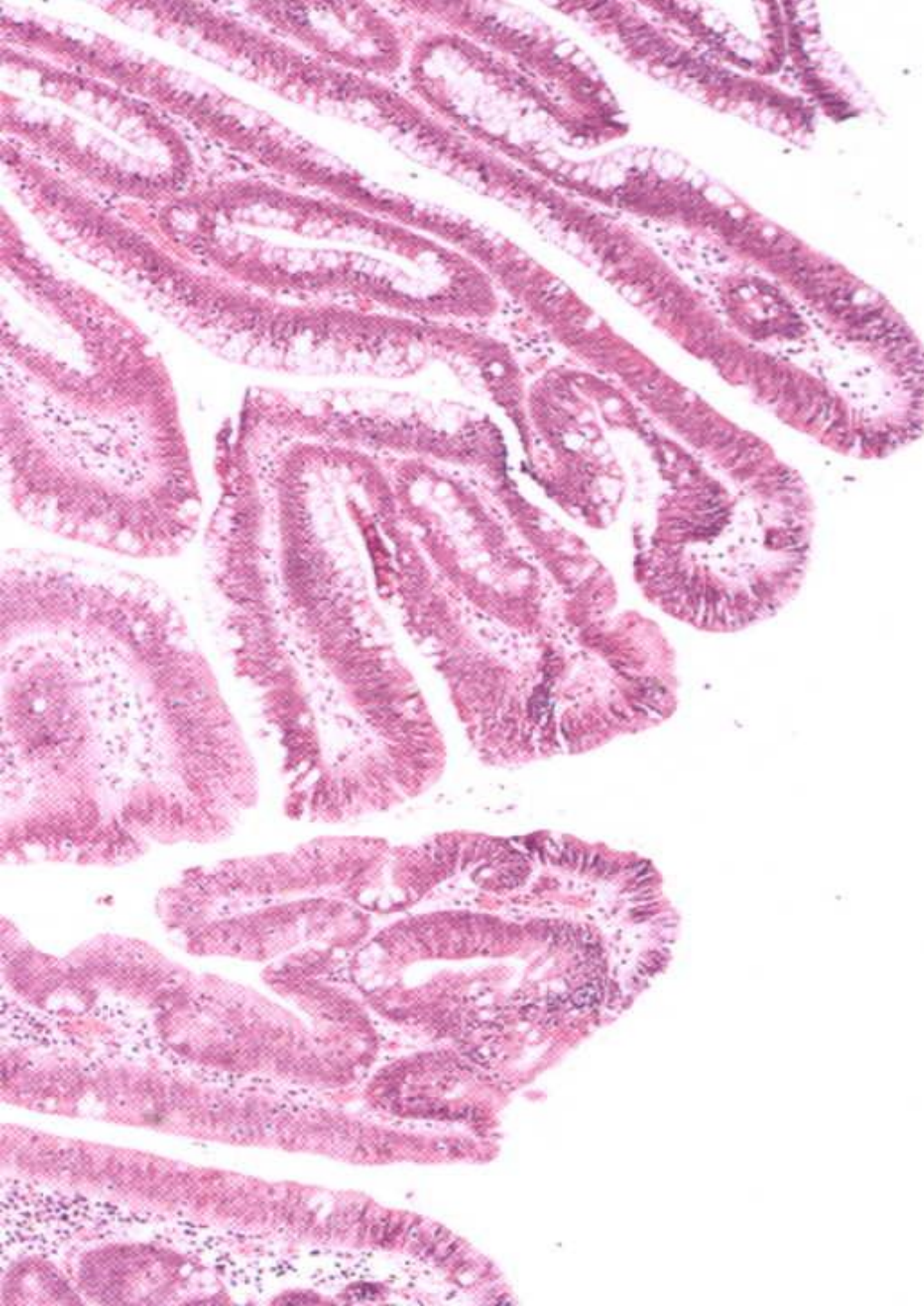
COLONIC ADENOMATOUS POLYPS

Villous Adenoma



**COLONIC
ADENOMATOUS
POLYP**

Tubulovillous Adenoma

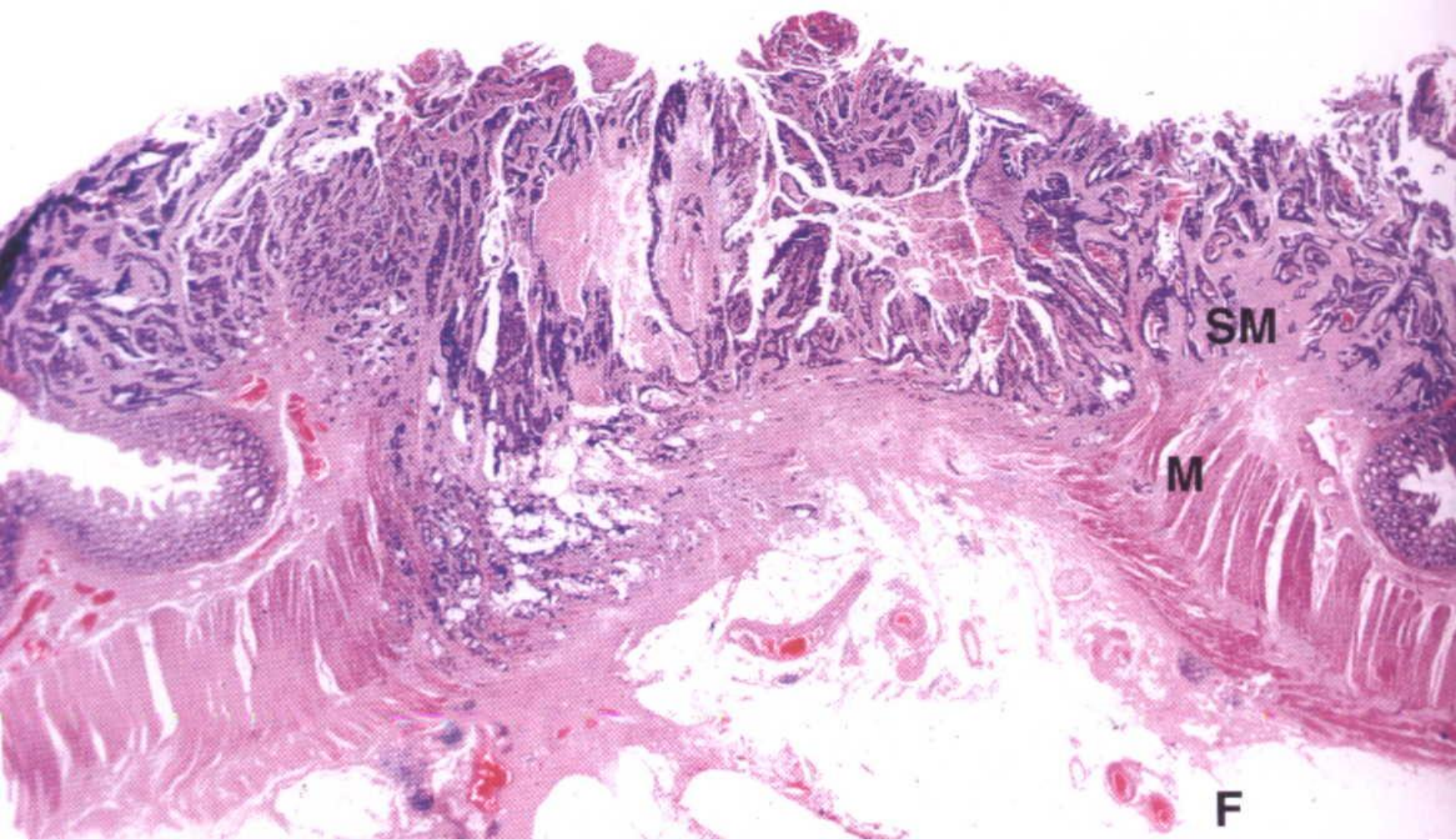


COLONIC ADENOMATOUS POLYP

Villous Adenoma

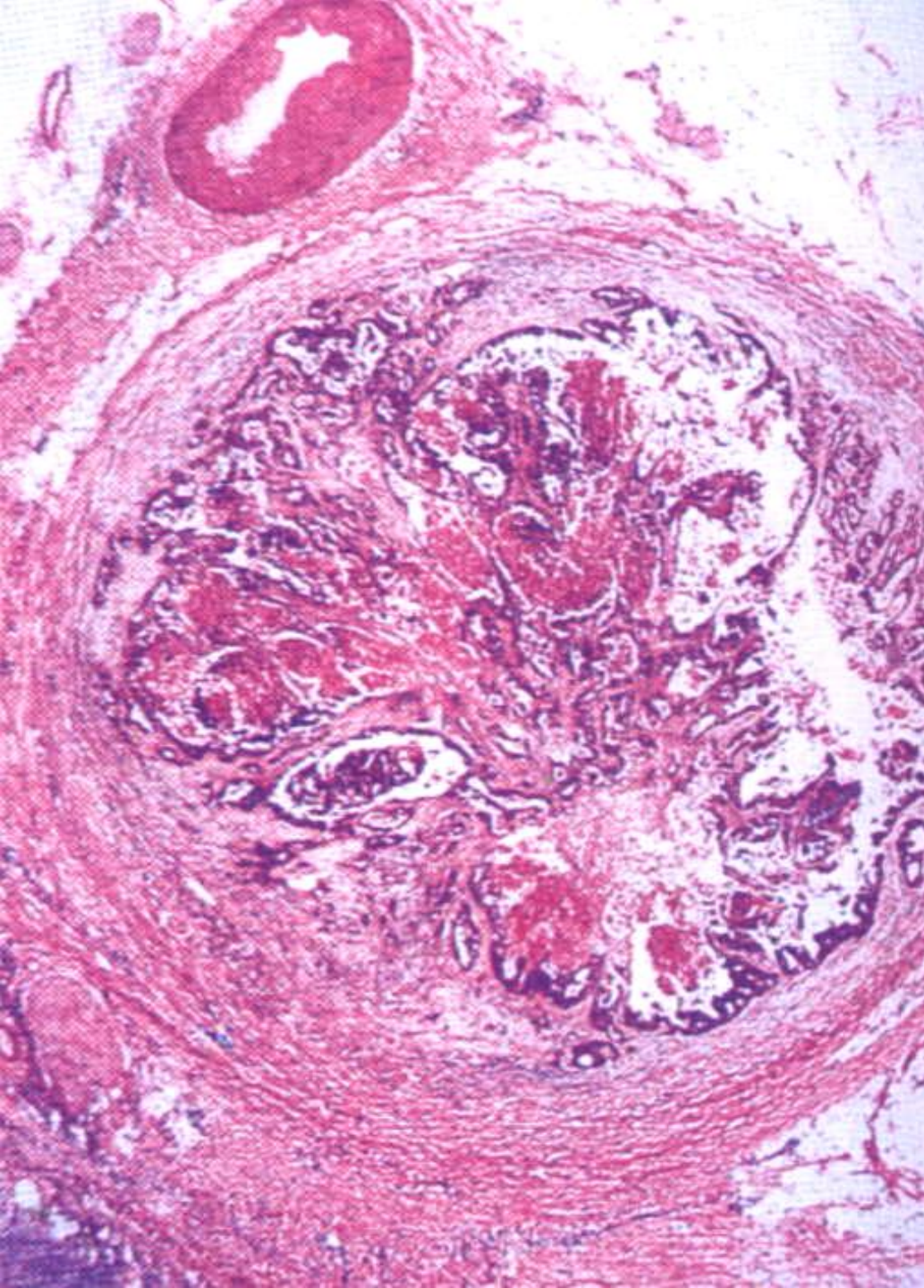
Dysplasia

1. Cells are enlarged and crowded with large pleomorphic nuclei,
2. Increased nuclear to cytoplasmic ratio,
3. Palisading of nuclei
4. Increased mitotic figures



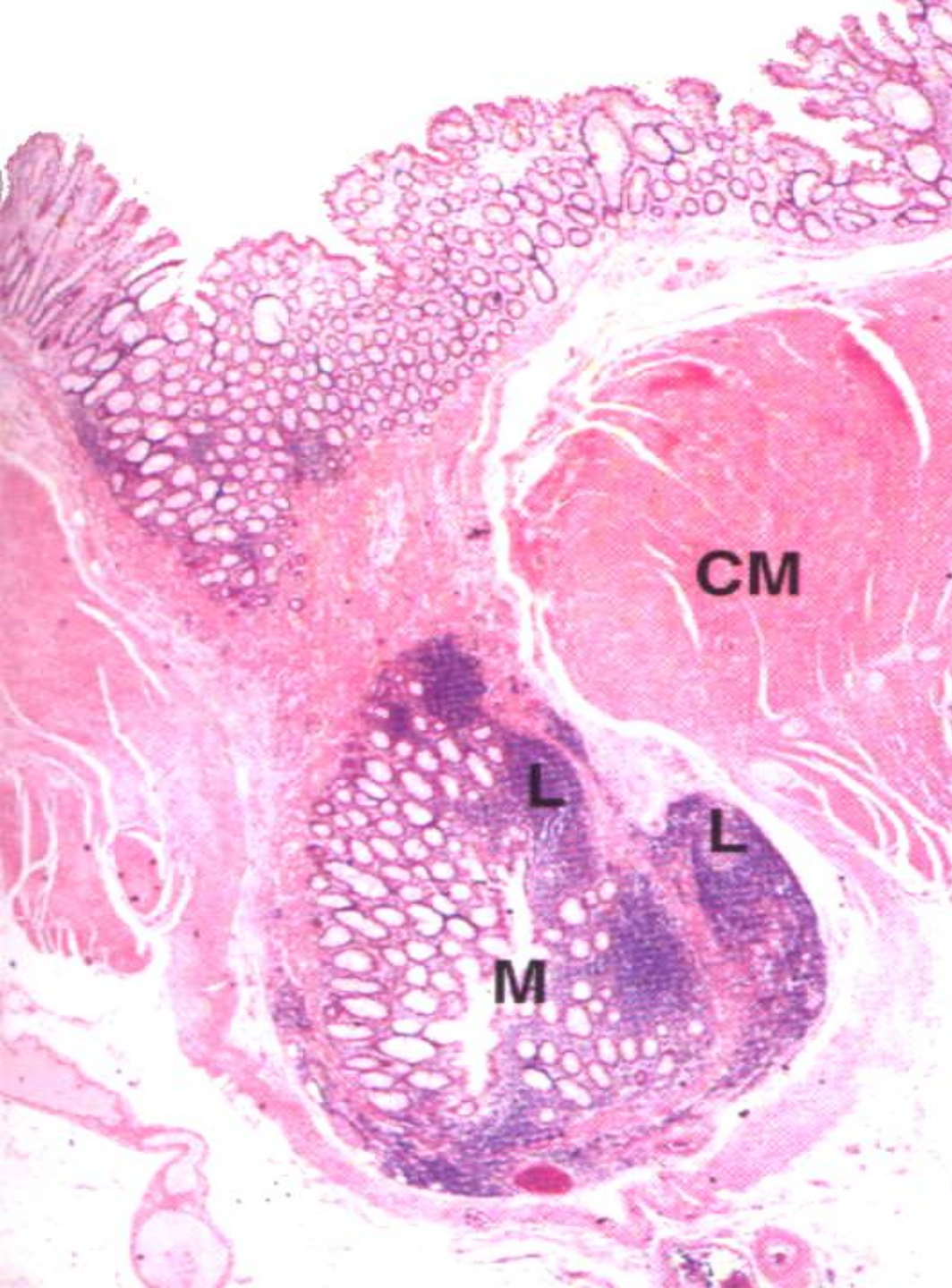
Adenocarcinoma of the colon (Invasive tumour)

F : Pericolic fat, M : Muscle, SM : Submucosa



Adenocarcinoma of the colon

A vein in the serosa of the colon, it is filled with colonic adenocarcinoma.



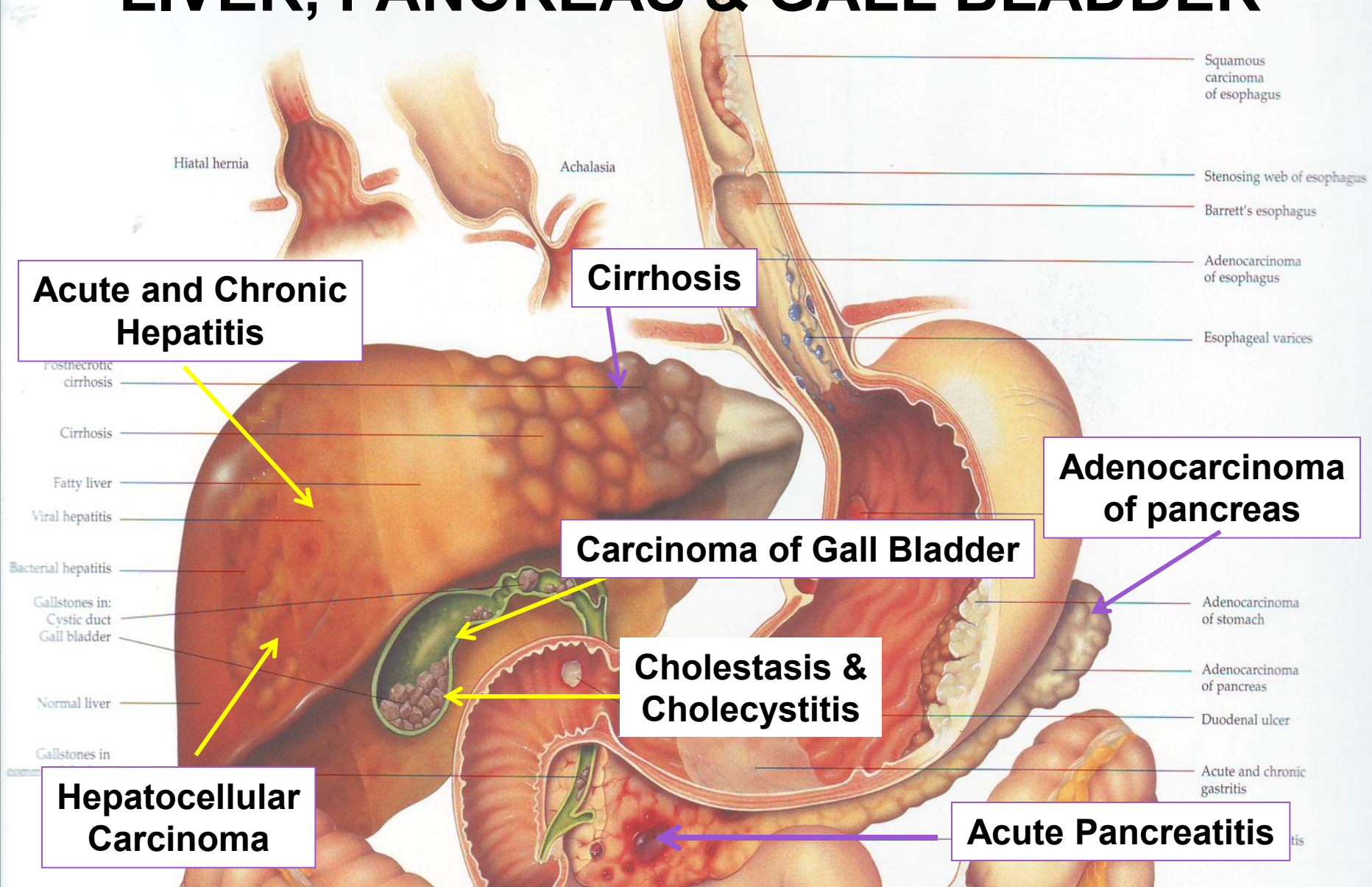
DIVERTICULAR DISEASE

- **Herniation** of pouches of the colonic mucosa (**M**) including mucosal lymphoid tissue (**L**), through unsupported areas of the circular muscle between taenia coli.
- **Marked hypertrophy** of the circular muscle layer (**CM**)

HEPATOBILLIARY SYSTEM AND PANCREAS

Diseases of the Digestive System

LIVER, PANCREAS & GALL BLADDER



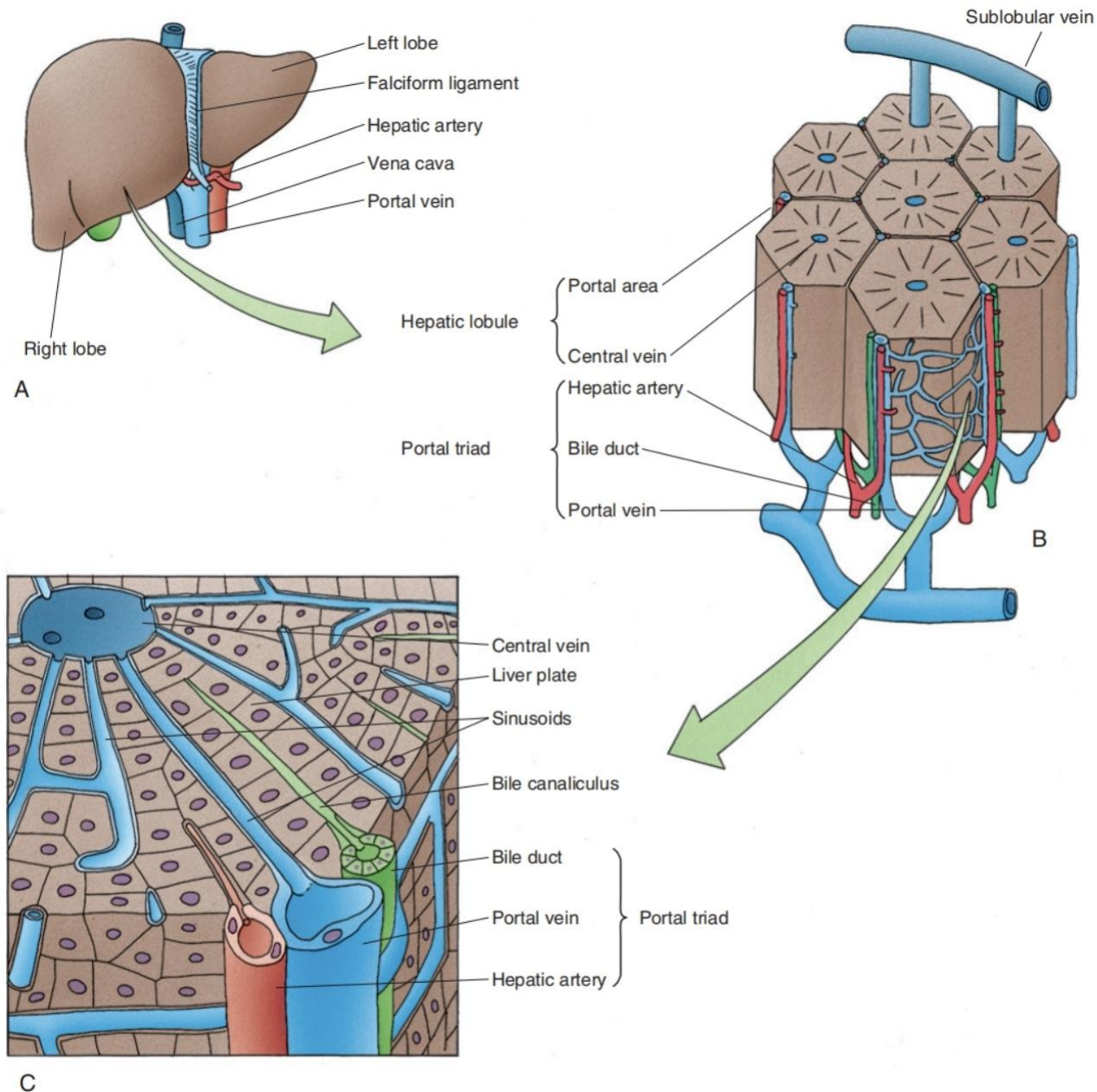
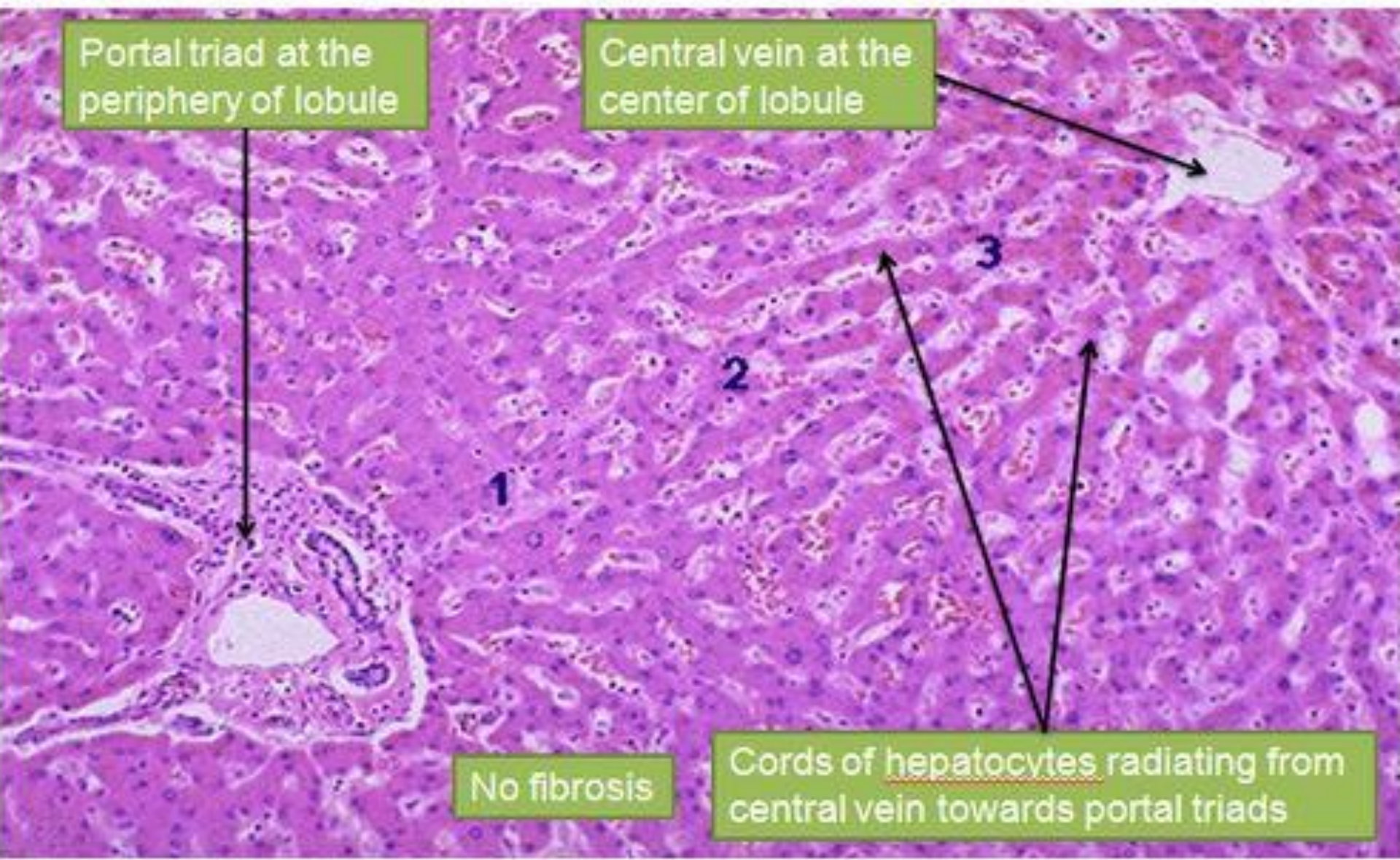


Figure 18-9 Schematic diagram of the liver. **A**, Gross anatomy of the liver. **B**, Liver lobules displaying the portal areas and the central vein. **C**, Portion of the liver lobule displaying the portal area, liver plates, sinusoids, and bile canaliculi.

Normal Liver: Microscopy



Gallbladder

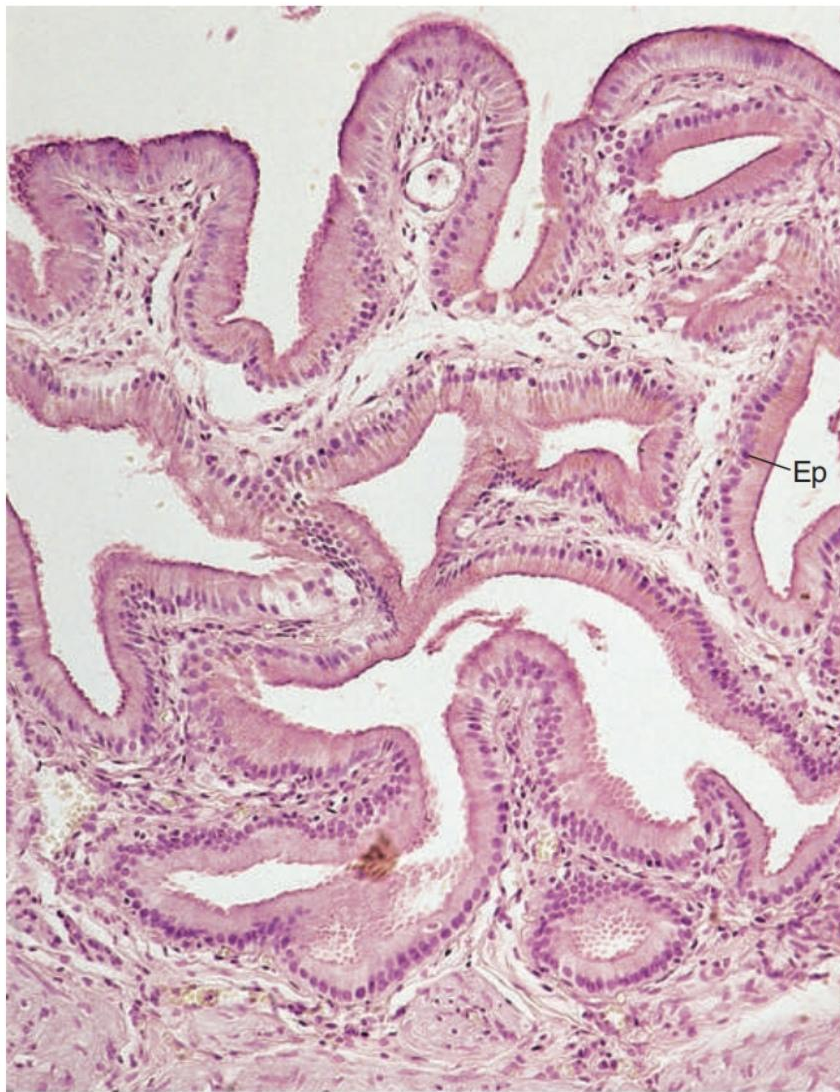


Figure 18-19 Photomicrograph of an empty gallbladder. Observe that the mucosa of the gallbladder is highly folded, indicating that it is empty. Note that the lumen of the gallbladder is lined by a simple columnar epithelium (Ep) ($\times 132$).

Pancreas

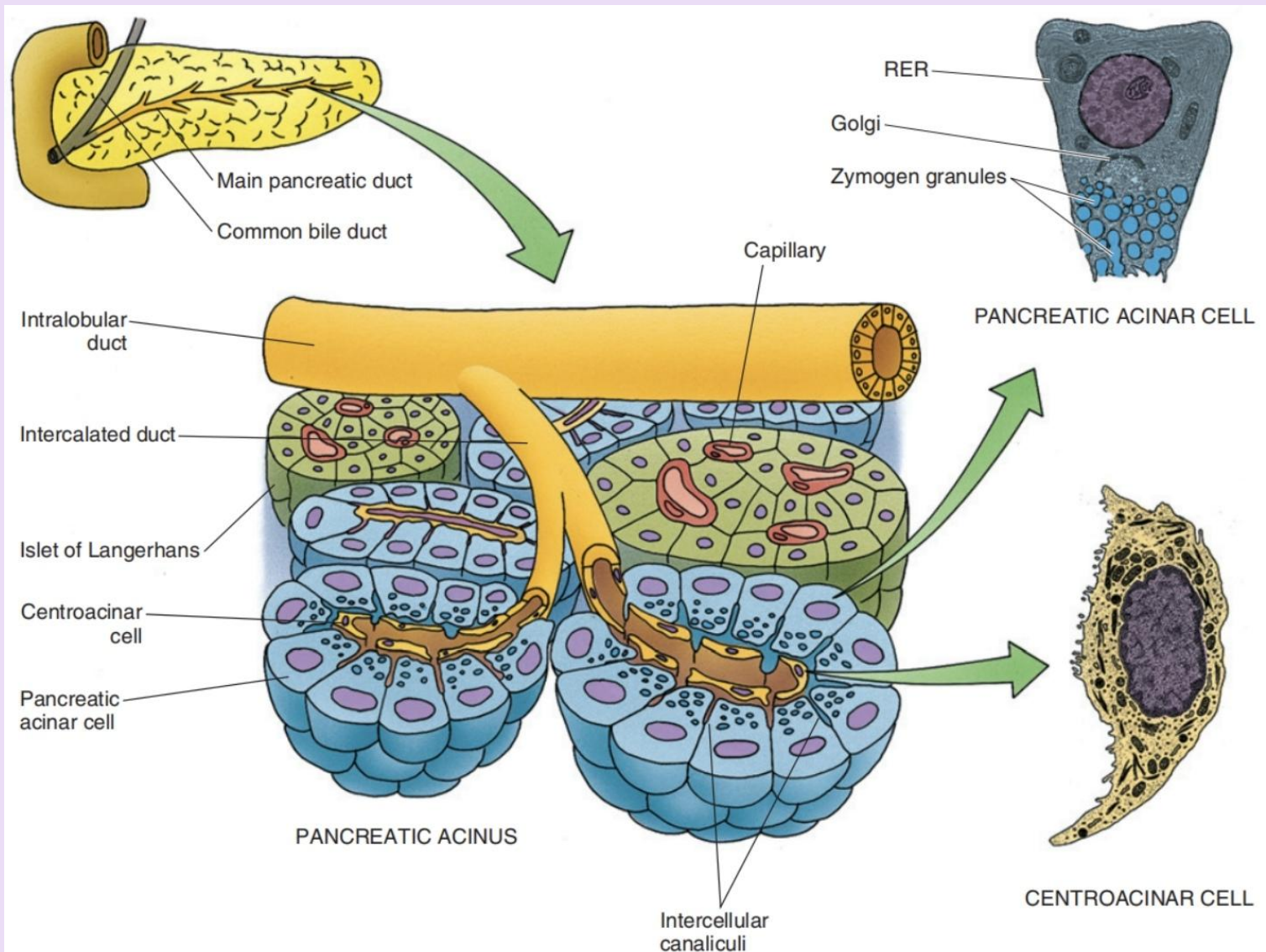
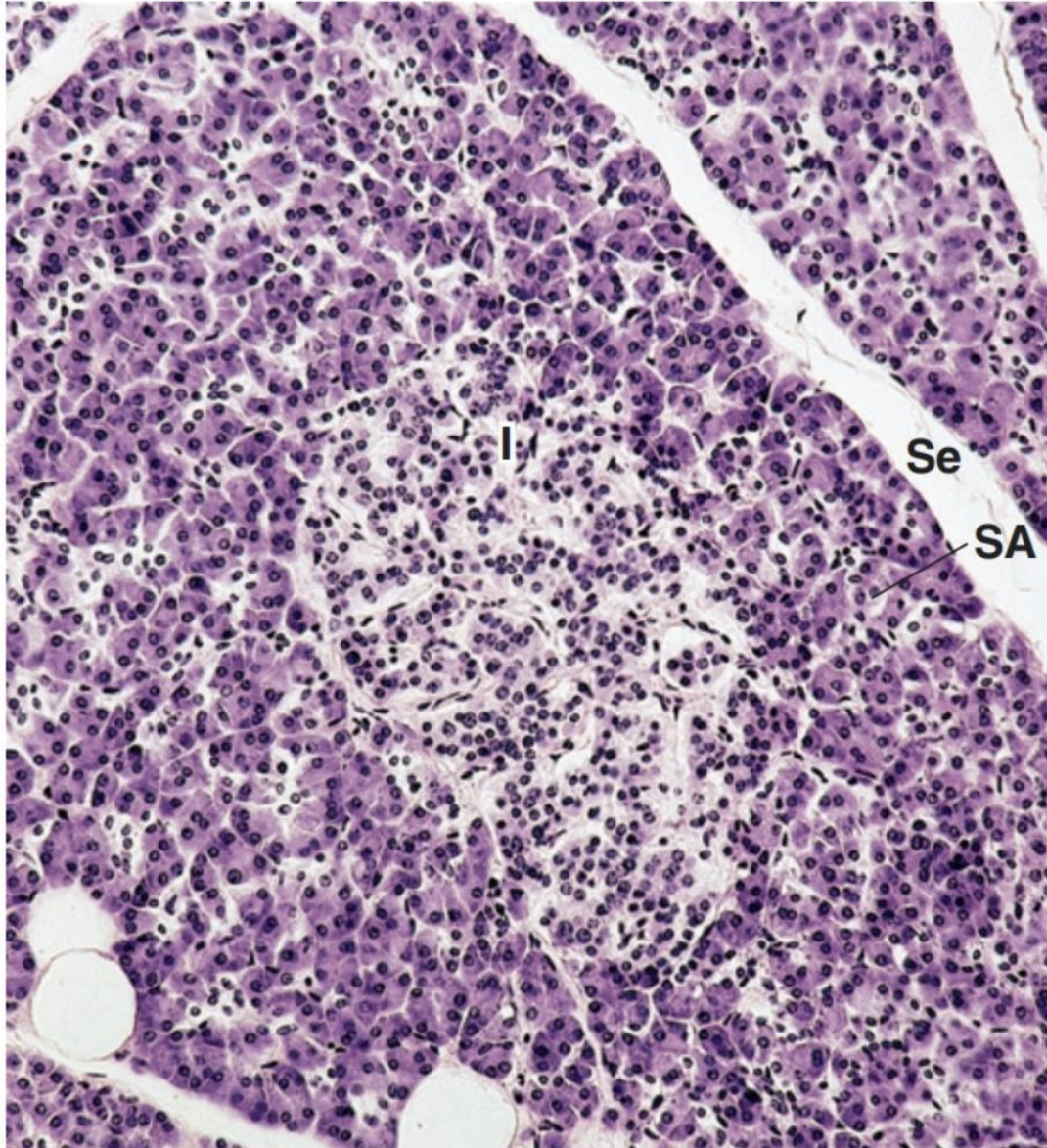


Figure 18-5 Schematic diagram of the pancreas displaying secretory acini, their cell types, and the endocrine islets of Langerhans. RER, Rough endoplasmic reticulum.

Pancreas



I = Islet of Langerhans

Se = Septum

SA = Serous Acinus

ACUTE HEPATITIS

Causes:

- Virus (A, **B**, C, D dan E),
- Toxins (alcohol),
- Drugs (halothane, isoniazid) and
- Systemic infection
(Leptospira, Toxoplasma).

ACUTE VIRAL HEPATITIS

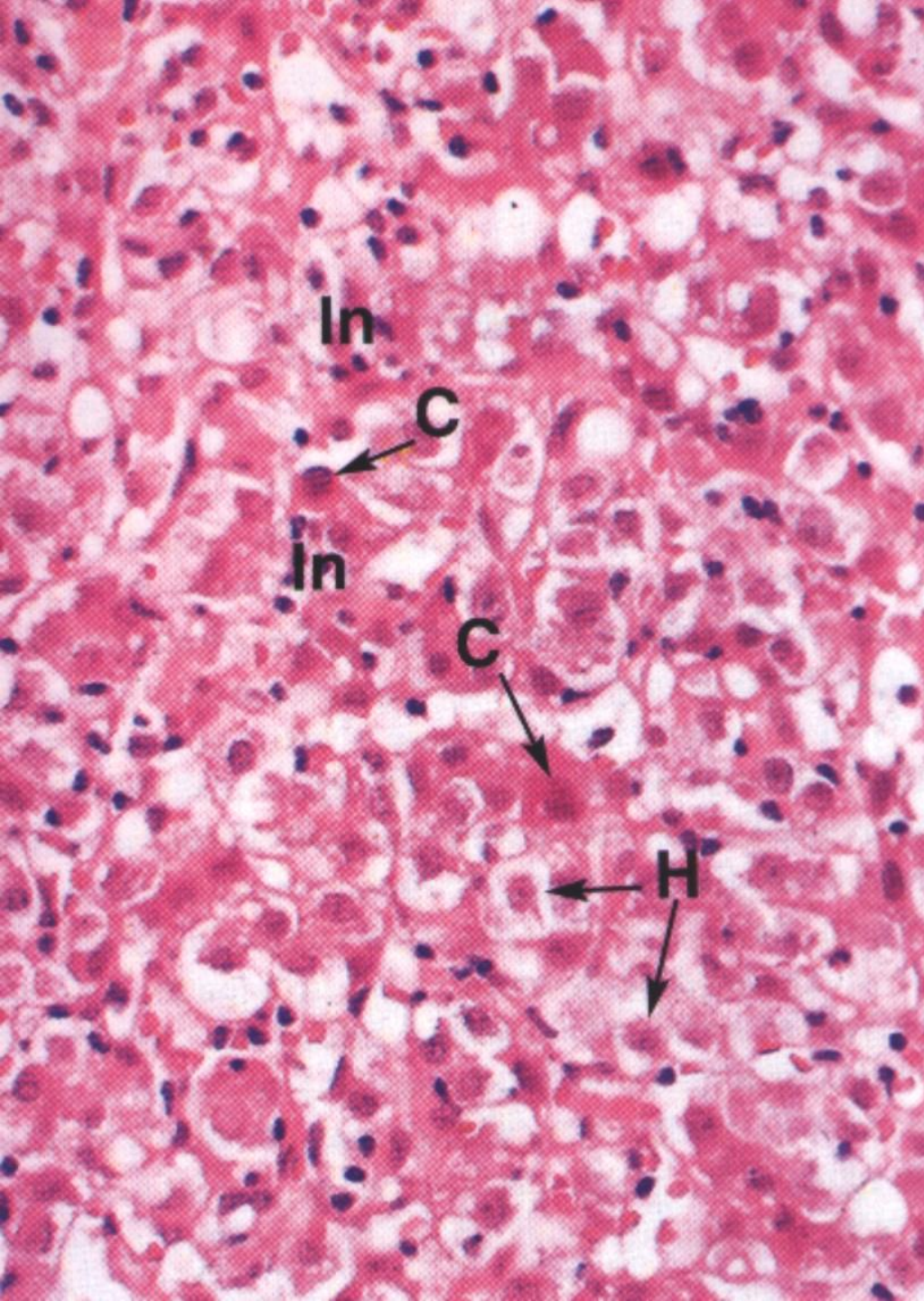
Histopathology:

- Widespread swelling and ballooning hepatocytes (hydropic degeneration) → focal or spotty necrosis
- Aggregates of inflammatory cells surrounding eosinophilic bodies (Councilman bodies), a cytoplasmic of necrotic liver cells.

ACUTE VIRAL HEPATITIS

Histopathology:

- **Kupffer Cells** are very active.
- A great number of **chronic inflammatory cells** within portal tract.
- **Regeneration** of dead hepatocytes, complete resolution (Hep A & E), persist or progress to chronic hepatitis (Vir B,C & D).



ACUTE VIRAL HEPATITIS

C : Councilman bodies

In : Inflammatory cells

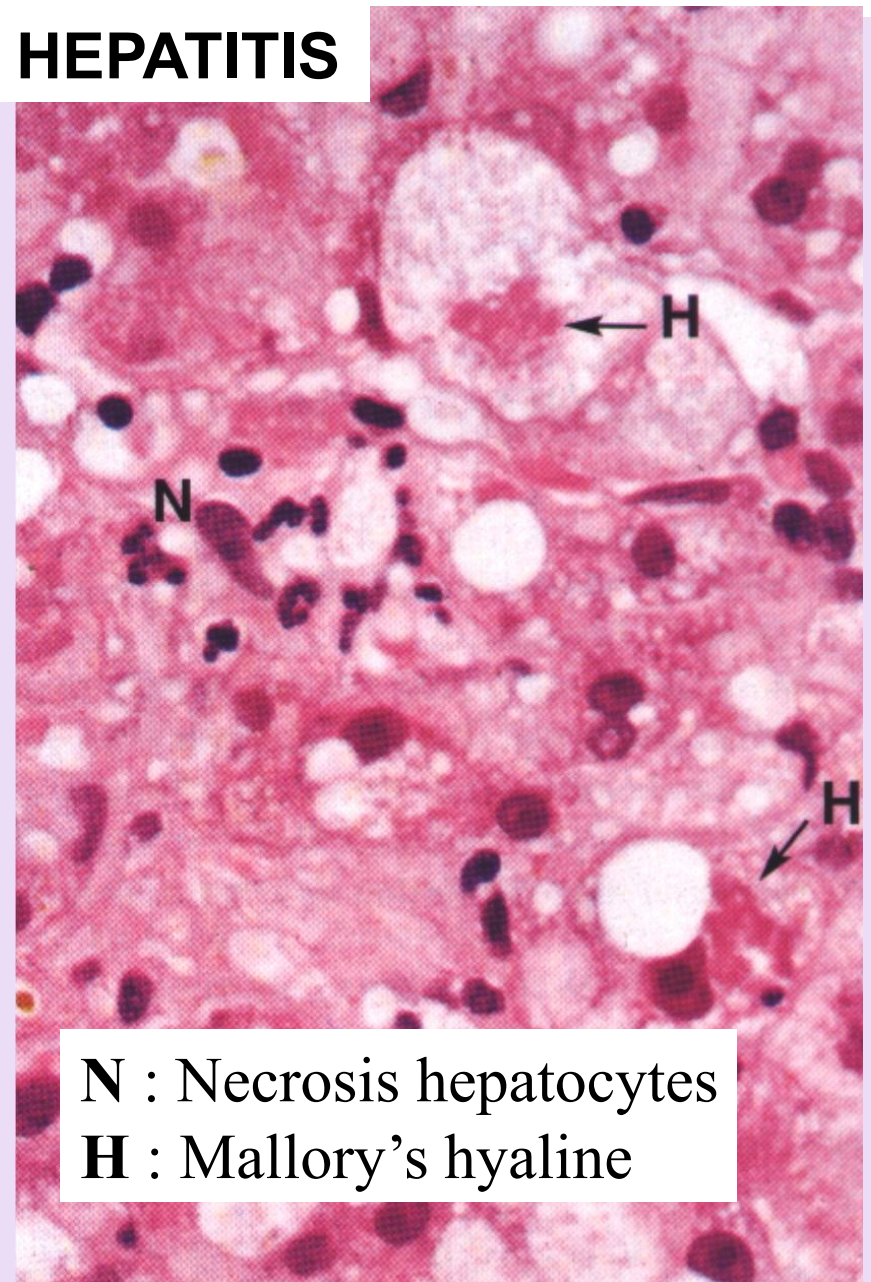
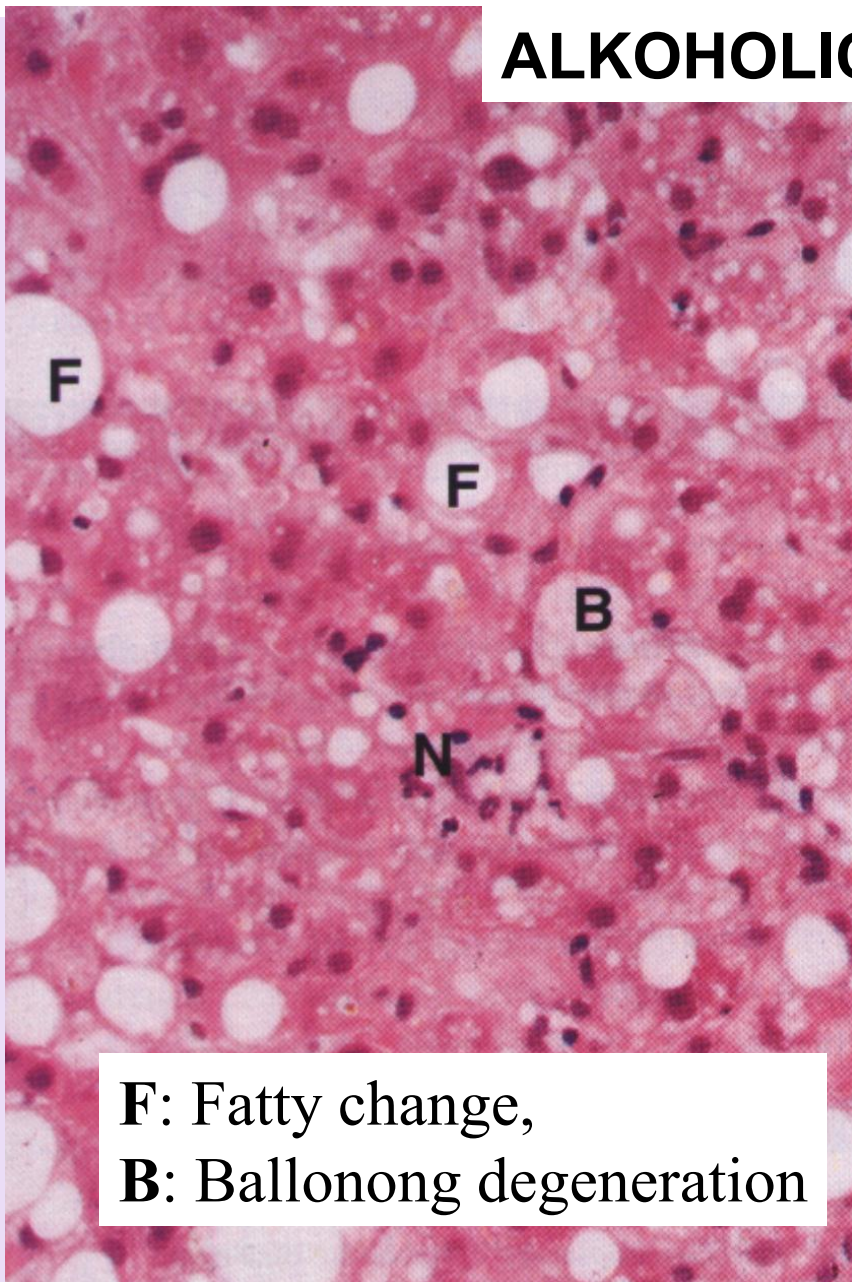
H : Hydropic degeneration
of the hepatocytes.

ALCOHOLIC HEPATITIS

Histopathology

- **Fatty change** (accumulation of lipid in the form of large cytoplasmic vacuoles within some hepatocytes).
- **Balloning degeneration** (swelling of hepatocytes)
- **Necrosis** of hepatocytes is marked by infiltration of neutrophil and lymphocytes.
- **Mallory's hyaline** (cytokeratin intermediate filaments)
- Hepatocytes around nentral vein most vulnerable
→ delicate fibrosis.
- Prolonged abuse → **Alcoholic Cirrhosis**.

ALKOHOLIC HEPATITIS



CHRONIC HEPATITIS

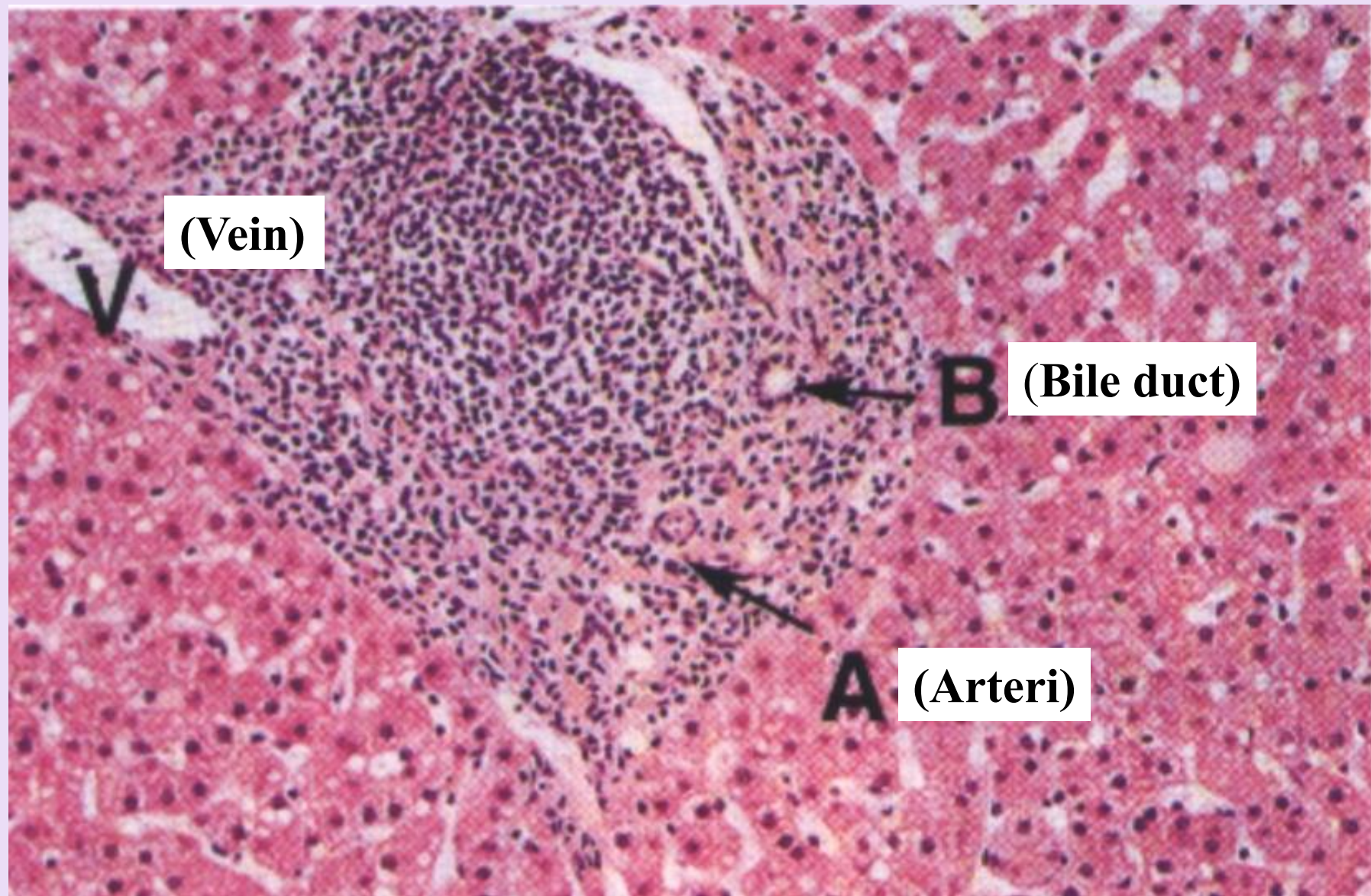
Causes:

1. **Viral Infection** (B, C & other hepatitis viruses).
2. **Autoimmune disease** (Autoimmune hepatitis, Primary biliary cirrhosis)
3. **Toxic/Metabolic:** (Wilson's disease, α_1 antitrypsin deficiency, drug hepatitis).

CHRONIC HEPATITIS

Histopathology.

- Necrosis of hepatocytes and the presence of inflammatory cells (either concentrated around portal tracts or scattered within the liver lobules, or both).
- The presence of fibrosis causing architectural distortion.

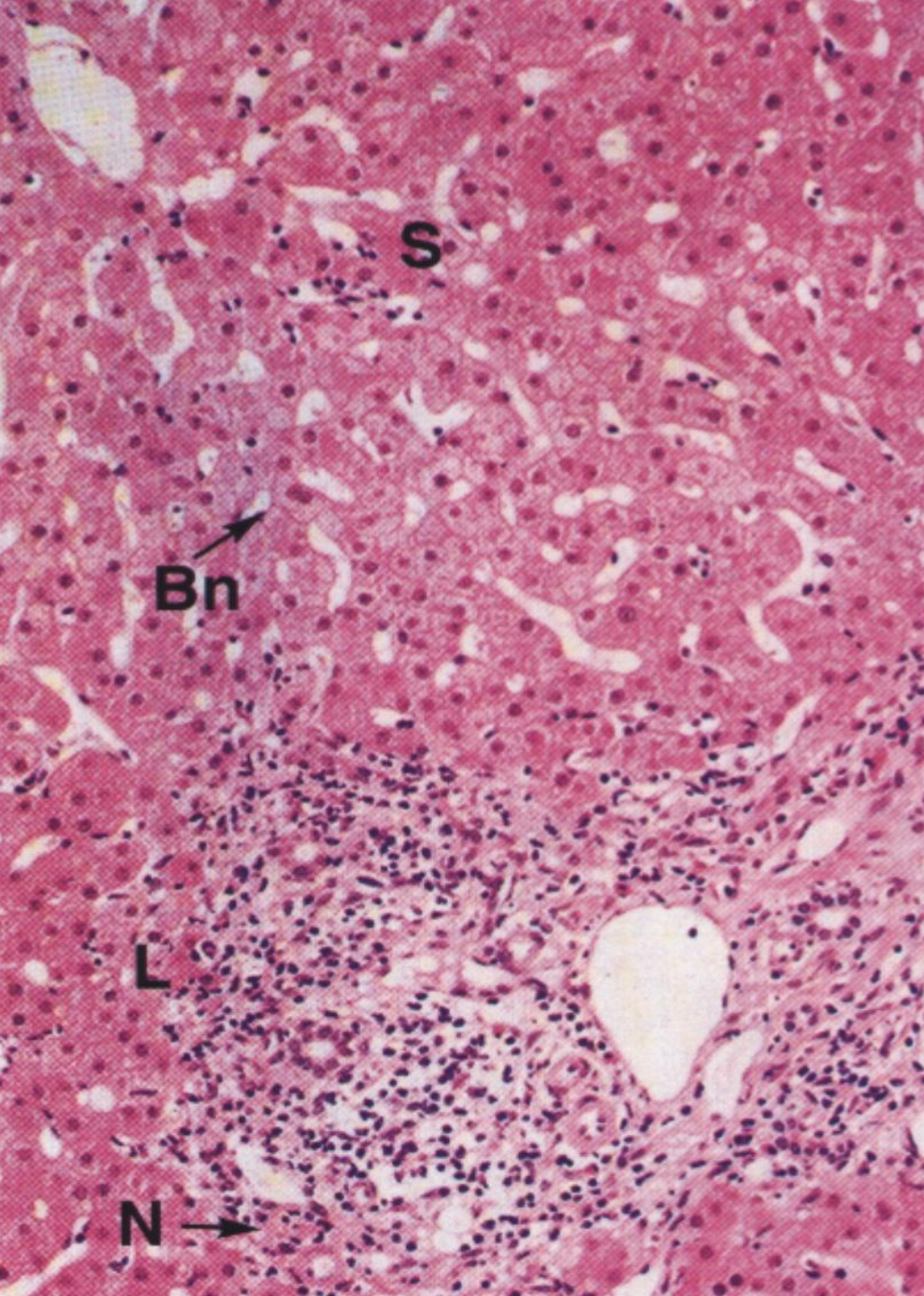


(Vein)

(Bile duct)

(Arteri)

CHRONIC HEPATITIS



CHRONIC HEPATITIS

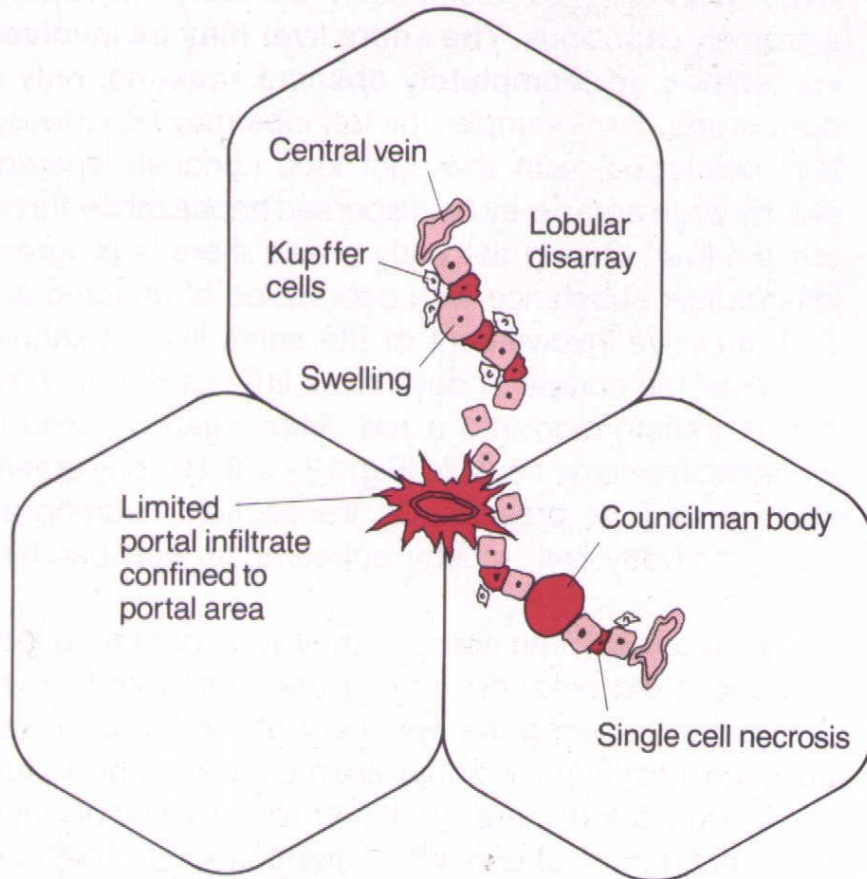
S : Spotty necrosis

Bn : Binucleate cells

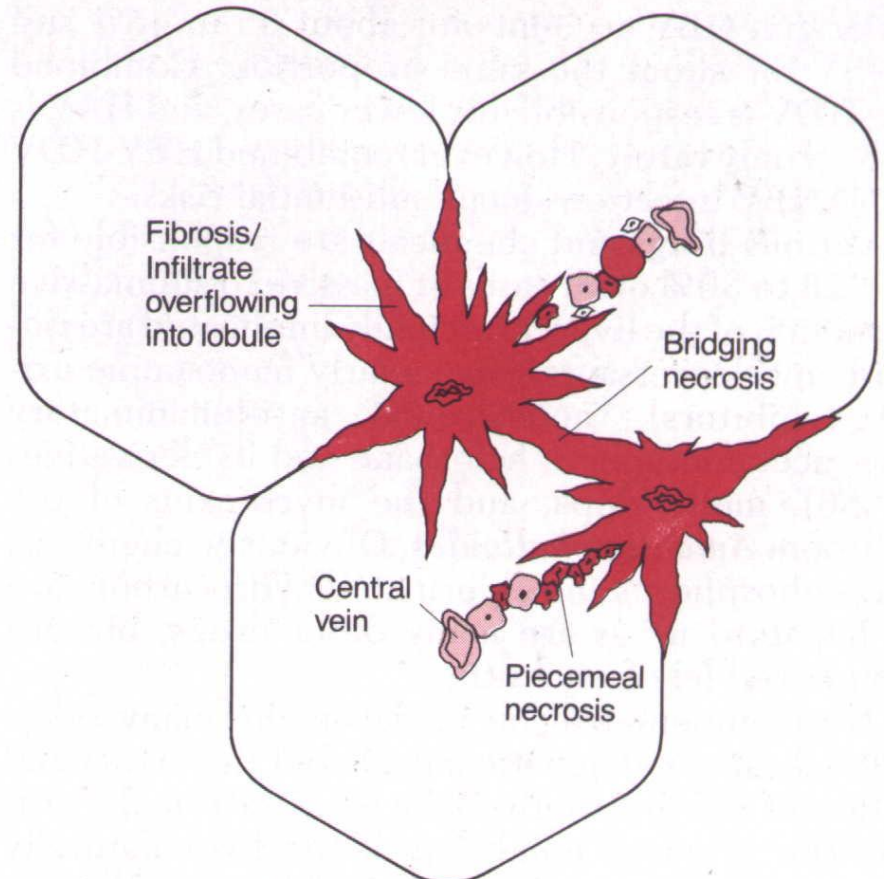
L : Periportal hepatocytes

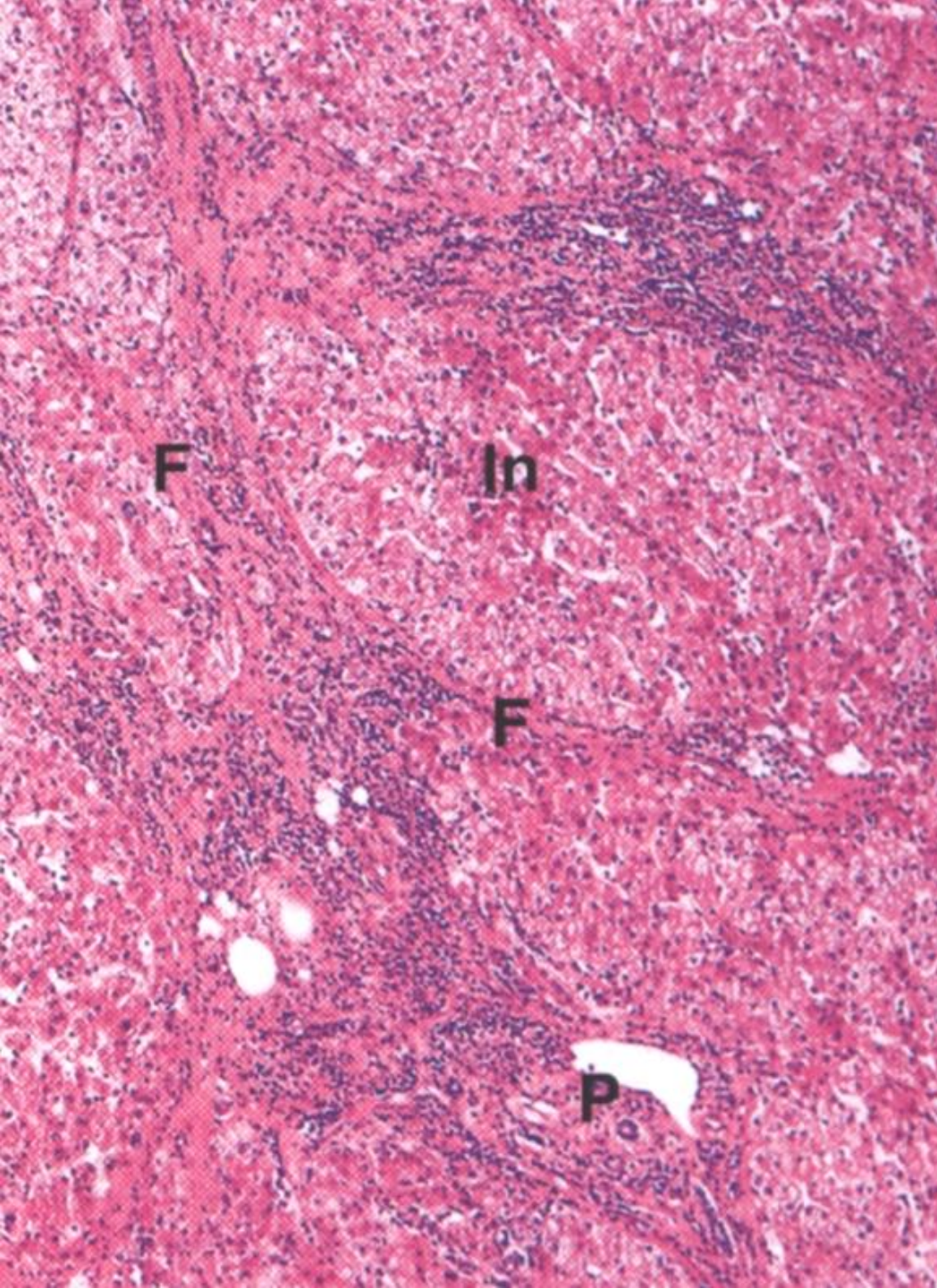
N : Necrosis.

ACUTE



CHRONIC ACTIVE





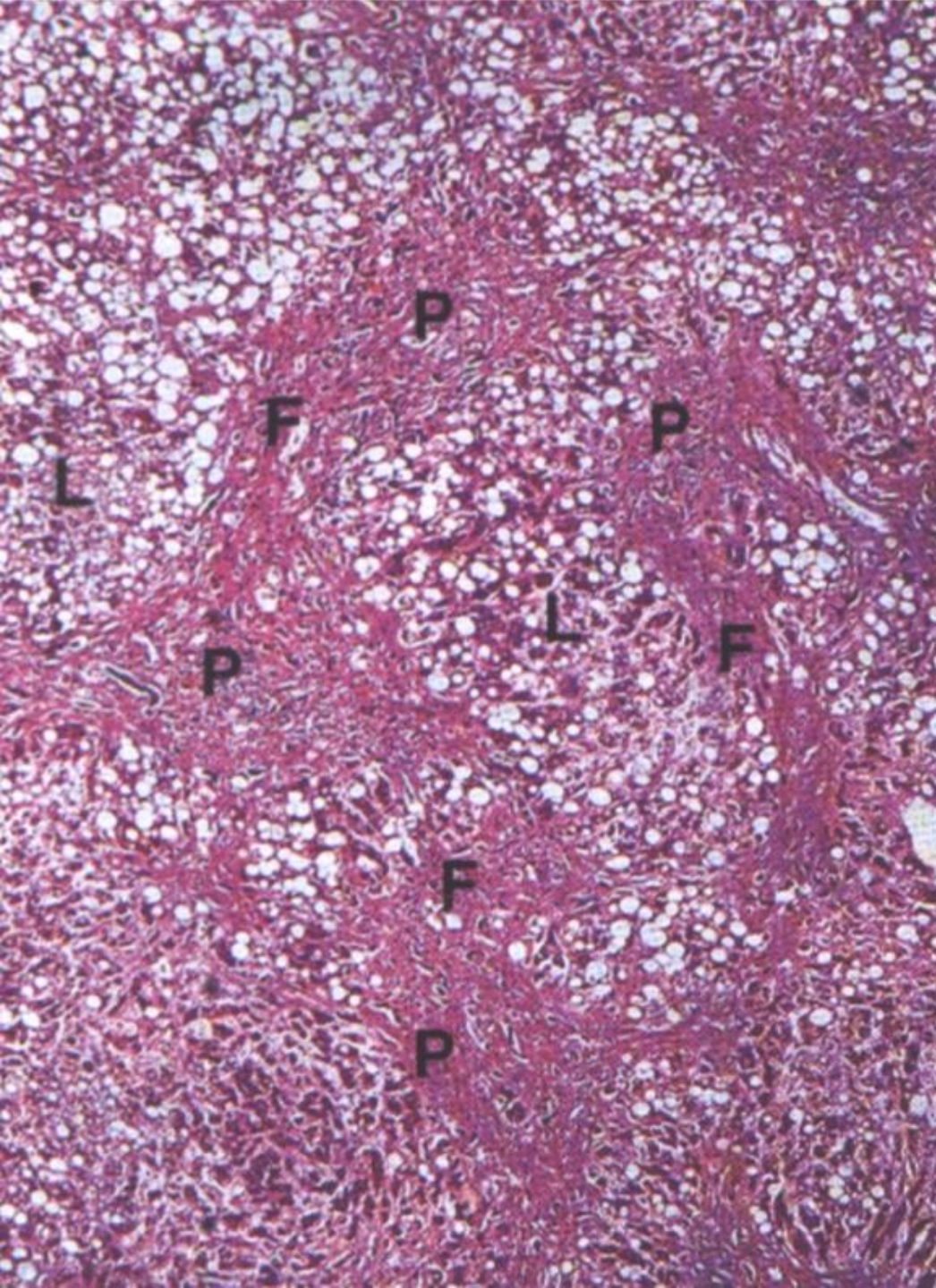
CIRRHOSIS

Cirrhosis due to progressive chronic hepatitis

F : Fibrosis

In: Inflammation.

P : Portal Tract.



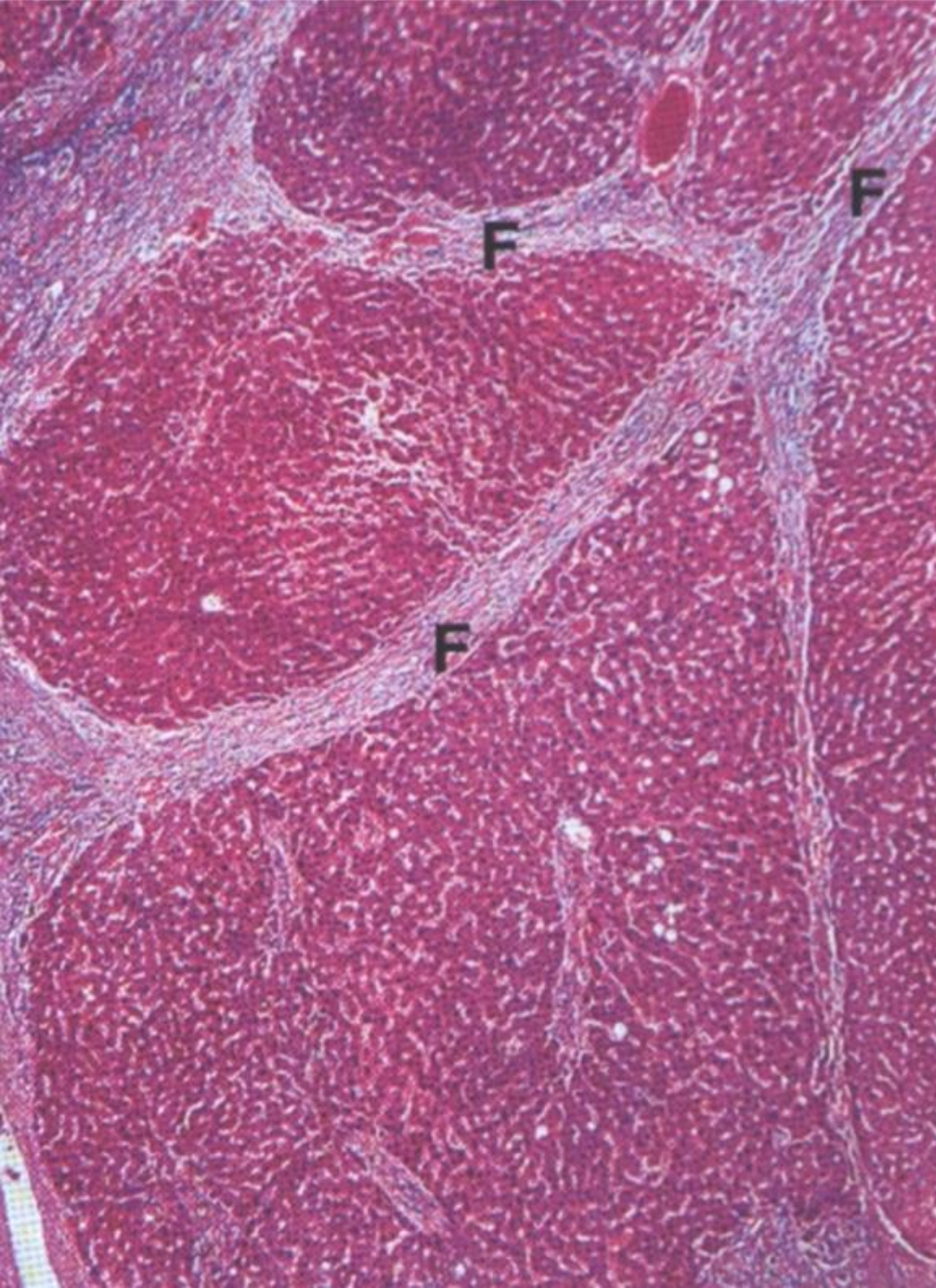
CIRRHOSIS

Alcoholic cirrhosis

F : Fibrosis

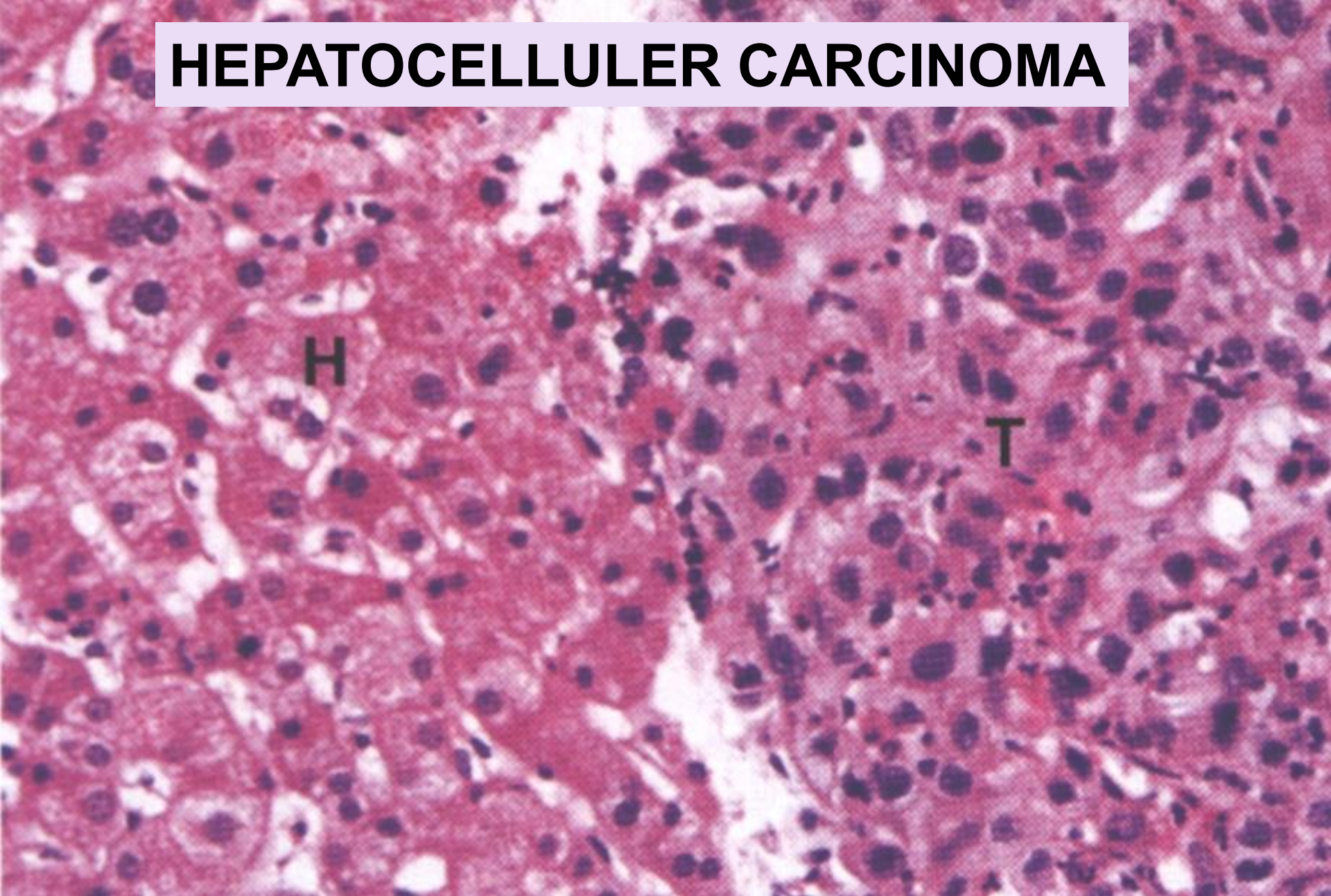
P : Portal areas

L : Nodule of liver cells
(fatty change)



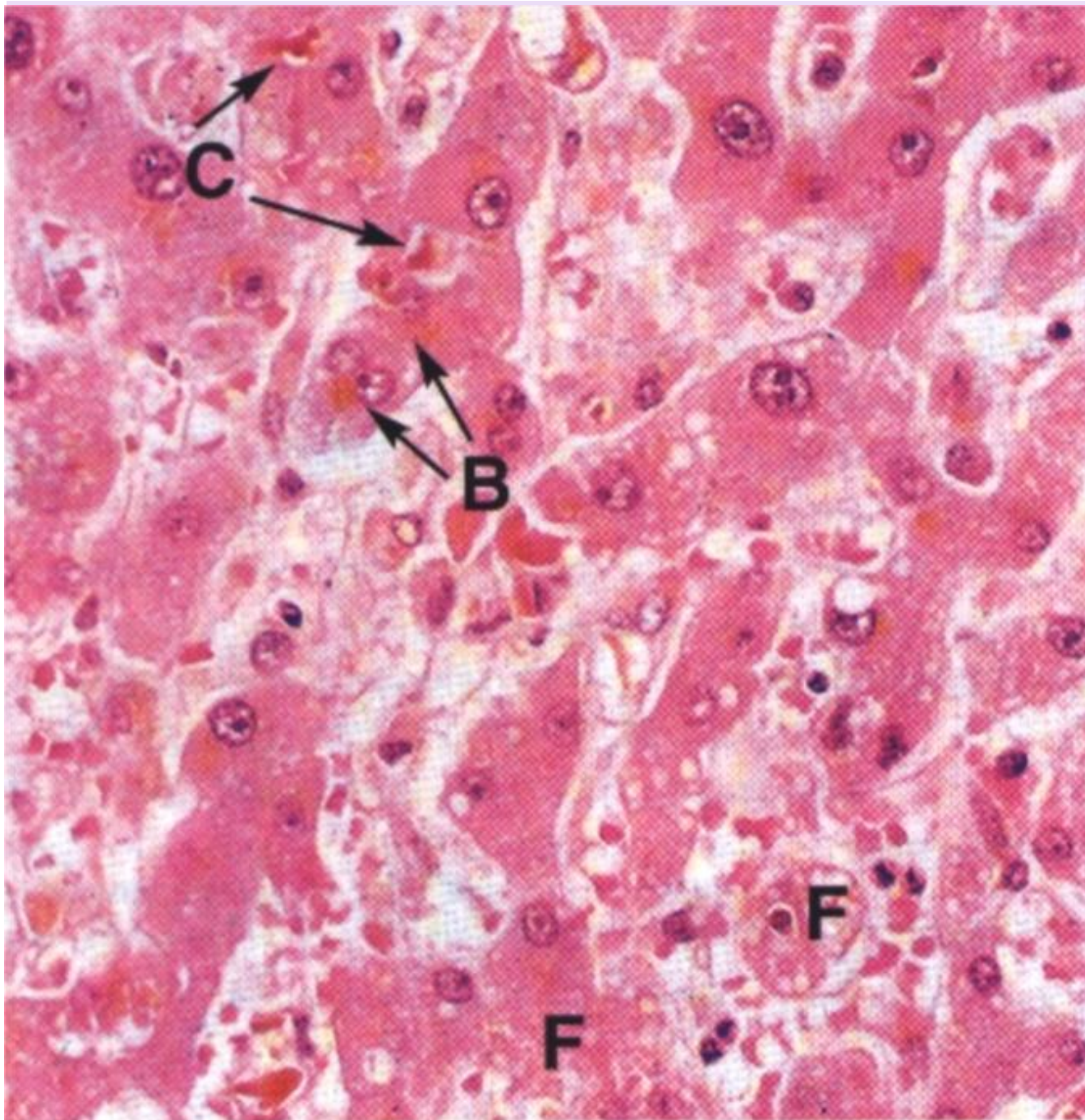
CIRRHOSIS

HEPATOCELLULAR CARCINOMA



H : Normal hepatocytes

T : Malignant hepatocytes



CHOLESTASIS

C : Canaliculi

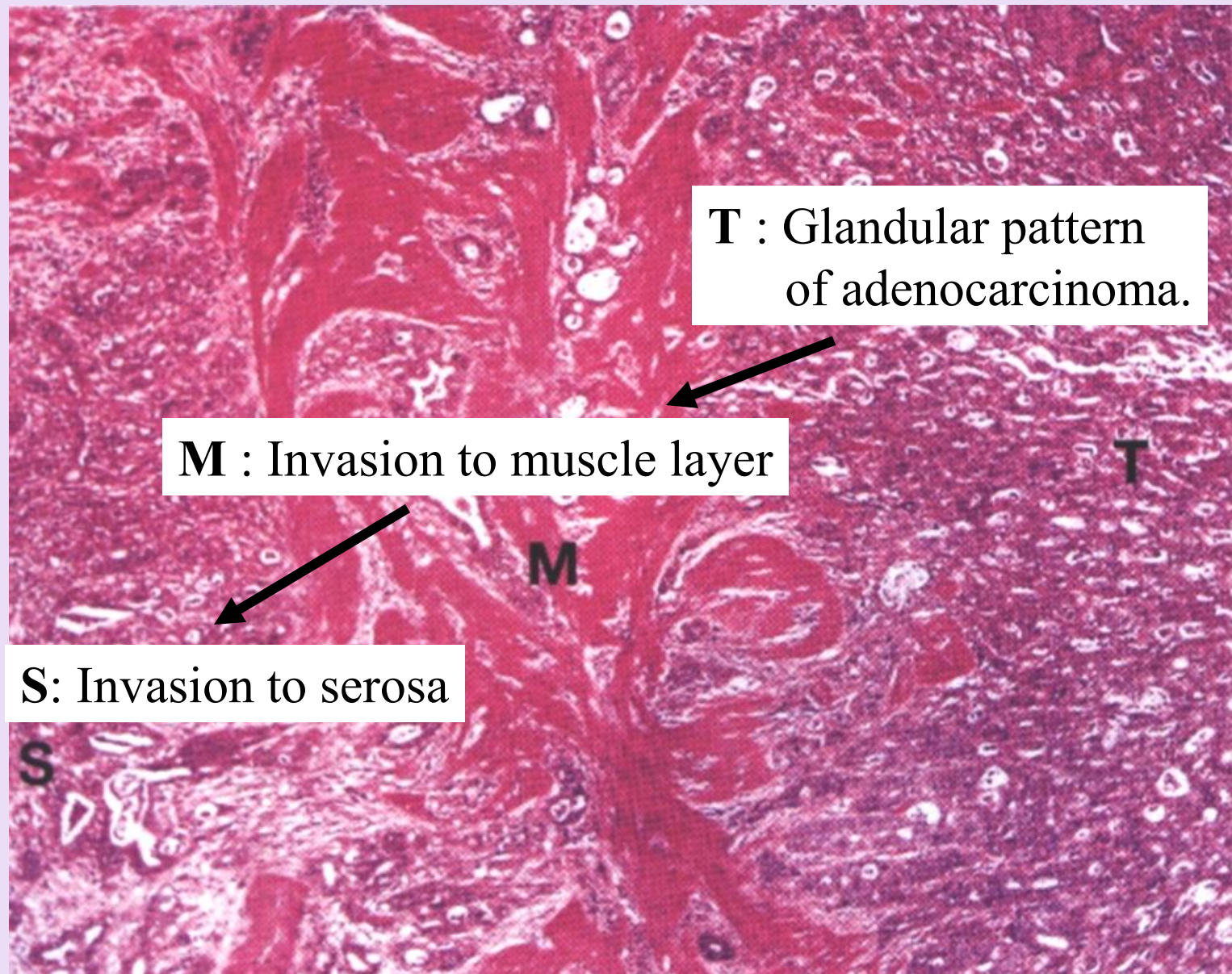
B : Bile

F : Feathery
degeneration
(foamy
cytoplasm)



CHRONIC CHOLECYSTITIS

- M** : Muscle hypertrophy
- R** : Rokitansky-Aschoff
sinuses
- G** : Bile granuloma
(aggregate of
histiocytes)
- F** : Fibrosis (serosa)

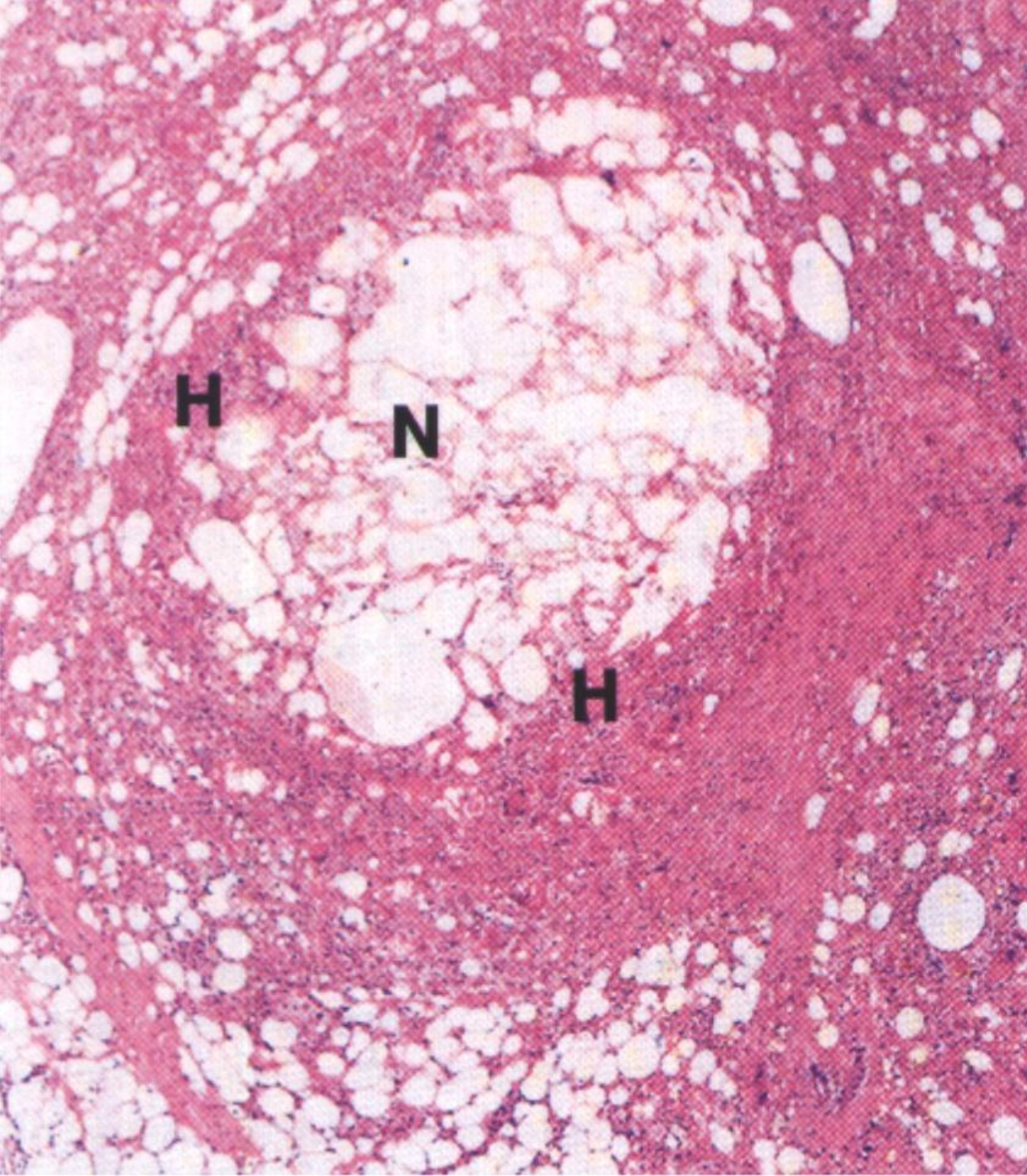


T : Glandular pattern
of adenocarcinoma.

M : Invasion to muscle layer

S: Invasion to serosa

CARCINOMA OF GALLBLADDER



FAT NECROSIS IN ACUTE PANCREATITIS

- Destruction of pancreas because of liberation of pancreatic enzymes
- Alcoholic abuse and biliary tract disease (stones)

H : Foamy histiocytes

N : Necrotic adipose tissue

FAT NECROSIS IN ACUTE PANCREATITIS

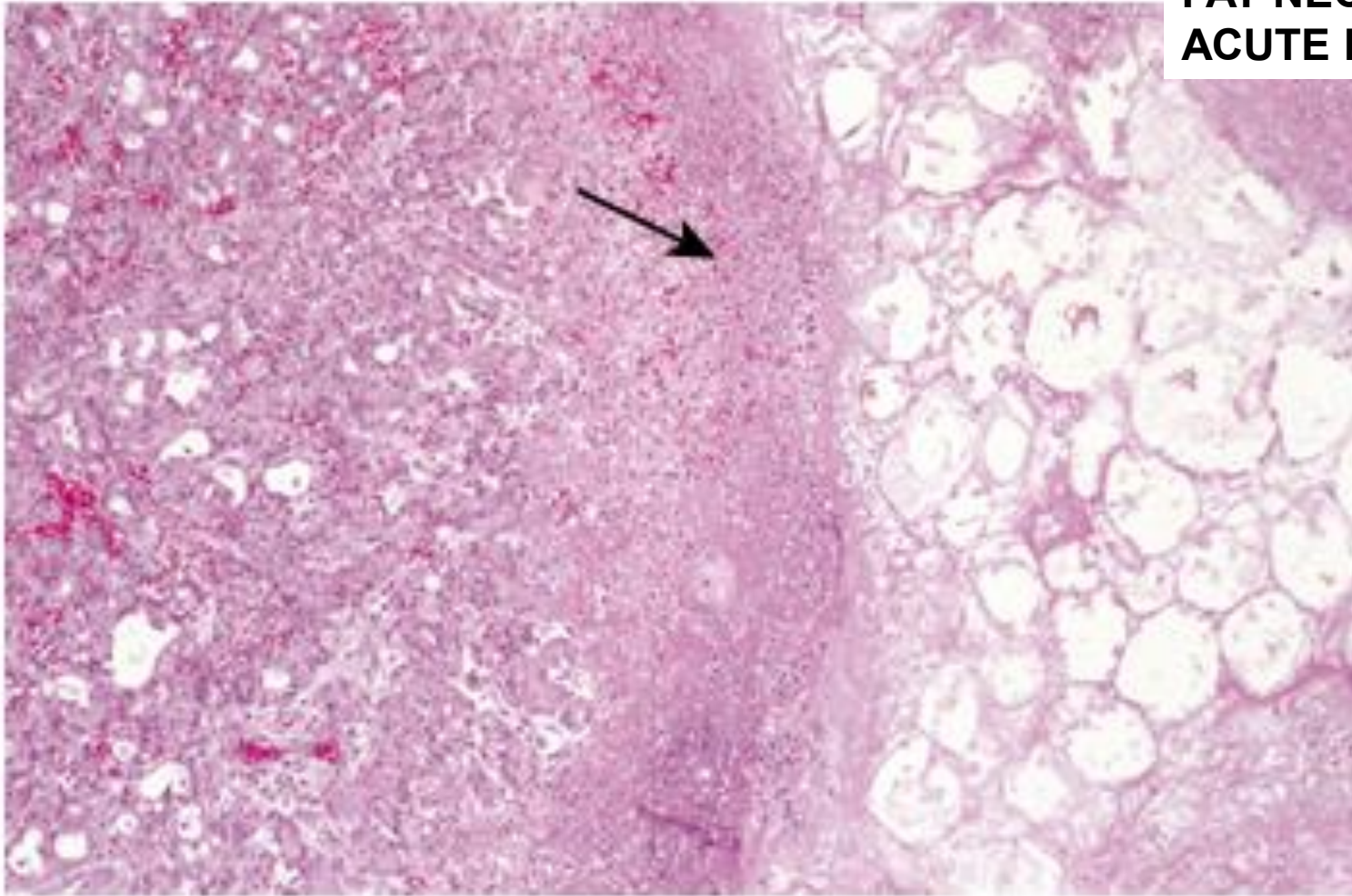
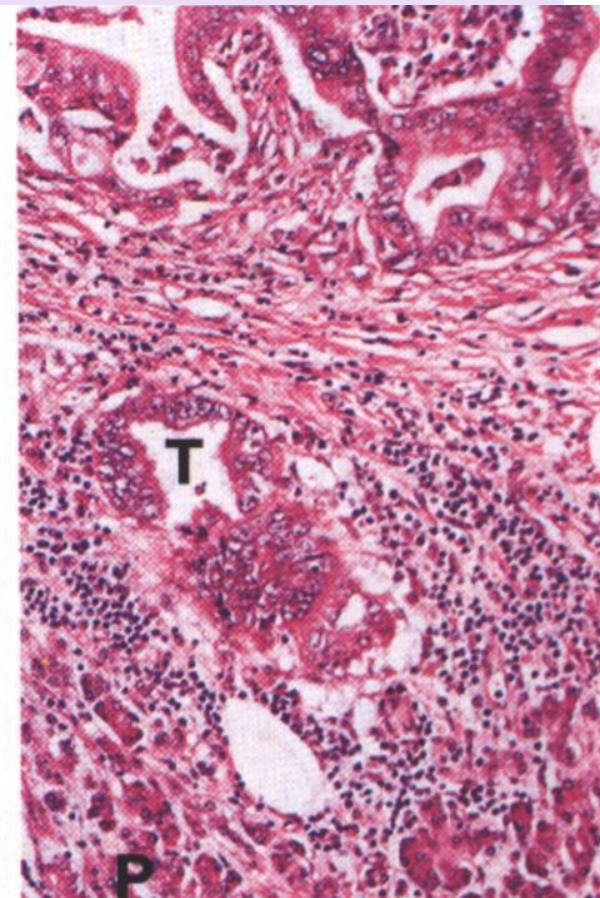
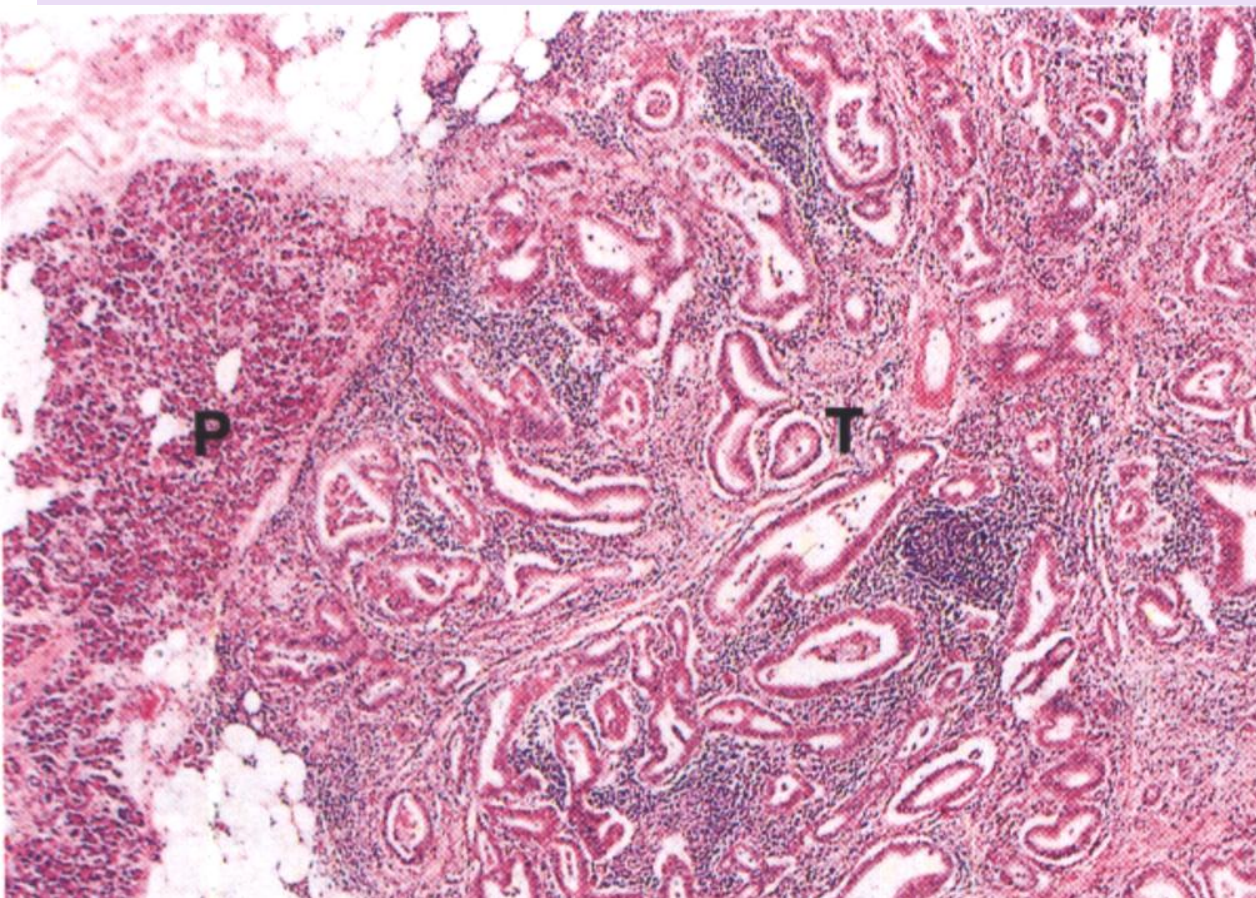


Fig. 19.3 Acute pancreatitis. The microscopic field shows a region of fat necrosis (*right*) and focal pancreatic parenchymal necrosis (*arrow*).

- Destruction of pancreas because of liberation of pancreatic enzymes
- Alcoholic abuse and biliary tract disease (stones)



ADENOCARCINOMA OF THE PANCREAS

T : Tumor cells with ductular pattern , **P** : Normal pancreas

**THANK
YOU!**

